



DOCTORAL THESIS

Formal modelling of cyclic biological behaviours with checkpoints: the regulation of the cell cycle

Déborah BOYENVAL

Laboratoire d'Informatique Signaux et Systèmes de Sophia Antipolis (I3S)
UMR7271 Université Côte d'Azur CNRS

**Submitted for the degree
of Doctor of Philosophy in Computer Sci-
ence
at Université Côte d'Azur**

Supervisor: Gilles BERNOT
Co-supervisor: Franck DELAUNAY
Additional supervisor: Jean-Paul COMET
Defence date: 15 December 2022

Examining committee:

François FAGES, Research Director, Inria
Saclay
Olivier ROUX, Professor, École Centrale de
Nantes
Laurence CALZONE, Research Engineer,
Institut Curie
Hélène COLLAVIZZA, Associate Professor,
Université Côte d'Azur
Clémence FRIIOUX, Research Scientist, In-
ria Bordeaux
Gilles BERNOT, Professor, Université Côte
d'Azur
Jean-Paul COMET, Professor, Université
Côte d'Azur
Franck DELAUNAY, Professor, Université
Côte d'Azur

**FORMAL MODELLING OF CYCLIC BIOLOGICAL BEHAVIOURS
WITH CHECKPOINTS: THE REGULATION OF THE CELL CYCLE**

***Formal modelling of cyclic biological behaviours with checkpoints:
the cell cycle regulation***

Déborah BOYENVAL



Examining committee:

Reviewers

François FAGES, Research Director, Inria Saclay
Olivier ROUX, Professor, École Centrale de Nantes

Examiners

Laurence CALZONE, Research Engineer, Institut Curie
Hélène COLLAVIZZA, Associate Professor, Université Côte d'Azur
Clémence FRIoux, Research Scientist, Inria Bordeaux
Gilles BERNOT, Professor, Université Côte d'Azur
Jean-Paul COMET, Professor, Université Côte d'Azur
Franck DELAUNAY, Professor, Université Côte d'Azur

Déborah BOYENVAL

Formal modelling of cyclic biological behaviours with checkpoints: the regulation of the cell cycle

xi+161 p.

Abstract

The cell cycle is a sequence of events *through* which a cell reproduces while preserving the integrity of its genome. This preservation property is ensured by a set of regulatory mechanisms called *checkpoints*. Their ability to arrest the cell cycle in response to DNA damage motivates extensive research in the biomedical field. This thesis argues that this arrest phenomenon, observed in a context of perturbation, is a corollary of more fundamental properties such as the temporal separation of phases and the irreversibility of phase transitions. In this context, a phase describes a particular aspect of a proliferating cell, for example the genome replication phase (S phase) or the cell division phase (M phase). The fundamental properties of checkpoints have never been formalised, essentially because phase modelling itself had never previously been addressed. The thesis presents a discrete model of cell cycle regulation and proposes a methodology for formalising a phase. A phase is initially described by a finite multiset of discrete events that occur from an initial state. By choosing René Thomas' modelling framework, we can apply genetically modified Hoare logic to compute the set of event sequences, exhibited by the initial multiset, that are compatible with a certain set of parameterisations of a regulatory graph. The set of event sequences characterising a phase temporally separates that phase from the others if there exists a parameterisation such that any event that can complete a phase is required before any event that can initiate the next phase. This problem has been specified using a Prolog predicate. The irreversibility of phase transitions is formalised using a temporal logic, and model checking identifies the parameterisations of a cell cycle model that satisfy this property. We show that this methodology for formalising and verifying the fundamental properties of checkpoints makes it possible to identify contradictions in a regulatory graph while avoiding certain arbitrary choices found in most studies of cell cycle modelling.

Keywords: Cell cycle modelling, phases, checkpoints, formalisation, René Thomas, Hoare logic, model checking, Prolog

Abstract

The cell cycle is a series of events by which a cell reproduces itself while maintaining the integrity of its genome. This property is ensured by a set of regulatory mechanisms called *checkpoints*. Their ability to stop the cell cycle in response to DNA damage has motivated numerous studies in the field of biomedicine. This thesis argues that the phenomenon of cell cycle arrest, which is observed in the context of perturbation, is a consequence of more fundamental properties such as the temporal separation of phases and the irreversibility of phase transitions. In this context, a phase describes a particular aspect of a proliferating cell, for example the phase of genome replication (S-phase) or that of cell division (M-phase). The fundamental properties of checkpoints have never been formalized, mainly due to the fact that phase modelling itself has never been tackled before. The thesis presents a discrete model of the regulation of the cell cycle and proposes a methodology for formalizing a phase. The latter is firstly characterized by a finite multi-set of discrete events which occur from an initial state. By opting for René Thomas' modelling framework, the genetically modified Hoare logic can be applied to compute the set of sequences of events, exhibited by the initial multiset, that are consistent with a certain set of parameterizations of a regulation graph. The set of event sequences that characterizes a phase temporally insulates it from the others if there is a parameterization such that any event that can complete a phase is required before any event that can trigger the next phase. This problem has been specified using a Prolog predicate. The irreversibility property of phase transitions is formalized using a temporal logic and model-checking identifies the parameterizations of a cell cycle model that satisfy this property. We show that this method for formalizing and verifying the fundamental properties of checkpoints allows us to target the inconsistencies of a regulatory graph while getting rid of certain arbitrary choices found in most cell cycle modeling studies.

Keywords: Cell cycle modelling, phases, checkpoints, formalization, René Thomas, Hoare logic, model-checking, Prolog

Contents

Notation	1
1 Introduction	3
1.1 Definition of the cell cycle	5
1.2 Cell cycle control	6
1.2.1 The key molecular players in cell cycle regulation	6
1.2.2 The multiple meanings of the term “checkpoint” in the context of the cell cycle	7
1.3 Motivations	8
1.4 Overview of the thesis	11
 State of the Art	
2 Formalisms for modelling the regulation of cyclic behaviours	15
2.1 An epistemological preamble: why formalise?	17
2.2 Systems of ordinary differential equations	17
2.3 From Boolean networks to multivalued networks	22
2.4 A multivalued approach with multiplexes: René Thomas’s modelling formalism	24
2.4.1 Abstract syntax: influence graph with multiplexes	24
2.4.2 From syntax to semantics: the notion of a dynamic parameter	29
2.4.3 Semantics: the asynchronous state transition graph	35
3 A review of landmark models of cell cycle regulation	39
3.1 Preamble	41
3.2 Pioneering continuous models	43
3.2.1 Identifying a regulatory structure: bistable switches	43
3.2.2 Two intrinsically discrete properties of the checkpoint phenomenon	45
3.3 Discrete models and the formalisation of the checkpoint phenomenon	46
3.3.1 An example of discrete modelling of the regulations involved in certain checkpoints	47
3.3.2 An example of formalising certain chronological constraints during the cell cycle	48
3.3.3 A first attempt to formalise the phases of the cell cycle	49
3.4 Formalising phases with checkpoints: the question remains open	51
3.4.1 Connecting the checkpoint mechanism to its two dynamic properties	52
3.4.2 Intrinsic and extrinsic checkpoints	53
3.5 Overview of checkpoint modelling	54

Formal Methods Used

4	Contribution of two formal methods for discrete regulatory networks	57
4.1	An epistemological preamble: conjunctions and refutations	59
4.2	Model checking of CTL formulae	60
4.2.1	CTL syntax	61
4.2.2	CTL semantics	61
4.2.3	Principle of model checking	64
4.2.4	CTL and fair paths	65
4.2.5	Some examples of properties formalised in (fair) CTL	66
4.3	Genetically modified Hoare logic	67
4.3.1	Syntax	68
4.3.2	Semantics	69
4.3.3	Inference rules	70
4.3.4	Computing the weakest precondition	72
4.4	Managing model exhaustiveness with TotemBioNet	73
4.4.1	Inputs and outputs	73
4.4.2	Workflow	74

Contributions of the Thesis

5	Our set of models for studying the phases of the cell cycle	79
5.1	Abstraction and modelling question	81
5.2	The variables and their discrete thresholds	81
5.3	Our initial conditions	84
5.4	Formalisation of the regulations involved in cell cycle progression	85
5.4.1	Our influence graph designed to define a canonical cell cycle	86
5.4.2	A canonical sequence of regulatory events during the phases of the cell cycle	91
5.4.3	Definition of multiplexes	95
5.5	Initial identification of parameter values	99
5.5.1	Verification of the stability of the quiescent phase	100
5.5.2	Formalisation of a canonical cell cycle as a cyclic attractor	101
5.5.3	Asking the right questions about checkpoints using TotemBioNet	101
5.5.4	Manual identification of certain parameters	103
6	Formalisation of cell cycle phases	109
6.1	Preamble	111
6.2	Syntactic definition of a canonical phase	112
6.3	Formalising a phase from its canonical definition	113
6.3.1	Definition of the hyperrectangle of a Hoare triple	113
6.3.2	Towards formalising a phase as a first-order predicate	115
6.4	Application to our cell cycle model	116
6.4.1	The canonical phases of our model	116
6.4.2	The hyperrectangles of our canonical phases	118

6.4.3	The phases of our model	121
7	Formalisation of a checkpoint between two phases	127
7.1	Preamble	129
7.2	The generic predicate <i>séparation – temporelle</i> (π_i, π_{i+1})	130
7.2.1	The predicates <i>peutClore</i> and <i>peutInitier</i>	130
7.2.2	The <i>estRequis</i> predicate	131
7.3	Irreversibility of phase transitions	133
7.4	Application to our model and our cell cycle phases	136
7.4.1	Implementation of the <i>séparation – temporelle</i> predicate in <code>Prolog</code>	136
7.4.2	Our regulatory graph and its phases with checkpoints	137
7.4.3	The proof of concept	139
8	Conclusion and perspectives	141
8.1	Assessment	141
8.1.1	Methodological contributions	142
8.1.2	Contributions to cell cycle modelling	142
8.2	Perspectives	143
8.2.1	Model inference using our predicates	143
8.2.2	On our modelling of mammalian cell cycle checkpoints	143
8.2.3	Some applications in biology	144

Appendices

A	Parameterisation of our cell cycle GRBM	157
----------	--	------------

List of Definitions

1	Hill functions	18
2	Heaviside step functions	21
3	Biological regulatory graph with multiplexes	25
4	Family of dynamic parameters	29
5	State	29
6	Applicable parameter	29
7	Biological regulatory network with multiplexes	30
8	Snoussi's condition	30
9	Asynchronous state transition graph	35
10	Non-functional edge	36
11	CTL formula	61
12	Kripke structure	62
13	Labelling function	62
14	Semantics of a CTL formula using a Kripke structure	63
15	Satisfaction relation for a CTL formula and a model	63
16	Assertion language	68
17	Path language	69
18	Hoare triple	69
19	Semantics of a path	70
20	Semantics of a Hoare triple	70
21	Increment and decrement axioms	71
22	Inference rule for sequential composition	72
23	Inference rule for the empty path	72
24	Bounds axioms	72
25	The weakest precondition <i>wp</i>	72
26	Elementary syntactic path	112
27	Elementary Hoare triple	112
28	Hyperrectangle of an elementary Hoare triple	113
29	Predicate <i>estAdmis</i>	116
30	Predicates <i>peutInitier</i> and <i>peutClore</i>	130
31	Predicate <i>séparation – temporelle</i>	131
32	Predicate <i>estRequis</i>	132
33	Predicate <i>irréversible</i>	133

Notation

CFCT	Terminal Strongly Connected Component
CPR	Pre-Replication Complexes
CTL	Computational Tree Logic
CYC/CDK	CYclin/Cyclin-Dependent Kinase Complex
EDO	Ordinary Differential Equations
GF	Growth Factors
GRBM	Biological Regulatory Graph with Multiplexes
GTA	Asynchronous Transition Graph
LHGM	Genetically Modified Hoare Logic
RRBM	Biological Regulatory Network with Multiplexes
SAC	Spindle Assembly Checkpoint

CHAPTER 1

Introduction

This thesis aims to formalise a key concept in the regulation of biological systems: checkpoints. The notion of a checkpoint suggests that a particular event is blocked until a set of tasks has been completed. The best-known biological checkpoints are those intrinsic to the process necessary for the survival of every species, namely the cell division cycle. This physiological system is among the most extensively studied, both experimentally and mathematically, leading us to navigate an ever-growing complexity. In this introduction, we choose to give readers who are not biologists an overview of the complexity of cell cycle checkpoints, which first requires defining the key concepts of cell cycle biology. We will then argue that a biological checkpoint is an intrinsically discrete process based on blocking, rather than delaying, a biological regulatory event.

1.1	Definition of the cell cycle	5
1.2	Cell cycle control	6
1.2.1	The key molecular players in cell cycle regulation	6
1.2.2	The multiple meanings of the term “checkpoint” in the context of the cell cycle	7
1.3	Motivations	8
1.4	Overview of the thesis	11

1.1 Definition of the cell cycle

Cellular reproduction is a fundamental characteristic of all living organisms. The cell cycle is a sequence of events through which cells reproduce while preserving the integrity of their genome. Evolution shows a tendency towards increasing complexity in this process, particularly in multicellular eukaryotes [89], making it more difficult to identify the determinants of this integrity. The cell cycle is generally described by the sequence of phases $G1 \rightarrow S \rightarrow G2 \rightarrow M$, where S and M are the two critical phases, *cf.* Figure 1.1. S phase (for “synthesis”) is a phase during which

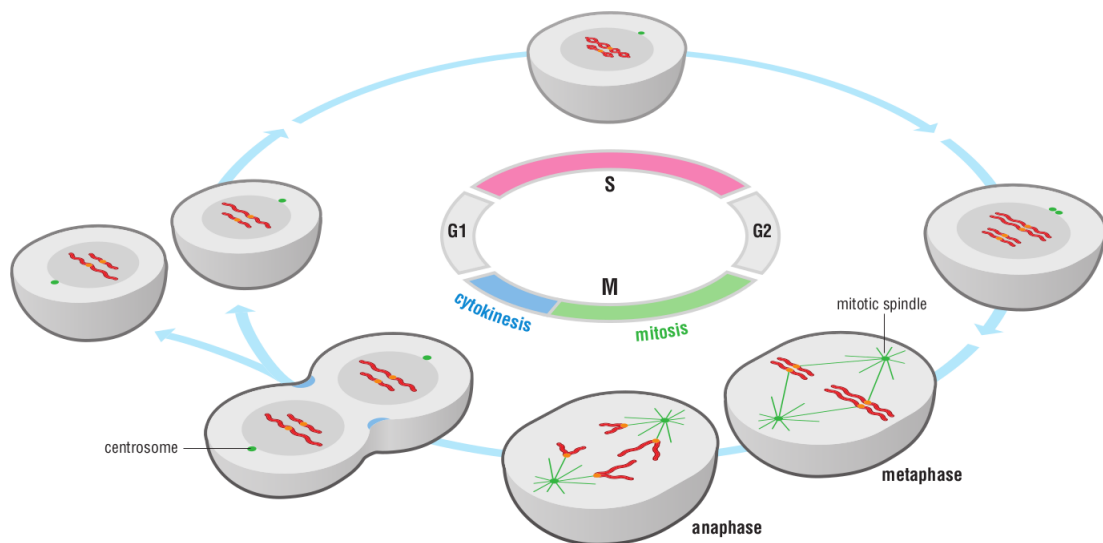


Figure 1.1: **The cell cycle and its four phases.** The DNA replication (S) and cell division (M) phases are the two critical phases of the cell cycle, which must necessarily take place separately. Figure taken from [33].

DNA is fully replicated. M phase (for “mitosis”) is the phase during which a mother cell divides into two daughter cells. Experimentally, the phase a cell is in can be identified by measuring its DNA content, for example using flow cytometry. A cell in G1 phase contains two copies of each chromosome. The cell is said to be diploid ($2n$). A cell in S phase sees its DNA content double ($4n$). At the end of mitosis, the two daughter cells resulting from the division of the mother cell are once again diploid.

M phase is subdivided into 4 subphases. The first is a phase of DNA condensation into chromosomes (prophase). The second coincides with the attachment of chromosomes to the mitotic spindle and their alignment along an imaginary line equidistant from the two opposite poles of the cell (metaphase). A chromosome consists of two sister chromatids (in red in Figure 1.1) joined by a structure called cohesin. At the very end of metaphase, cohesion between the two chromatids is destroyed, leading to the separation of the sister chromatids (anaphase). The sister chromatids are then moved, each to an opposite pole, to a structure called the centrosome (in green in Figure 1.1), and the nuclear envelope reforms around the DNA of the two future daughter cells (telophase). The end of telophase is commonly marked by the division of the mother cell into two daughter cells (cytokinesis).

S and M phases are both preceded by a preparatory phase called a “gap”. G1 is the phase preceding replication and G2 the phase preceding mitosis. Finally, in unfavourable growth conditions or in

the presence of inhibitory signals from other cells, a cell may arrest for long periods in a prolonged non-dividing state.

1.2 Cell cycle control

The cell cycle process is subject to stringent genetic controls that ensure the faithful transmission of DNA from one generation of cells to the next. Maintaining genome integrity throughout the cell cycle is of great importance for organismal development and the homeostasis of tissues and organs. Any cell cycle abnormality can have consequences for these fundamental processes. Microscopic observation of the cell cycle shows that these two processes follow one another without any overlap during a non-pathological cell cycle [55]. Furthermore, when these two processes are not separated in time, DNA integrity is compromised for future generations. Here are two examples. A cell that enters mitosis prematurely, that is, before the entire genome has been replicated, inherits an incomplete set of chromosomes. A cell that correctly enters mitosis, that is, once the whole genome has been replicated, but then carries out replication a second time may also inherit an incomplete set of chromosomes if chromosome segregation occurs before the end of the second round of replication [56]. The cell cycle must be controlled “internally” to ensure strict alternation of its two critical phases and thus preserve DNA integrity. Here we present the main control mechanisms, brought to light over the past thirty years, that:

1. make S-phase and M-phase events interdependent so that the mitotic process cannot begin until replication is complete,
2. arrest the cell cycle and repair DNA during genotoxic stress caused by replication or mitosis themselves, or by exogenous factors.

1.2.1 The key molecular players in cell cycle regulation

Cell cycle progression is controlled by cyclins (abbreviated *cyc*), which form molecular complexes with cyclin-dependent kinases (abbreviated *cdk*) [14]. These complexes are denoted *cyc/cdk* and are assigned distinct roles during the cell cycle. The *cycD/cdk4-6* and *cycE/cdk2* complexes trigger entry into the cell cycle itself from the quiescent phase, the *cycE-A/cdk2* complexes contribute to triggering S phase, and the *cycA-B/cdk1* complexes contribute to initiating M phase (Figure 1.2). Indeed, the *cycE/cdk2* and *cycA/cdk1* complexes are required for the G1/S and G2/M transitions,

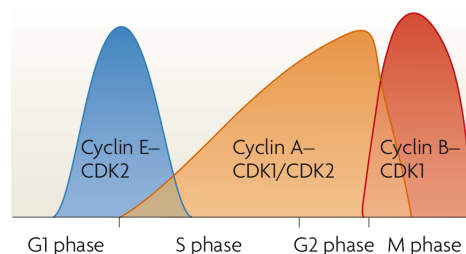


Figure 1.2: **The *cyc/cdk* complexes are the key molecular players in cell cycle regulation.** The activity of the *cyc/cdk* complexes, shown on the vertical axis, is restricted to specific time windows. Figure adapted from [40].

respectively. Loss-of-function experiments or chemical inhibition of the *cycE*/cdk2 and *cycA*/cdk1 complexes in human cell lines lead to arrest in G1 and G2 phase, respectively [40, 55]. Cell cycle control is based on the sequential and transient activation of these main *cyc*/cdk complexes and, by extension, of the processes under their control. For example, activation of M phase by the mitotic *cycB*/cdk1 complex is conditional on the completion of replication.

1.2.2 The multiple meanings of the term “checkpoint” in the context of the cell cycle

If the cell fails to complete DNA replication, a signal blocks the initiation of mitosis. The ongoing replication process generates a signal resembling replication stress, but under non-pathological conditions. This signal is detected by replication surveillance proteins that *block* activation of the molecular complex initiating mitosis until replication is complete. These “surveillance” proteins are ATR and Chk1. Specifically, ATR inhibits the *cycA*/cdk1 complex, which is involved in preparing M phase. Indeed, activation of this complex is sufficient to trigger autonomous activation of the *cycB*/cdk1 complex at the G2/M transition [40, 85, 91]. ATR activates Chk1 downstream, a kinase involved in particular in stabilising replication forks in S phase. Furthermore, [91] suggest that the ATR/CHK1 pathway postpones activation of the *cycB*/cdk1 complex *via* Cdc25/Wee1, that is, in a manner similar to the DNA damage response. In fact, [85] observe an initial accumulation of cyclin B in S phase in cells treated with an ATR or Chk1 inhibitor and synchronised at the beginning of S phase. In healthy, unperturbed cells, the increase in cyclin B concentration (and, by extension, in *cycB*/cdk1 complex activity) is observed during preparation for mitosis in G2 phase. Inhibiting the control of *cycA*/cdk1 complex activation exerted by ATR and Chk1 during replication causes a real identity crisis during the cell cycle, in which S and G2 phases overlap. This overlap leads to premature mitosis, insufficient replication and DNA damage. We have just given a specific example of a molecular mechanism that preserves DNA integrity, a property attributed to cell cycle *checkpoints*. In everyday language, the term “checkpoint” first refers to *a point of passage that can be crossed only if a condition is satisfied*. The corollary of this initial description of a checkpoint is the stability of a phase as long as the checkpoint condition is not satisfied [30, 87]. The checkpoint phenomenon is therefore intrinsically *discrete*. The checkpoint given as an example is S/G2, but historically the first cell cycle checkpoints identified were those located at the G1/S and G2/M transitions.

The G1/S checkpoint ensures that DNA replication takes place only once per cycle. G1 is a phase of preparation for replication, which can begin only if (condition) all pre-replication complexes (hereafter abbreviated CPR) have been recruited to the replication origins. This molecular complex consists of enzymes that catalyse the unwinding of DNA strands, the helicases. This step is essential for the initiation and elongation stages of DNA replication. Helicase loading and activation are crucial events that must take place before the initiation of replication, which is itself characterised by the recruitment of DNA polymerase α to single-stranded DNA [26]. Activation of the CPR is under the control of the *cycE*/cdk2 complex. In S phase, that is, once the replication process has been initiated, the proteins required for CPR assembly are inhibited. This process is under the control of the *cycA*/cdk2 complex, the inhibitor of the CPR-activating *cycE*/cdk2 complex. [57]. Inhibition of the CPR is an essential step in S phase to prevent multiple rounds of replication and depletion of nucleotide resources. The start of the process of CPR inhibition by *cycA*/cdk2 must naturally occur before the end of the process of CPR assembly and activation by

cyclin E and cdk2 to ensure complete genome replication. In light of this information, the term checkpoint evolves: *late cell cycle events (i.e. mitosis) depend on the completion of earlier cell cycle events (i.e. replication)* [7]. In this specific case, the checkpoint is said to be *intrinsic* to the cell cycle. The *Spindle Assembly Checkpoint* (hereafter abbreviated SAC) is another example. This checkpoint coordinates two phases of mitosis: metaphase and anaphase. Alignment of the chromosomes along the metaphase plate (metaphase) must be fully complete before the two sister chromatids of the chromosomes separate (anaphase). The protein responsible for triggering anaphase (APC-cdc20) is inhibited until all chromosomes are attached to the mitotic spindle during metaphase [54].

The G2/M checkpoint is an example of an additional process controlling entry into mitosis. The activity of the cycB/cdk1 complex is repressed during replication *via* ATR-dependent inhibition of its activator, the cycA/cdk1 complex. This inhibition of the complex is reinforced by the protein Wee1, a kinase that inhibits cycB/cdk1 complex activity in S phase and also in G2 phase [99]. Wee1's function in the G2/M checkpoint becomes clear when cells sustain DNA damage. It represses the activity of cyc/cdk complexes, particularly the cycB/cdk1 complex, the main driver of entry into mitosis, in response to single- or double-strand breaks generated, for example, after an error during replication or exposure to mutagenic agents. Wee1 contributes to cell cycle arrest in G2, allowing DNA repair before chromosome segregation [11]. A cell may sustain DNA damage caused by genotoxic environmental factors, for example ionising radiation such as UV. This damage generally consists of DNA breaks detected by biosensors that activate DNA repair pathways. ATR and Chk1 are among the key players in DNA damage response pathways. Their role here is to stabilise a phase until the damage is repaired [30]. In this case, Chk1 activates Wee1 (the inhibitor of cycB/cdk1) and ATR inhibits Cdc25 (the activator of cycB/cdk1), and the transition to mitosis is suspended. The necessary condition for this transition is directly the absence of DNA damage. The concept of a cell cycle checkpoint is expanded: *a checkpoint blocks cell cycle progression in response to exogenous DNA damage by directly targeting the regulators of cell cycle progression*. We refer to an *extrinsic* cell cycle checkpoint.

Whether intrinsic or extrinsic, checkpoints ensure that potentially lethal genetic errors will not be transmitted to the cell's descendants. One final checkpoint remains: the restriction point, which stabilises G1 phase *via* growth factor concentration. It is also possible to synchronise cells in the quiescent phase (G0) or early G1 phase through growth factor deprivation. The restriction point marks the moment when a cell becomes insensitive to growth factors and commits irreversibly to the cell cycle (Figure 1.3). This commitment is directly influenced by extracellular signals, and the *restriction checkpoint* therefore ensures control of proliferation. The action of the restriction checkpoint is limited in time to remove any possibility for growth factors to control the progression of S and M phases, and thus potentially interrupt them while they are in progress [15]. In other words, regardless of growth factor concentration, S and M phases necessarily occur once the cell passes the restriction point.

1.3 Motivations

An in-depth understanding of the mechanisms underlying checkpoints is motivated by the search for therapeutic strategies based on checkpoint activation by cytostatic or cytotoxic agents, to

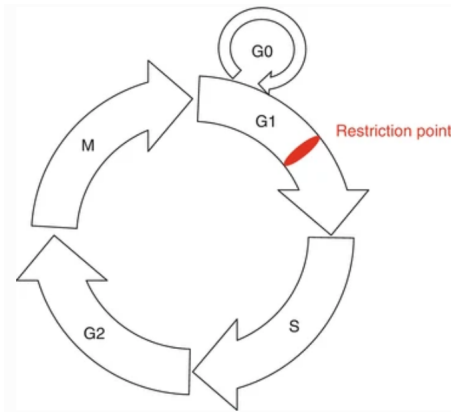


Figure 1.3: **The restriction point, canonically located in G1.** It corresponds to the boundary separating a portion of the cell cycle that is sensitive to growth factor deprivation from a portion that is insensitive to it. Figure adapted from [62].

counteract the uncontrolled proliferation of cancer cells [58, 104]. In basic research, several experimental studies show a bidirectional link between the cell cycle (and inseparably its checkpoints) on the one hand, and the circadian clock [66, 103] or metabolism [32] on the other. Here is a striking result. The tumour suppressor protein p53 is a key player in activating the G1/S checkpoint in the event of DNA damage. It controls cell proliferation upstream of S phase in the event of DNA damage, and disruption of this checkpoint (and presumably of the others) is associated with uncontrolled proliferation and genomic instability, two characteristics of cancer [58]. p53 is found to be under the control of the circadian clock, raising the question of the impact of circadian clock disruption (*e.g.* shift work) on cancer. Indeed, the gene encoding the p53 protein is mutated in more than 50% of human tumours. Furthermore, it is now accepted that disruption of the circadian rhythm plays a key role in the development of cancerous tumours and, conversely, that cancer directly disrupts circadian rhythms [95].

The predominant role of checkpoints in both the emergence and treatment of cancers makes it urgent to address the question of checkpoint formalisation. An already substantial and growing body of biological knowledge is accumulating under the term cell cycle checkpoint. An approach based on mathematical modelling is essential to cope with the complexity generated by the exploding number of molecules and regulatory pathways. By way of indication, for *Homo sapiens*, at least 1244 genes are annotated with the term “cell cycle checkpoint” at the time of writing this thesis [106], and at least 274 proteins are involved in cell cycle checkpoints [109]. This does not include the links with other physiological systems. We have identified five cell cycle checkpoints: the G1/S, S/G2 and G2/M checkpoints, the SAC and the restriction checkpoint. They form the intrinsic cell cycle checkpoints, and the first four can be activated by extrinsic checkpoints in response to genotoxic stress. The G1/S, S/G2 and G2/M checkpoints and the SAC share a property that we generalise through the following proposition: *an intrinsic checkpoint prevents any event initiating a given phase from taking place until all the events of the preceding phase have been correctly completed.* This is the minimal property that intrinsic cell cycle checkpoints must generate to ensure, in this case, temporal separation of S and M phases.

Only mathematical modelling can predict whether a set of regulatory pathways, assumed to be involved in cell cycle checkpoints, can generate dynamics that reproduce at least this fundamental checkpoint property. Through its rigorous approach, a modelling formalism suited to the research question can reveal contradictions among the formalised biological knowledge. Contradictions in a given model form the basis for predicting new experiments that will help elucidate new mechanisms involved in checkpoints. Moreover, the challenge of checkpoint modelling lies less in formalising the regulatory pathways involved than in formalising cell cycle phases and checkpoints. In practice, cell cycle modelling studies concentrate their efforts on designing regulatory pathways and simulating or verifying their dynamical properties in order to reproduce the expected asymptotic behaviours. Typical examples include the presence of a cyclic attractor characteristic of the cell cycle under a constant supply of growth factors, arrest in the quiescent phase in the absence of these factors, or observation of a trajectory in which cyc/cdk complexes are activated and inactivated in an expected order [29, 43, 76]. Yet, to our knowledge, no study has so far been devoted to *generalising the dynamical constraints generated by checkpoints* along the cell cycle. One of the main reasons is that no study has addressed *the mathematical representation of a cell cycle phase*, an object inseparable from checkpoints.

1.4 Overview of the thesis

Chapters 2 and 4 define the theoretical foundations of the thesis. Chapter 2 is devoted to an account of the different modelling formalisms used to represent the dynamics of biological regulatory networks. Chapter 4 defines our formal approach within theoretical computer science, applied to the formal specification and verification of cell cycle checkpoints.

Chapter 3 presents the state of the art in mathematical modelling of cell cycle checkpoints. Eight models have been selected to illustrate how the scientific community approaches the question of checkpoints. Although the pioneering studies historically relied on continuous formalisms, the most recent studies tend towards a discretisation of the problem. The review by [41], recently updated [83], was foundational for this thesis. The authors conclude that a checkpoint is an intrinsically discrete phenomenon structurally based on the coupling of *bistable switches*. A bistable switch is a positive feedback loop that generates distinct stable states and contributes to separating certain cell cycle events (*e.g.* those of replication) from others (*e.g.* those of mitosis). This separation phenomenon enables *temporal separation* of S and M phases. The study by [87] shows, through an example, that if a bistable switch characteristic of a phase can be assumed, then coupling several bistable switches in series can generate irreversible commitment from one phase to the next. We will call this phenomenon *the irreversibility of phase transitions*. In light of the state of the art presented in this chapter, two challenges clearly emerge. The first is to define a cell cycle phase correctly, a definition that will necessarily be included in that of the two dynamical properties of a checkpoint. The second challenge is methodological. How can we prove that a particular checkpoint mechanism, *e.g.* a coupling of bistable switches involving cell cycle regulators, generates a particular checkpoint phenomenology, *e.g.* the phenomena of temporal separation of phases and irreversibility of phase transitions?

Chapter 5 is devoted to formalising static and dynamical knowledge of the cell cycle. This is carried out using René Thomas’ modelling formalism. The regulations considered essential to the biological question are formalised by an influence graph inspired by those of [41] and [71]. Such a graph covers, *a priori*, a large set of different models because the strengths of activations and inhibitions are unknown. We use a family of dynamical parameters that encodes these strengths [8] and avoids an arbitrary choice of an activation function for each variable, and therefore an arbitrary choice of model. In the 1970s, qualitative models based on discrete mathematics proved useful for understanding the main causal relationships governing observed phenotypes, and René Thomas’ multivalued framework became a classic for biological regulatory networks. It gained in power with the introduction of formal methods in the early 2000s [21]. This chapter proves that the classical dynamical properties of the cell cycle observed experimentally (*e.g.* quiescence or oscillations of cyc/cdk complexes) can be formalised using two temporal logics central to theoretical computer science and adapted to Thomas’ formalism: computation tree logic [21] and genetically modified Hoare logic [86]. The TotemBioNet software platform is chosen to connect static knowledge (*i.e.* the influence graph) with the classical dynamical phenomena of the cell cycle. TotemBioNet identifies the parameterisations satisfying all the formalised dynamical properties [97]. The resulting set of models encodes the experimental data against which our definition of a phase is tested in Chapter 6.

Chapter 6 is devoted to generalising the concept of a phase independently of the definition

of a cell cycle model. Three generic definitions are proposed, differing in their degree of freedom. Defining a *canonical phase* is the step at which the assumptions are fixed. A phase is a multiset of biological regulatory events, which can be regarded as discrete events, required for a cell cycle phase to take place. The definition of the *hyperrectangle of a canonical phase* represents the domain of the phase, within which no arbitrary knowledge of the order of events is presupposed. A *phase* is finally defined as the result of confronting the hyperrectangle of a canonical phase with experimental data. This comparison is based on applying genetically modified Hoare logic.

Chapter 7 is the central chapter, in which the two fundamental checkpoint properties are formalised. The *temporal separation* predicate is introduced and ensures that any event that can complete a phase cannot be permuted with any event that can initiate the next phase. A constraint is generated to select the cell cycle models satisfying this property. Verification of *the irreversibility of phase transitions* is encoded directly by a CTL formula that uses the characteristic formula of a phase to check that no visit to other phases is allowed from any state other than the final state of a phase. This conceptualisation of the two checkpoint properties is finally validated using the set of models inferred in Chapter 5: the proof of concept is successful if there exists at least one parameterisation satisfying both intrinsic checkpoint properties.

State of the Art

CHAPTER 2

Formalisms for modelling the regulation of cyclic behaviours

2.1	An epistemological preamble: why formalise?	17
2.2	Systems of ordinary differential equations	17
2.3	From Boolean networks to multivalued networks	22
2.4	A multivalued approach with multiplexes: René Thomas's modelling formalism	24
2.4.1	Abstract syntax: influence graph with multiplexes	24
2.4.1.1	On the origin of discrete biological regulatory thresholds	25
2.4.1.2	On the usefulness of multiplexes	27
2.4.2	From syntax to semantics: the notion of a dynamic parameter	29
2.4.2.1	Defining the family of dynamic parameters from a GRBM	29
2.4.2.2	Meaning of dynamic parameter values	29
2.4.2.3	Snoussi's condition for constraining dynamic parameter values	30
2.4.2.4	A thought experiment for constraining dynamic parameter values	31
2.4.3	Semantics: the asynchronous state transition graph	35

2.1 An epistemological preamble: why formalise?

A major question asked by the molecular biology research community is the following: which biological regulatory mechanisms *induce* a given biological behaviour? Experimental biology attempts to answer this question through an *abductive* approach that extracts a set of hypotheses about the causes of the biological behaviour under study from *unexpected* correlations. The abductive approach is a form of reasoning that does not follow a rigorous framework and particularly exposes researchers to errors in interpreting experimental results.

Living systems are complex and operate at multiple scales, making the task of investigating their mechanisms almost impossible without establishing general theoretical frameworks from particular observations through an *inductive* approach. Formal frameworks for modelling biological regulatory networks are a typical example. They allow biological concepts to be formalised and *deductive* approaches to be applied to them, establishing proofs of the form: *given a biological regulatory network, certain biological behaviours are necessarily admitted*. This step of formalising knowledge and a deductive approach are also used when biological experiments cannot directly test hypotheses about the causes of the biological behaviour under study in living systems.

Ultimately, formalising biological knowledge is an essential step in ensuring the correctness of reasoning, and a *reasoned* choice of modelling formalism is equally essential. Any formalism must be suited to the biological question posed in order to answer it. There are two major strands of modelling. A quantitative approach continuously represents the concentration of a biochemical species by real numbers, and a qualitative approach abstracts its concentration by a non-negative integer representing the action of the biochemical species on one or more of its targets. In Section 2.2, we shall present the quantitative formalism most commonly used to model biological regulatory networks, namely systems of differential equations. Similarly, for the qualitative approach, we shall focus on Boolean networks in Section 2.3. We shall position each of these formalisms in relation to the biological questions they can address and the assumptions they admit. Then, in Section 2.4, we shall position the multivalued formalism for modelling biological networks introduced by René Thomas and Houssine Snoussi [4, 9, 36].

2.2 Systems of ordinary differential equations

Systems of ordinary differential equations constitute a classical formalism for modelling the dynamics of a biological regulatory graph in purely quantitative terms. Many cell cycle models use this formalism, which is why we devote an entire section to it. An ordinary differential equation (abbreviated ODE) typically represents the variation over time in the concentration of a biological entity denoted x_i , with i ranging from 1 to n (e.g. *mRNA*, a protein, a metabolite, a biological species). An ODE relates the derivative $\frac{dx_i}{dt}$ to a function f_i describing the biological mechanisms influencing x_i , a function that is almost always *nonlinear*. In general, one ODE is used for each biological species participating in the biological regulatory network. Formally, for a system of ODEs of dimension n [18]:

$$\frac{dx_i}{dt} = f_i(x), \quad 1 \leq i \leq n$$

where x is a vector of non-negative concentrations $(x_1, \dots, x_n)$. The function $f_i : \mathbb{R}_+^n \rightarrow \mathbb{R}_+$ defines the regulatory mechanisms of x_i :

$$f_i(x) = r(\bar{x}_i) - \delta_i x_i$$

where $\bar{x}_i$ is the vector of concentrations of the predecessors of x_i in the regulatory graph, δ_i is the degradation rate of x_i , and $r : \mathbb{R}^m \rightarrow \mathbb{R}$ is most often a sum of regulatory functions, each representing the effect of one of the m regulators of x_i . This sum assumes that the x_j act independently of one another, and the regulatory functions classically used are *Hill functions* (Definition 1).

A Hill function approximates a sigmoidal curve characteristic of a phenomenon observed in biochemistry: *positive cooperativity* (binding of a regulatory molecule x to a target molecule y increases the affinity of x for y). The regulatory functions h^+ and h^- in Definition 1 have a *cooperativity parameter* denoted n , which gives these functions a sigmoidal shape when $n > 1$, illustrated in Figure 2.1.

Definition 1: Hill functions

Given an *activator* x and its target y , the Hill function h^+ describing the process of activation of y by x (denoted $x \xrightarrow{+} y$) is defined by:

$$h_{\theta,n}^+(x) = \frac{x^n}{x^n + \theta^n}$$

Given an *inhibitor* x and its target y , the Hill function h^- describing the process of inhibition of y by x (denoted $x \xrightarrow{-} y$) is defined by:

$$h_{\theta,n}^-(x) = \frac{\theta^n}{x^n + \theta^n}$$

n is called the *cooperativity* parameter and θ the *threshold* parameter.

The larger the cooperativity parameter, the steeper the sigmoid. As the number of molecules x increases, the effect on y increases until a limiting value is reached. This value reflects saturation of the molecule y , which has only a finite number of receptors for x . The effect on y of regulator x in the case of activation (resp. inhibition), denoted $h_{\theta,n}^+$ (resp. $h_{\theta,n}^-$), converges to 1 (resp. 0) as x tends to $+\infty$. The regulatory functions h^+ and h^- also have a *threshold parameter* denoted θ , which, as n tends to $+\infty$, defines a *value of x* from which the effect of regulation on y becomes maximal.

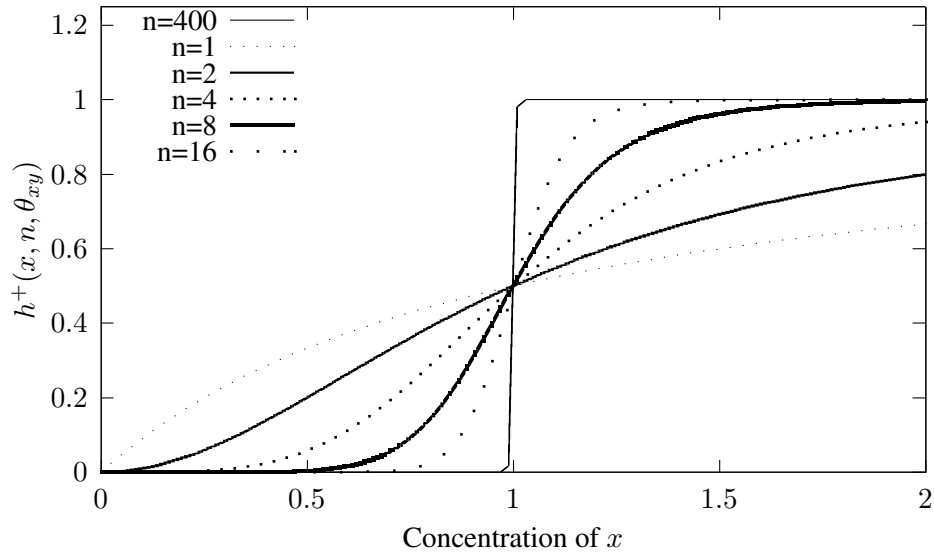
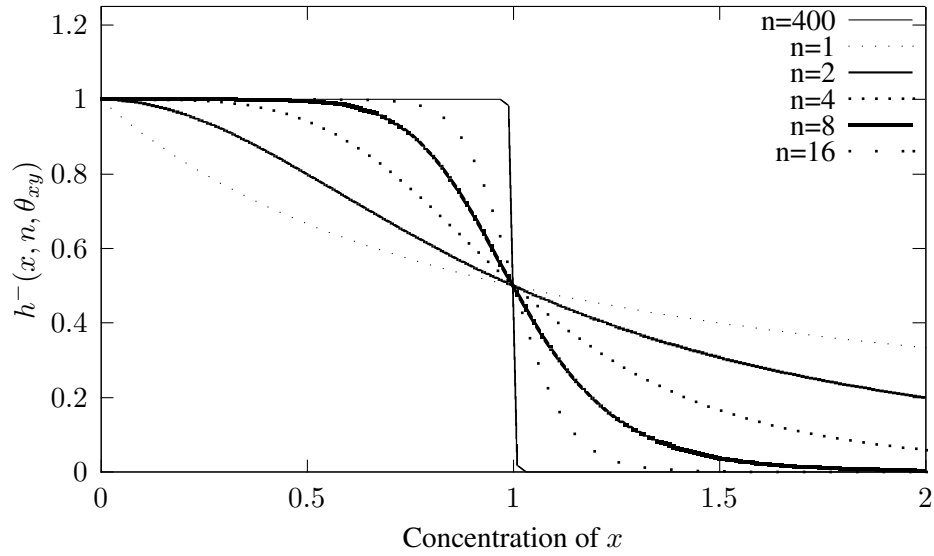
(a) The positive regulatory function $h^+(x, \theta_{xy}, n)$ (b) The negative regulatory function $h^-(x, \theta_{xy}, n)$

Figure 2.1: **The Hill functions h^+ and h^- .** They associate the concentration of a regulator x with the effect of its regulation on y , with parameter $\theta_{xy} = 1$ and for $n = 1, 2, 4, 8, 16$ and 400. θ is the inflection point. When $n = 1$, the function h^+ equals the expression given by Michaelis–Menten, where θ_{xy} represents an affinity constant.

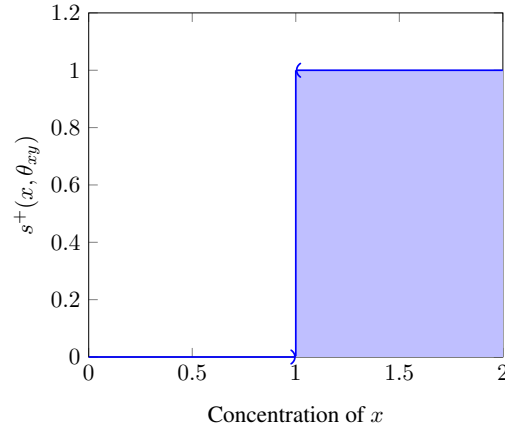
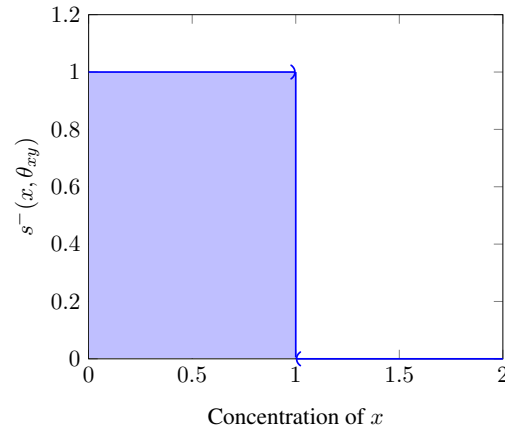
(a) The positive regulatory function $s^+(x, \theta_{xy})$ (b) The negative regulatory function $s^-(x, \theta_{xy})$

Figure 2.2: **The Heaviside step functions s^+ and s^- with parameter $\theta_{xy} = 1$** They logically describe the effect of regulation by x on y . The value $y = 1$ represents its normalised maximum value.

The Heaviside step function, given in Definition 2, is a discrete approximation of the Hill function as n tends to $+\infty$ and provides qualitative information: x is inactive below the value θ_{xy} , resulting in no effect on y ; otherwise, x has its maximal effect on y from the value θ_{xy} onwards. An illustration of the Heaviside step function is given in Figure 2.2. This qualitative view also echoes the initial intuition behind René Thomas’s qualitative formalism (*cf.* Section 2.4), used in the modelling work of this thesis.

Definition 2: Heaviside step functions

Given an *activator* x and its target y , the step function s^+ is defined by:

$$s^+(x, \theta) = \begin{cases} 0 & \text{if } x < \theta_{xy} \\ 1 & \text{if } x \geq \theta_{xy} \end{cases}$$

Given an *inhibitor* x and its target y , the step function s^- is defined by:

$$s^-(x, \theta) = \begin{cases} 1 & \text{if } x < \theta_{xy} \\ 0 & \text{if } x \geq \theta_{xy} \end{cases}$$

θ_{xy} is the *threshold* parameter of the step function.

In general, systems of ODEs abstracting biological regulatory mechanisms cannot be solved analytically, given the nonlinearity of the function r . Nevertheless, numerical solution methods can obtain approximate solutions starting from an initial condition.

Consider the example of a basic molecular oscillator with two variables x and y . Each variable has a basal synthesis rate (respectively σ_x and σ_y) and a degradation rate (respectively δ_x and δ_y). Consider a negative feedback loop: x activates y and y inhibits x . The two-dimensional system of ODEs that formalises this regulatory network, using a Hill function, is as follows:

$$\begin{cases} \frac{dx}{dt} = \sigma_x \times \frac{x^{n_x}}{x^{n_x} + \theta_{xy}} - \delta_x x \\ \frac{dy}{dt} = \sigma_y \times \frac{\theta_{yx}}{\theta_{yx} + x^{n_x}} - \delta_y y \end{cases}$$

In addition to the analytical solution, the *kinetic* parameters of this system of ODEs are unknown, namely the parameters σ_i , δ_i , the threshold parameters θ_{xy} , θ_{yx} and the cooperativity parameters n_x and n_y . Estimating these parameters encounters limitations.

We shall first mention the need for a large quantity of experimental data to address the problem of parameter identifiability. In general, obtaining these data is experimentally costly in both time and funding. A parameter associated with a variable v is said to be structurally unidentifiable if varying its value has no effect on the trajectory of v . The value of an unidentifiable parameter estimated from experimental data would therefore have no biological interpretation. The choice of variables to make explicit within the system of ODEs thus always ends up limiting model size. This limitation is perfectly illustrated by Jon Von Newmann: “*With four parameters I can fit an elephant, and with five I can make him wiggle his trunk*”.

The continuity assumption admits that biochemical species are present in infinite quantities and entails generating a large quantity of experimental data on the activity of the biochemical species of interest. Moreover, this assumption does not allow the use of knowledge about the heterogeneity of cell populations that analysis of single-cell sequencing data can provide. Qualitative models (Boolean or multivalued) constitute a simplifying abstraction of continuous models but allow the satisfiability of asymptotic behaviours characteristic of fundamental aspects of living systems to

be proved, such as limit cycles (*e.g.* homeostatic processes) and multistationarity (epigenetic phenomena or cell differentiation). The abstraction power of the discrete framework opens the door to formal methods from theoretical computer science, both to formalise and to verify purely qualitative biological properties (*e.g.* cell cycle checkpoints) that must fundamentally be studied before refining the question towards more quantitative aspects, *e.g.* the dose–response effects of a chemical substance on living systems or the synchronisation of the timing of biological regulations.

2.3 From Boolean networks to multivalued networks

Introduced in the late 1960s by Kauffman [3], Boolean networks are now commonly used to study the qualitative dynamic properties of a biological influence network. We mentioned above the asymptotic behaviours ubiquitous in living systems, such as cyclic attractors and multistationarity. Each modelled biochemical entity is represented by a Boolean variable v . When its concentration is sufficient (resp. insufficient) to act on its targets, it is characterised by $v = 1$ (resp. $v = 0$). The state space of a Boolean network with n variables is therefore defined by $\{0, 1\}^n$, and each variable v is equipped with a logical activation function f_v to construct transitions between these states. This function characterises the nature of the cooperation between variables to activate v . It is expressed in propositional logic restricted to conjunction, disjunction and negation. Consider the following logical activation functions, associated with each of the two Boolean variables x and y :

$$\begin{aligned} f_x &: \neg y \\ f_y &: x \wedge y \end{aligned}$$

The function f_x expresses that activation of x occurs when y is inactive, and f_y formalises activation of y by the complex formed by x and y . We can also abstract the following influence graph, where nodes represent variables and edges between nodes represent the positive or negative influences a variable exerts on the synthesis of another. The labels $+$ and $-$ represent activation and inhibition, respectively:

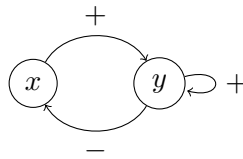


Figure 2.3: The influence graph abstracting the Boolean network defined by f_x and f_y

This representation avoids considering a model *a priori*, which is defined by a state transition graph from the set of activation functions f_v . Two interpretations of Boolean networks are classically contrasted. Modelling that adopts an asynchronous strategy assumes that each variable that must activate in a state is updated independently of the others. The synchronous strategy consists in updating all variables that must activate at the same time. Intermediate strategies have since been proposed. For example, [29] propose assigning priority classes to Boolean variables, producing a transition system that combines synchronous and asynchronous transitions. More recently, more permissive semantics were introduced by [101], taking into account transient states of variables undergoing activation or inhibition that can be interpreted as either 0 or 1. It should

also be noted that the choice of semantics for a Boolean network affects the reachability and stability properties it may admit. The choice of semantics is therefore a modelling assumption that should be discussed in any conclusion of a modelling study based on Boolean networks. A synchronous Boolean network *à la Kauffman* does not account for the fact that the total order in which variables are updated is almost unknown in biology. In biology, it is implausible that two variables always cross their activation (resp. inactivation) thresholds at exactly the same time. Asynchronous semantics *à la Thomas* account for all possible update orders, and thus all possible execution paths from a state. These semantics are considered more realistic from a biological perspective; however, they result in an over-approximation, rather than an under-approximation, of the reachable states relative to biological knowledge.

When a variable activates two targets, it is unlikely that the two targets are nevertheless activated at exactly the same time. One way to address this is to consider a multivalued network. These allow the activity of a variable to be expressed more finely. A variable is equipped with a bound $b_v \in \mathbb{N}^*$. It is therefore possible to define the order in which a variable v will activate or inhibit its targets by equipping it with several distinct regulatory thresholds. This is what René Thomas's multivalued approach, presented in Section 2.4, proposes; it proves the most appropriate for explicitly representing the different qualitative levels of cyclin/cdk complexes according to distinct phosphorylation states.

Analysis of (synchronous or asynchronous) state transition graphs can identify a terminal strongly connected component (CFCT) characteristic of a cyclic attractor (e.g. the cell cycle), or several such components characteristic of multistationarity. The main limitation lies in analysing the biological relevance of these CFCT. For example, the cell cycle is characterised by a CFCT of a Boolean network by [88] and [29]. However, identifying biologically admissible paths within this component is laborious, given that the number of states increases exponentially with the number of variables. This task is all the more laborious because biological knowledge about the behaviour of a system is often partial.

Furthermore, when the logical rules for updating variables are unknown, a logical learning problem arises: *which models satisfy a given set of properties?* Typically, the *Answer Set Programming* method can characterise an exhaustive set of logical models associated with a signalling network [61], or identify a set of Boolean models that satisfy formalised biological properties, for example [70, 98]. Houssine Snoussi, meanwhile, introduces the notion of a resource and a family of logical parameters denoted $K_{v,\omega}$, which associates each variable with all possible resource configurations of v from an influence graph. Here, a resource denoted ω is abstracted as the presence of an activator and the absence of an inhibitor. For example, the influence graph 2.3 admits that variable y can *potentially* be activated by the presence of x or the absence of y . Variable y has 4 parameters governing its dynamics: $K_{y,\emptyset}$, $K_{y,\{x\}}$, $K_{y,\{y\}}$, $K_{y,\{x,y\}}$, and x has two: $K_{x,\emptyset}$ and $K_{x,\bar{y}}$. In the Boolean framework, $K_{v,\omega}$ is also a Boolean variable. The set of these symbolic parameters associated with a variable v avoids presupposing any update rule f_v . Once each parameter is instantiated, the truth table of the parameters $K_{v,\omega}$ gives the expression of the function f_v . For example, the following truth table allows $f_y = x \wedge y$ to be deduced:

y is activated only when both its resources are activated. [21] introduced symbolic model learning based on model checking of CTL formulae from an exhaustive set of models deduced from an influence graph and the family of parameters $K_{v,\omega}$. The aim of learning is to identify the models (that is, the functions f_v in the Boolean framework) that satisfy temporal properties expressed

x	y	ω	$K_{y,\omega}$
0	0	$\emptyset$	0
1	0	x	0
0	1	y	0
1	1	$\{x, y\}$	1

Table 2.1: **Inferring a variable’s activation rule from the truth table of its parameters** $K_{v,\omega}$. This parameter set is none other than the truth table of the conjunction $x \wedge y$.

in CTL (e.g. reachability and stability), without arbitrarily presupposing parameter values when biological knowledge is lacking.

2.4 A multivalued approach with multiplexes: René Thomas’s modelling formalism

This formalism mathematically defines a set of biological regulations as a *biological regulatory graph with multiplexes*, introduced in Section 2.4.1. Its dynamics are defined in Section 2.4.3 by an *asynchronous state transition graph*, for which the behaviour of each variable, subjected to all possible regulatory configurations, is defined by a family of *dynamic parameters* introduced in Section 2.4.2. This section offers two levels of description: an intuitive level illustrating the need for formalisation in biology, and a purely formal level provided in the definition boxes.

2.4.1 Abstract syntax: influence graph with multiplexes

The abstract syntax of the formalism allows regulations between biological variables to be specified as a biological regulatory graph with multiplexes (Definition 3). Biological variables represent molecules (or groups of molecules) in living systems, typically genes, transcripts, proteins, molecular complexes of various kinds, or abiotic variables such as temperature, pH or oxygen level. The set of interactions between the different variables is represented mathematically by a directed graph whose vertices are either variables (graphically represented by circles) or multiplexes (graphically represented by rectangles), as in Figure 2.5. The latter formalise biological regulatory processes of varying complexity using a subset of first-order logic in which each variable, denoted v , represents a biological entity interpreted by a discrete concentration level.

The introduction of multiplexes [36] into the syntax of René Thomas’s formalism enriches an initial multivalued approach [9] that introduces the fundamental notion of a *discrete regulatory threshold*. There are finitely many of these thresholds and we use no quantifiers; consequently, the number of atoms is finite and multiplex logic is propositional. We first propose to explain where these thresholds come from, allowing the basic building blocks of biological regulations (literals) to be defined. Finally, we shall show the expressive power of multiplexes for describing a broad range of complex regulations through combinations of these literals, following propositional logic.

Definition 3: Biological regulatory graph with multiplexes

A *biological regulatory graph with multiplexes* (GRBM) is a triple $\mathcal{G} = (V, M, E)$ where:

V is a finite set of variables v equipped with a bound $b_v \in \mathbb{N}^*$

M is a finite set of symbols m equipped with a formula φ_m over the language $\mathcal{L}$ defined inductively by:

for every variable $v \in V$ and every integer n such that $n \in \llbracket 0, b_v \rrbracket$, $(v \geq n) \in \mathcal{L}$

if $(\varphi, \psi) \in \mathcal{L}^2$, then $\neg\varphi \in \mathcal{L}$, $\varphi \wedge \psi \in \mathcal{L}$ and $\varphi \vee \psi \in \mathcal{L}$

E is the set of edges, where $E \subseteq M \times V$.

We denote an element of E by $(m \rightarrow v)$, and $E^{-1}(v)$ is the set of predecessors of v .

2.4.1.1 On the origin of discrete biological regulatory thresholds

In 1973, René Thomas proposed a purely Boolean approach [4] alongside Kauffman. The **founding idea** is the same:

$x = 0$ represents an *interval* of concentrations of x too low to produce an effect on y , not to be confused with a zero concentration of x .

$x = 1$ represents an *interval* of concentrations of x for which an effect of regulation on y is observed.

Regulation of gene transcription by a transcription factor is a typical example in biology. In this case, the variable x describes the concentration of the transcription factor and y the concentration of the product of the transcription process of gene y , namely its messenger RNA.

René Thomas's Boolean approach is based on discretising the continuous concentration axis of each variable x , on which the threshold θ_{xy} is placed. Given a system with n variables, the n -dimensional concentration space is divided into 2^n hyperrectangles within which each variable acts uniformly on its targets. In practice, estimates of experimentally measured concentration thresholds θ_{xy} are very rarely available. Generally, experimental results instead provide qualitative information, as illustrated by René Thomas himself: “*Note that biologists known to be allergic to any formal treatment nevertheless invariably say (and write): “in the absence of immunity, this, in the presence of immunity, that; at low temperature, this, at high temperature, that ...”. This is a typically Boolean attitude.*” [9].

This purely Boolean approach has the drawback of greatly limiting the expressive power of regulations between several biological variables, because a variable can act on several targets at thresholds that must be distinguished, or because a variable can act differently on the same target at different concentrations. The cell cycle offers a typical example: the cyclinE/Cdk2 complex (abbreviated *cycE/cdk2* and introduced in the previous chapter). This biological complex has two distinct thresholds of action on its targets, giving rise to three concentration values: *cycE/cdk2* = 1 corresponds to a *monophosphorylated* state of the complex, *cycE/cdk2* = 2 to

a *hyperphosphorylated* state, and finally the complex has no significant effect on any target when $cycE/cdk2 = 0$. In the first case, the $cycE/cdk2$ complex can only activate its own phosphorylation, whereas in the second it can, *in addition*, activate cyclin A transcription.

In 1991, Houssine Snoussi and René Thomas therefore proposed a **multivalued** formalism [9] for modelling biological regulatory networks. If a variable has n targets, it is associated with a finite discrete domain whose maximum bound is $b_v = n$ in the worst case. It is possible that v regulates several targets *at the same threshold*, in which case b_v is strictly less than n . When no biological information distinguishes the different thresholds of action of a variable, it is tempting to choose the maximum number of thresholds, but it is often preferable to adopt the simplest hypothesis according to Occam's razor: choose the minimum number of thresholds in the absence of experimental data.

Given a set of variables V , discretisation of the concentration space of the continuous system results in a division into $\prod_{v \in V} (b_v + 1)$ hyperrectangles within which the regulations are constant. An example is provided in Figure 2.4. Consider a two-dimensional system where variable x has

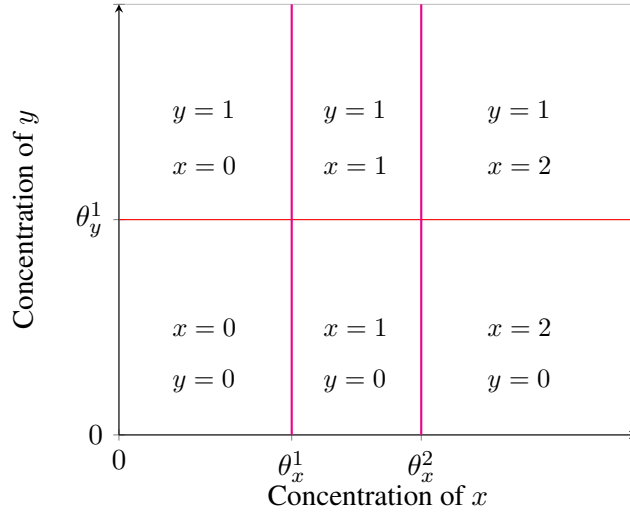


Figure 2.4: **Example of discretisation of a two-dimensional concentration space from the single regulatory threshold of y (denoted θ_y^1) and the two thresholds of x (denoted θ_x^1 and θ_x^2).** The continuous concentration space is divided into 6 states in which the action of each variable on its targets is uniform and each state abstracts a region of the continuous concentration space.

two regulatory thresholds, denoted θ_x^1 and θ_x^2 respectively (a multivalued variable), and variable y has only one, denoted θ_y^1 (a Boolean variable). Variable x can therefore take only three discrete values:

$$\begin{cases} x = 0 & \text{if } x < \theta_x^1 \\ x = 1 & \text{if } \theta_x^1 \leq x < \theta_x^2 \\ x = 2 & \text{if } x \geq \theta_x^2 \end{cases}$$

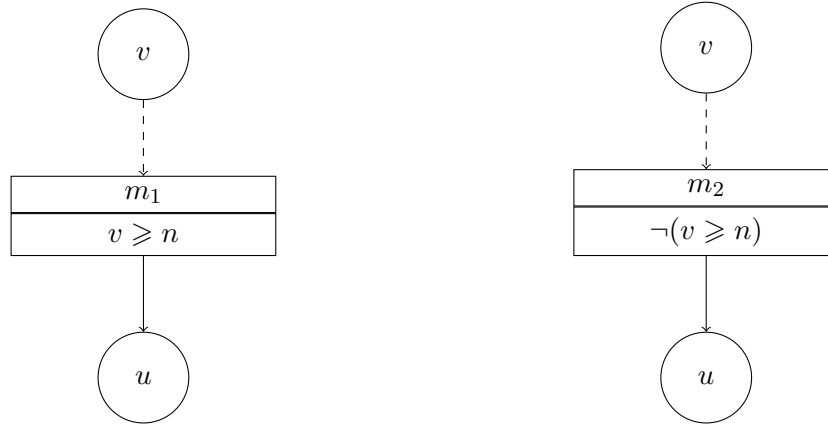
Variable y reduces to the Boolean case discussed at the beginning of this section:

$$\begin{cases} y = 0 & \text{if } y < \theta_y^1 \\ y = 1 & \text{if } y \geq \theta_y^1 \end{cases}$$

Within the continuous concentration space of a biological system, the precise location of each regulatory threshold of each variable is *ultimately* a minor concern in René Thomas's formalism. If we are interested only in the qualitative behaviour of the system, what matters is solely *the order in which each variable crosses its regulatory thresholds*.

2.4.1.2 On the usefulness of multiplexes

A multiplex is defined by a pair consisting of an identifier denoted m and a propositional logic formula denoted φ_m . When φ_m reduces to an atom (without negation), it defines the positive action exerted by a variable on one or more targets.



(a) Multiplex m_1 represents the presence of activator v : it reaches or exceeds its threshold n .

(b) Multiplex m_2 represents the absence of inhibitor v : it does not reach threshold n .

Figure 2.5: Literals of the multiplex language. The edge $m_1 \rightarrow u$ (resp. $m_2 \rightarrow u$) represents activation (resp. inhibition) of u by v . Formally, the edges from v to m_1 (resp. to m_2) are superfluous because v appears explicitly in the formula of multiplex m_1 (resp. m_2).

In a GRBM, an edge of the form $m \rightarrow u$ such that $\varphi_m = (v \geq n)$ represents activation of u by v at threshold n , cf. Figure 2.5a. To describe inhibition, it suffices to use the negation of the atomic formula, namely $\neg(v \geq n)$, cf. Figure 2.5b. A multiplex whose formula is an atom (resp. the negation of an atom) therefore describes *the presence of an activator* (resp. *the absence of an inhibitor*) at a given threshold. The expressive power of multiplexes comes from their ability to combine several literals (atoms or their negations) using conjunction and disjunction from propositional logic to represent cooperation/competition phenomena. Here are some examples:

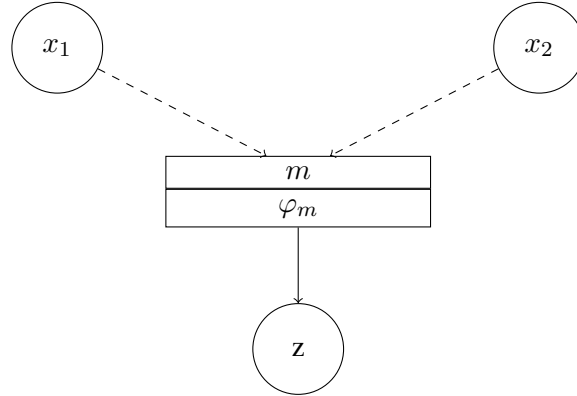


Figure 2.6: **Combining the action of two regulators x_1 and x_2 on a common target z .** Multiplex m with formula φ_m formalises the nature of cooperation/competition phenomena between the x_i .

Complex formation: Formation of a complex is classically described at the syntactic level by a *conjunction* of several literals. For example, suppose in Figure 2.6 that x_1 and x_2 are two activators of z , but only in the form of a complex x_1x_2 . The resulting complex is formalised by $\varphi_m \equiv (x_1 \geq 1) \wedge (x_2 \geq 1)$. To describe the action of an inhibitory complex, it suffices to define a conjunction of negative literals.

Inhibition of regulation: A conjunction also allows inhibition of the action of one regulator by another regulator to be described. In the case of two regulations, this inhibition is generally formalised by the conjunction of a literal and a negative literal. This time, suppose in Figure 2.6 (with $n = 2$) that x_2 inhibits activation of z by x_1 . This inhibition is formalised by $\varphi_m \equiv (x_1 \geq 1) \wedge \neg(x_2 \geq 1)$.

Grouping regulations of the same kind: An *inclusive disjunction* of several literals allows the action of several regulators to be grouped when they act *independently* on a common target. If x_1 and x_2 act independently on z , we write $\varphi_m \equiv (x_1 \geq 1) \vee (x_2 \geq 1)$.

Competition phenomena: An *exclusive disjunction* allows competition phenomena to be formalised. If x_1 and x_2 compete to activate z , we write $\varphi_m \equiv (x_1 \geq 1 \vee x_2 \geq 1) \wedge \neg(x_1 \geq 1 \wedge x_2 \geq 1)$. This formula means that x_1 and x_2 can act positively on z only if they are not active at the same time. A typical example is when x_1 and x_2 are two genes producing two proteins X_1 and X_2 with high affinity for each other, and the X_1 - X_2 complex can no longer act on z .

Multiplexes can abstract a wide range of genetic regulatory mechanisms, typically transcription, translation, (de)phosphorylation and proteolytic degradation. We refer the reader to Chapter 5 for more sophisticated examples with our regulatory graph with multiplexes of mammalian cell cycle progression.

In the four preceding examples, variable z has a single predecessor, whereas a conventional representation would give it two predecessors. A direct consequence is a reduction in the number of dynamics that a GRBM can generate, cf. Section 2.4.2. More generally, when a variable v has n

predecessors, it is *potentially* subject to the influence of 2^n possible regulatory configurations. The syntactic level does not determine whether a multiplex affects the dynamics of variable v , that is, the value towards which v is attracted.

2.4.2 From syntax to semantics: the notion of a dynamic parameter

2.4.2.1 Defining the family of dynamic parameters from a GRBM

The GRBM provides a syntactic description of the system in which each variable v is subject to the influence of a set of predecessor multiplexes. The notion of *dynamic parameters* is the junction between a syntactic and a dynamic description: the parameters are denoted $K_{v,\omega}$, where ω is a set of *multiplexes* that are predecessors of v . The set of parameters associated with a variable can be extracted directly from the GRBM by exhibiting all possible combinations of the predecessor multiplexes of v . Each variable v has 2^n parameters, where n is the in-degree of v in a GRBM.

Definition 4: Family of dynamic parameters

Given a GRBM $\mathcal{G} = (V, M, E)$, we denote by $\mathcal{K}$ the family of *dynamic parameters* consisting of symbols of the form $K_{v,\omega}$, where v ranges over V and ω ranges over the power set of $E^{-1}(v)$.

$K_{v,\omega}$ is called the parameter of v for the resource set ω .

2.4.2.2 Meaning of dynamic parameter values

The dynamics of a GRBM are defined by transitions between states, a state being defined by the current value of each variable in the graph:

Definition 5: State

Given a GRBM $\mathcal{G}$, a *state* is a substitution $\eta : V \rightarrow \mathbb{N}$ such that, for every $v \in V$, $\eta(v) \in \llbracket 0, b_v \rrbracket$. The set of states is denoted ζ .

In each state denoted η , every variable v is substituted by a value belonging to its discrete domain: for every variable v , we can therefore identify the unique set of regulations to which it is *effectively* subject. Any predecessor multiplex of v whose characteristic formula φ_m is satisfied in η is called a *resource* of v in η . Thus, a variable has only one applicable parameter per state, *cf.* Definition 6.

Definition 6: Applicable parameter

Given a GRBM $\mathcal{G}$ and a state η , a *parameter* $K_{v,\omega} \in \mathcal{K}$ is said to be *applicable* in η if ω is the set of $m \in E^{-1}(v)$ such that φ_m is satisfied under substitution η .

The integer value of parameter $K_{v,\omega}$ defines the value towards which a variable v is attracted when ω is its resource set. The codomain of $K_{v,\omega}$ is therefore finite and restricted to the discrete domain of v . The *local dynamics* in η are determined by an instantiation of the applicable parameter $K_{v,\omega}$

for every v . We speak of a biological regulatory network with multiplexes when each parameter $K_{v,\omega}$ is substituted by a value belonging to its discrete domain.

Definition 7: Biological regulatory network with multiplexes

A *biological regulatory network with multiplexes* (RRBM) is a pair $\mathcal{R} = (\mathcal{G}, \sigma)$ where $\mathcal{G}$ is a GRBM and $\sigma : \mathcal{K} \rightarrow \mathbb{N}$ is a substitution such that, for every $K_{v,\omega} \in \mathcal{K}$, $\sigma(K_{v,\omega}) \in \llbracket 0, b_v \rrbracket$.
 σ is called a parameterisation of the GRBM.

In this thesis, an instance of a pair $(\mathcal{G}, \sigma)$ is called a *model*.

The substitution σ will allow the *global dynamics* of the system to be formalised by a *state transition graph*, where the set of vertices represents the set of discrete states of the system and transitions between states are defined from the values $\sigma(K_{v,\omega})$, cf. Section 2.4.3.

The critical step in modelling biological regulatory networks is *identifying the parameters* $K_{v,\omega}$, that is, finding the substitution or substitutions σ consistent with biological knowledge (if they exist). Introducing the family of parameters $\mathcal{K}$ makes it possible to use classical formal methods from theoretical computer science, such as *model checking* or *Hoare logic*, to carry out symbolic learning of the values of parameters $K_{v,\omega}$ from biological knowledge; cf. Chapter 4 for an introduction to these formal methods. The number of parameterisations increases exponentially with the size of the GRBM. More precisely, this number equals $\prod_{v \in V} (b_v + 1)^{2^{d^-(v)}}$, where b_v is the maximum bound of v and $d^-(v)$ its in-degree. Thus, *reasonable* use of these theoretical tools relies on reducing the domain of parameters $K_{v,\omega}$ as much as possible.

2.4.2.3 Snoussi's condition for constraining dynamic parameter values

Several methods exist for constraining the domain of parameters $K_{v,\omega}$ before using model checking and Hoare logic. First, in 1989 Houssine Snoussi introduced the following monotonicity condition [8]: any resource added to a variable's resource set cannot decrease its focal point. In other words: the absence of a resource must be an absence, and the presence of a resource must be a presence and must not become an absence.

Definition 8: Snoussi's condition

An RRBM $\mathcal{R} = (\mathcal{G}, \sigma)$ satisfies *Snoussi's condition* for a variable v if:
 $\forall \omega \in E^{-1}(v), \forall \omega' \in E^{-1}(v), \omega \subseteq \omega' \Rightarrow \sigma(K_{v,\omega'}) \geq \sigma(K_{v,\omega})$

Not every variable is necessarily subject to Snoussi's condition. There are cases where adding a resource to a variable v decreases its focal point because an interaction between several of its resources, not anticipated when designing the GRBM, renders it inactive.

2.4.2.4 A thought experiment for constraining dynamic parameter values

Furthermore, an empirical method for constraining the domain of parameters $K_{v,\omega}$ has emerged from several decades of modelling work by the *Bio-info Formelle* team at I3S, addressing varied biological questions. This is a pure **thought experiment**, with the following scenario:

Given a variable v receiving *as input* a combination of resources ω assumed constant *ad vitam eternam*, we ask which target or targets v can regulate *as output*. This leads us to consider at least the following two cases:

When v is inactive ($v = 0$). This case reveals whether the resources ω are sufficient for v to act on one or more targets.

When v is saturated ($v = b_v$). This case reveals whether v can maintain its maximum value.

Snoussi's condition can also assist the procedure. This scenario is repeated for every ω .

Note that asking which value v converges towards, with ω fixed, amounts to asking which thresholds v can cross and therefore which target or targets v can act on, given ω .

This experiment requires *biochemical* expertise that cannot be provided directly by a biological experiment. Experimental biology is necessarily systemic. It is therefore impossible to fix and isolate the action of a regulator on a variable, if only because experimental results take into account all the cell's biological processes. It is therefore not possible to render a regulator completely inactive and fix ω as the context of the experiment. To answer the thought experiment, an expert contributes biochemical expertise, while a modeller conducts the experiment and ensures that the expert does not reason systemically.

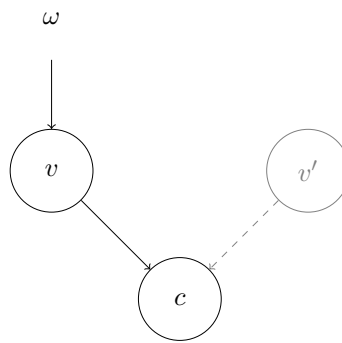


Figure 2.7: **An example of systemic reasoning during the thought experiment.** Here, the error consists in considering that a target c can effectively be regulated by a regulator v , given its resources ω , while implicitly including another variable v' that also regulates c .

As an example, consider the GRBM in Figure 2.8, which represents a positive loop and a negative cycle involving x . Variable x is multivalued and can take three values (0, 1 or 2); y is a Boolean variable. This GRBM generates exactly 324 possible models.

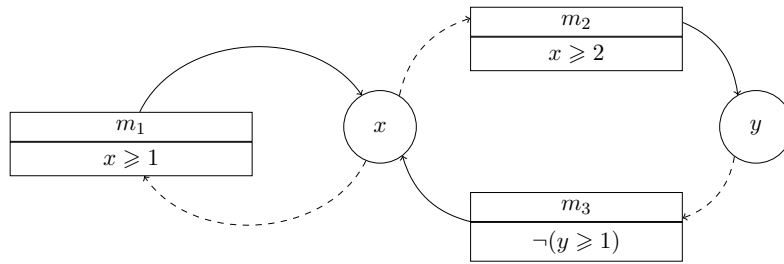


Figure 2.8: **An example of a GRBM to illustrate the thought experiment** It formalises self-activation of x at threshold 1, activation of y by x at threshold 2, and inhibition of x by y at threshold 1.

Suppose the aim of the experiment is to determine the effect of each of these cycles on the qualitative behaviour of x . First, the GRBM identifies 6 dynamic parameters. Four parameters govern the dynamics of x : $K_{x,\{m_1,m_3\}}$, $K_{x,\{m_1\}}$, $K_{x,\{m_3\}}$ and $K_{x,\emptyset}$. Two parameters govern the dynamics of y : $K_{y,\{m_2\}}$ and $K_{y,\emptyset}$. Let us now refocus the thought experiment on variable x and consider all its possible combinations of influences as 4 scenarios illustrated in Figure 2.9.

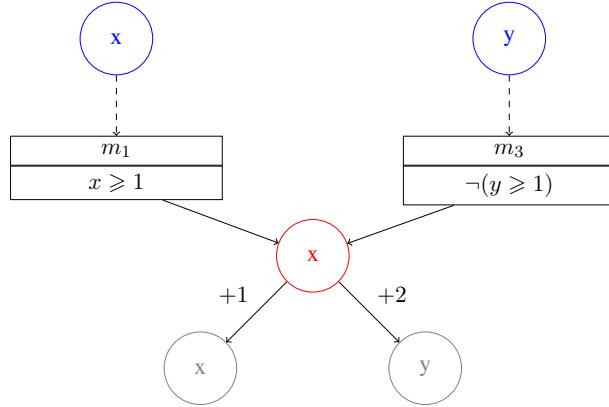
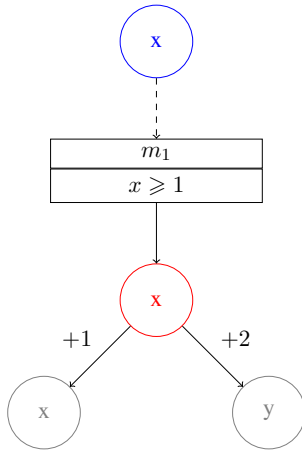
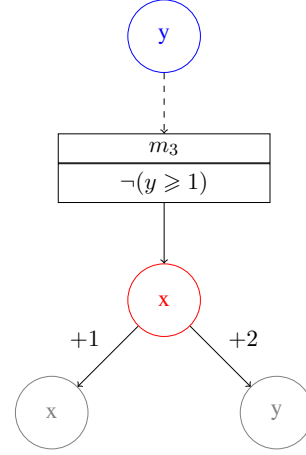
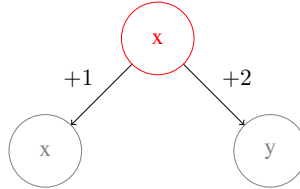
(a) $\omega = \{m1, m3\}$: the case where x is subject to its self-activation and inhibition by y .(b) $\omega = \{m1\}$: the case of effective self-activation of x alone.(c) $\omega = \{m3\}$: the case of effective inhibition of x by y alone.(d) $\omega = \emptyset$: the case where x is subject to no influence.

Figure 2.9: **Illustration of the framework of the thought experiment for constraining the domain of Thomas parameters $K_{x,\omega}$ for every ω of x .** Variable x is placed in the centre in red, its regulators above in blue, and its targets below in grey. The labels on edges preceding the targets of x represent either activation (sign $+$) or inhibition (sign $-$) at a given discrete threshold of x . In cases *a* and *b*, variable x is represented three times: in blue as a *resource*, in red as the *target of the resource*, and in grey as the *target of regulation by x* . The subtlety of the experiment lies in the fact that x is considered active *ad vitam eternam* only in its role as a resource.

$\mathbf{K}_{x,\{m_1,m_3\}}$: This parameter describes the context in which x self-activates (at threshold 1) and is *not* inhibited by y (at threshold 1), *cf.* Figure 2.9a. The question here is which value x is attracted towards in the presence of all its resources, considering an initial condition in which x is inactive ($x = 0$). The *same question* is asked considering an initial condition in which x is saturated ($x = 2$). Suppose our expert biologist tells us in the first case that x has enough resources to reach at least its first regulatory threshold. The modeller formalises this information by the constraint $K_{x,\{m_1,m_3\}} \geq 1$. Suppose in the second case that the resources of x allow x to remain in its saturated state. This information is formalised by the constraint $K_{x,\{m_1,m_3\}} = 2$. In this example, we choose the stronger constraint, namely $K_{x,\{m_1,m_3\}} \geq 2$.

$\mathbf{K}_{x,\{m_1\}}$: This parameter describes the context in which both activator x and inhibitor y are present, implying that ω does not include resource multiplex m_3 , *cf.* Figure 2.9b. Suppose a procedure similar to the preceding scenario gives us the constraint $K_{x,\{m_1\}} = 1$.

$\mathbf{K}_{x,\{m_3\}}$: This parameter describes the context in which both activator x and inhibitor y are absent, implying that ω does not include resource multiplex m_1 , *cf.* Figure 2.9c. This context is called *the basal case* of regulation of x , since no inhibitory or activating regulation is effective. Suppose a procedure similar to the first scenario gives us the constraint $K_{x,\{m_3\}} = 1$.

$\mathbf{K}_{x,\emptyset}$: Finally, this parameter describes the context in which no multiplex is a resource; in other words, no activator of x is present and no inhibitor of x is absent, *cf.* Figure 2.9d. Suppose a procedure similar to the preceding scenario gives us the constraint $K_{x,\emptyset} = 0$.

Finally, the dynamics of x are determined by the set of fixed parameters $K_{x,\{m_1,m_3\}} = 2$, $K_{x,\{m_1\}} = 1$, $K_{x,\{m_3\}} = 1$ and $K_{x,\emptyset} = 0$, allowing us to conclude that:

Cooperation between resources m_1 and m_3 is sufficient to activate x at its second regulatory threshold.

Resource m_3 is necessary for x to reach its first regulatory threshold, but is not sufficient to cross its second threshold, preventing inhibition of y by x .

The same holds for resource m_1 .

From the parameter values, we deduce that activation of x at threshold 1 is characterised by the logical function $\varphi_{m_1} \vee \varphi_{m_3}$, to be distinguished from activation of x at threshold 2, which is characterised by $\varphi_{m_1} \wedge \varphi_{m_3}$. The truth table provided by the parameters $K_{v,\omega}$ therefore gives the update function of a variable, at distinct thresholds in the multivalued case.

Ultimately, since all the dynamic parameters of x have been instantiated and none of those of y have been, 4 possible models remain. This thought experiment accumulates information about the *local dynamics* of the system under study by directly constraining the domain of parameters $K_{v,\omega}$ manually. We denote the conjunction of all these biological constraints by $\phi_{K_{v,\omega}}$. $\phi_{K_{v,\omega}}$ depends on neither v nor ω : it is the conjunction of them all.

2.4.3 Semantics: the asynchronous state transition graph

As mentioned in the preceding section, the asynchronous state transition graph is a mathematical object *deduced from an RRB* and represents its global dynamics. The vertices are the states of the system, and the edges represent transitions between states constructed from a parameterisation σ of the $K_{v,\omega}$.

Definition 9: Asynchronous state transition graph

Given an RRB $\mathcal{R} = (\mathcal{G}, \sigma)$, its asynchronous state transition graph (GTA) is defined as follows:

the set of vertices is the set of possible states ζ of $\mathcal{R}$,

a transition $\eta \rightarrow \eta'$ exists if there is a variable $v \in V$ such that $\sigma(K_{v,\omega}) \neq \eta(v)$, where $K_{v,\omega}$ is the applicable parameter of v in η , and:

1. $\eta'(v) = \eta(v) + 1$ if $\sigma(K_{v,\omega}) > \eta(v)$
2. $\eta'(v) = \eta(v) - 1$ if $\sigma(K_{v,\omega}) < \eta(v)$
3. for every $v' \in V$, $v' \neq v \Rightarrow \eta'(v') = \eta(v')$

The semantics chosen in René Thomas's formalism have two characteristics: they are **asynchronous** (condition 3 of Definition 9) and **unitary** (conditions 1 and 2 of Definition 9).

The first characterises the desynchronisation of variable evolution: when one variable evolves, no other variable evolves simultaneously. The intuitive idea is that the probability of two variables reaching their regulatory thresholds *simultaneously* is zero, particularly given the imprecision of instruments measuring continuous concentrations. Discretisation of the concentration space therefore consists not only in dividing it into hyperrectangles (within which the resource set is constant), but also in desynchronising continuous dynamics.

The second characteristic requires that any edge of a GTA can connect only two neighbouring vertices differing in a single component. Thus, when a variable v is attracted towards a value $\eta'(v)$ that is not its current state $\eta(v)$ and differs by more than one unit, for example $\eta'(v) = \eta(v) + 2$, v must necessarily pass through the state $\eta''(v) = \eta(v) + 1$, in which it is potentially subject to other resources whose influence does not necessarily allow the state $\eta'(v)$ to be reached.

As an example, consider the interaction graph in Figure 2.8, denoted $\mathcal{G}_1$, and the following two parameterisations:

$$\sigma_1 = \{K_{x,\emptyset} = 0, K_{x,m1} = 0, K_{x,m3} = 1, K_{x,m1,m3} = 2, K_{y,\emptyset} = 0, K_{y,m2} = 1\}$$

$$\sigma_2 = \{K_{x,\emptyset} = 0, K_{x,m1} = 0, K_{x,m3} = 1, K_{x,m1,m3} = 2, K_{y,\emptyset} = 0, K_{y,m2} = 0\}.$$

Each parameterisation takes into account the results of the thought experiment on the dynamic parameters of x carried out in Section 2.4.2, except for $K_{\{x,m1\}}$, which takes the value 0 to illustrate the asynchrony of the dynamics in the state characterised by $x = 1 \wedge y = 1$. All else being equal, the two parameterisations differ in a single parameter value: $K_{y,m2}$.

The models $(\mathcal{G}_1, \sigma_1)$ and $(\mathcal{G}_1, \sigma_2)$ allow two asynchronous state transition graphs of order 6 to be constructed, shown in Figure 2.10, producing two qualitatively distinct behaviours. Model $(\mathcal{G}_1, \sigma_1)$ generates an *oscillation* around the thresholds $x = 2$ and $y = 1$, whereas model $(\mathcal{G}_1, \sigma_2)$ generates a *stable state* characterised by $x = 2 \wedge y = 0$.

We observe that parameterisation σ_2 of the regulatory graph in Figure 2.8 shows that adding resource m_2 (presence of activator x at threshold 2) to any resource set of y has no effect on y . In this case, the edge $m_2 \rightarrow y$ is said to be *non-functional*.

Definition 10: Non-functional edge

Given an RRBm $\mathcal{R} = (\mathcal{G}, \sigma)$, an *edge* of $\mathcal{G}$ of the form $m \rightarrow v$ is *non-functional* if:
 $\forall \omega \in E^{-1}(v) \setminus m, \sigma(K_{v, \{\omega, m\}}) = \sigma(K_{v, \omega})$

Dynamic properties of the biological system can be observed directly from an asynchronous state transition graph generated by a model $(\mathcal{G}, \sigma)$, classically including reachability of a set of states, existence of a stable state or cycle, or the presence of one or more basins of attraction.

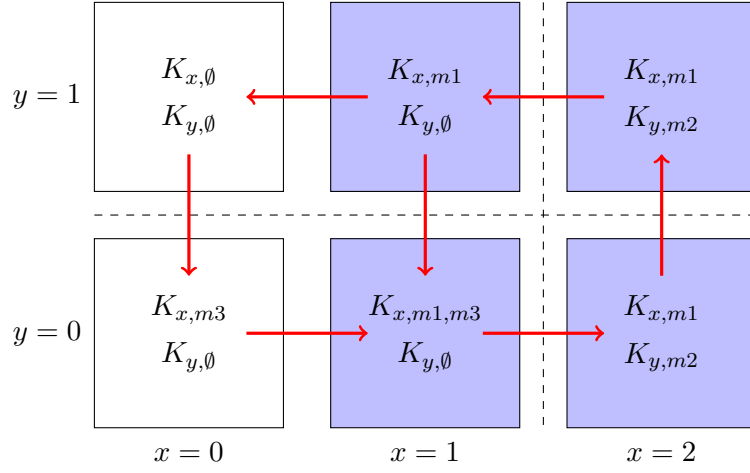
In René Thomas’s multivalued formalism, two *general properties* of RRBm can be deduced directly from the dynamics of an RRBm:

Concerning the presence of several stable states in a GTA: a necessary condition for this observation is the presence of at least one positive cycle in the RRBm. A positive cycle contains an *even* number of inhibitions (with all its inhibitory and activating edges functional). This property is of particular interest because it applies to cell differentiation phenomena (*e.g.* haematopoiesis, embryonic development, plant morphogenesis) or epigenetic phenomena (*e.g.* the case study of *pseudomonas aeruginosa* [21]).

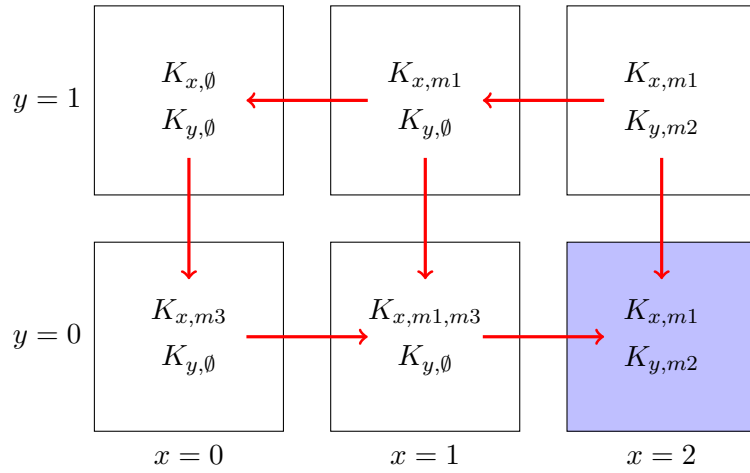
The second property states that the presence of a negative cycle in an RRBm is a necessary condition for the presence of an oscillation in a GTA. A negative cycle contains an *odd* number of inhibitions (again, with all its inhibitory and activating edges functional). Oscillating processes are ubiquitous in biology and characteristic of homeostasis. The cell cycle itself is an example.

These two conjectures were proved by [34] and [50], respectively.

Ultimately, René Thomas’s formalism represents the temporal dimension *chronologically*: only successions of states are represented. Information about measured time therefore cannot be used in this formalism, and all questions about regulatory delays are delegated to dedicated formalisms: either purely quantitative, as discussed in Section 2.2, or hybrid, such as the hybrid extension of René Thomas’s multivalued approach itself.



(a) Asynchronous state transition graph generated from model $(\mathcal{G}_1, \sigma_1)$. An oscillation is observed on either side of the thresholds $x = 2$ and $y = 1$, conceptually represented by the dashed lines.



(b) Asynchronous state transition graph generated from model $(\mathcal{G}_1, \sigma_2)$. The oscillation arising from parameterisation σ_1 does not exist in this model because the activating edge from x to y in the regulatory graph of Figure 2.8 is non-functional. Instead, this model admits a stable state characterised by the formula $x = 2 \wedge y = 0$.

Figure 2.10: Two asynchronous state transition graphs generated from the interaction graph in Figure 2.8 and two parameterisations. States are represented by squares, and transitions by arrows directed from the source state to the target state. Since the modelled system is two-dimensional, each state has only two parameters: one applicable to x and one to y .

CHAPTER 3

A review of landmark models of cell cycle regulation

3.1 Preamble	41
3.2 Pioneering continuous models	43
3.2.1 Identifying a regulatory structure: bistable switches	43
3.2.2 Two intrinsically discrete properties of the checkpoint phenomenon	45
3.3 Discrete models and the formalisation of the checkpoint phenomenon	46
3.3.1 An example of discrete modelling of the regulations involved in certain checkpoints	47
3.3.2 An example of formalising certain chronological constraints during the cell cycle	48
3.3.3 A first attempt to formalise the phases of the cell cycle	49
3.4 Formalising phases with checkpoints: the question remains open	51
3.4.1 Connecting the checkpoint mechanism to its two dynamic properties	52
3.4.2 Intrinsic and extrinsic checkpoints	53
3.5 Overview of checkpoint modelling	54

3.1 Preamble

“Essentially, all models are wrong, but some are useful.” - Georges Box.

The cell cycle is a well-characterised process from a phenomenological perspective. Its different phases (G1, S, G2 and M) are described at the cellular scale by observing the structures formed by the cell and its ploidy, using microscopy or cell-sorting techniques such as flow cytometry. Each phase can also be described by a succession of molecular events (*e.g.* activation or inhibition of biochemical regulators) that correlate with cellular events in the cycle and occur in a precise order to preserve DNA integrity. A typical example is preparation for the S phase. During the G1 phase, pre-replication complexes (abbreviated CPR) are exhaustively recruited to their respective origins of replication. This recruitment is initiated by cyclinE/cdk2 complexes. The genetic integrity of the cell is preserved when exhaustive recruitment of the CPR occurs *before* their activation, which in turn is activated by the cyclinA/cdk2 complex; otherwise, the cell is exposed to deleterious mutations. These preservation mechanisms, called *checkpoints*, ensure that cell cycle events proceed correctly in a precise order, so that DNA is fully duplicated, no more than one copy is made per replication cycle, and DNA is distributed equally between the two daughter cells.

These control mechanisms are encoded in the structure of regulations involving the main cell cycle regulators and intricately coupled feedback loops [45]. Cell cycle regulation has been extensively studied since the discovery of cyclin/cdk complexes by Tim Hunt and Paul Nurse in 2001, Nobel laureates in Physiology or Medicine. Experimentally, the challenge is to identify the main actors in cell cycle regulation that can be correlated with cellular phenomena, such as passage through the restriction point or the G1/S transition. The modelling challenge lies in managing complexity and characterising the regulatory structures involved in the emergence of checkpoint phenomena.

This chapter provides an overview of mathematical models that address cell cycle control and how they approach the question of checkpoints. Historically, modelling used systems of ordinary differential equations (Section 3.2), but this approach soon encountered the costly process of fitting parameters to experimental data. Qualitative approaches followed and demonstrated their ability to reproduce the main behaviours of the cell cycle in a purely discrete universe (Section 3.3). These approaches focus on studying interaction motifs that are necessary or sufficient to obtain fundamental behaviours. In light of these approaches, we shall demonstrate that purely discrete approaches prove the most suitable for formalising and verifying the dynamic properties characteristic of checkpoints (Section 3.4). Table 3.1 provides an overall view.

Model	Formalism	Formalisation of a phase	Formalisation of an intrinsic checkpoint
[87]	Boolean	Conditionally stable motif (CSM) of an expanded network	Sufficient condition for the stability of the CSM (propositional logic restricted to conjunction and negation.)
[88]	Boolean	Partition of a summary graph of an asynchronous state transition graph	Irreversibility of transitions between two adjacent phases
[76]	Multivalued	Stable state (quiescence only)	CTL formula encoding the non-existence of certain paths in a state transition graph
[71]	Multivalued then hybrid	An arbitrary path in genetically modified Hoare logic	Not addressed
[64]	Multivalued	Not addressed	The ATM/ATR-p38MAP pathway in the case of DNA damage (extrinsic checkpoint).
[43]	Continuous (ODEs)	(Transiently) stable state (G1, S and M)	ATR/Chk1 pathway (DNA replication checkpoint)
[41]	Continuous (ODEs)	(Transiently) stable state (G1 and S/G2/M only)	Bistable switch
[19]	Piecewise linear equations	Not addressed	A CTL formula that verifies obligatory passage through a checkpoint state before the transition to the next phase.

Table 3.1: Overview of the formalisation of phases and their checkpoints in landmark models for understanding the phenomenon and its causes. The formalisation of a phase is inseparable from that of checkpoints. The representation and biological validation of the domain of a phase remain open questions. Moreover, to our knowledge, no study has examined the characterisation of all the phases of the cell cycle (G1, S, G2 and M) and the dynamic constraints underlying checkpoints.

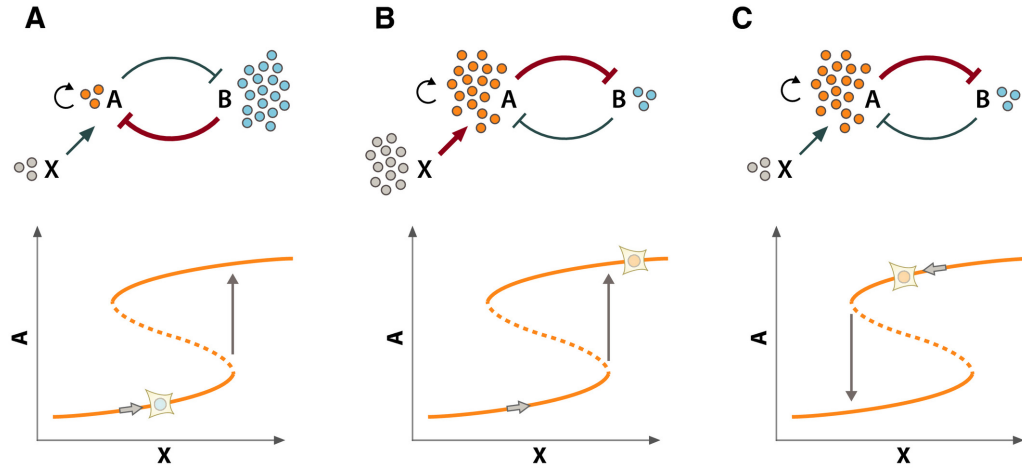


Figure 3.1: **The stable switch mechanism and its hysteresis property: a first step towards understanding checkpoints.** Variables A and B have an activation threshold which, when crossed, allows the system to stabilise exclusively in one or the other stable state. The first stable state is characterised by inactive A and active B (panel A). In this example, activation by X allows A to reach a concentration sufficient to inhibit B , placing the system in the stable state characterised by the inactivation of B and activation of A (panel B). Adding the activation signal from X to A switches the system *irreversibly* into one of the stable states exhibited by the bistable switch. In the chosen example, the self-activation of A and the inhibition of B by A are sufficient to maintain A in a stably active state, regardless of the concentration of X (panel C of the figure). This figure is adapted from [94].

3.2 Pioneering continuous models

Modelling with systems of ordinary differential equations (abbreviated ODEs) was initially the most common approach and provided a structural explanation for the emergence of the dynamic properties of the cell cycle. We shall mention the temporal separation of replication and mitosis and the unidirectionality of the cell cycle. A system of ODEs continuously represents changes in the concentrations of the biochemical entities involved in the biological question as a continuous function of the concentrations of their regulators, *cf.* Chapter 2, Section 2.2.

3.2.1 Identifying a regulatory structure: bistable switches

The landmark models are [24, 41, 43]. They show that the regulation of the main cyclin/cdk complexes is nonlinear and relies on a *stable switch* mechanism. A bistable switch is a positive feedback loop (an even number of inhibitions) that generates two *distinct* stable states. Figure 3.1 is a toy example in which two distinct stable states are generated by mutual inhibition between two variables denoted A and B . The model in [43] also exhibits two distinct stable states: a quiescent cell and a proliferating cell. Here, a quiescent cell is associated with the constant inactivation of the various cyclin/cdk complexes, namely cyclinD/cdk4-6, cyclinE/cdk2, cyclinA/cdk2 and cyclinB/cdk1. A proliferating cell is associated with a self-sustained oscillation of these complexes. The transition from a quiescent state to a self-sustained cell cycle is abrupt and is controlled by a bistable switch involving growth factors, *cf.* Figure 3.2. The stable switch mechanism is system-

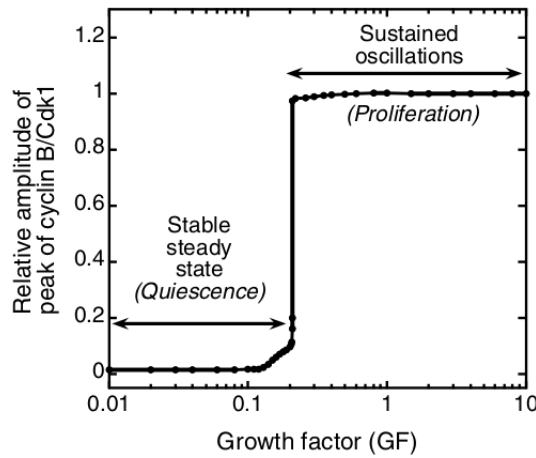


Figure 3.2: **The switch-like activity of cell cycle regulatory complexes illustrated with the cyclinB/cdk1 complex.** The effect of growth factors (GF) on the activity of the cyclinB/cdk1 complex can be approximated by a step function (e.g. Figure 2.2) or a logical function. This figure is adapted from [43].

atically modelled *via* a Hill function (Definition 1, page 18), and an appropriate parameterisation reproduces a phenomenon of *ultrasensitivity*. This phenomenon reflects the fact that a regulator has a concentration threshold beyond which its target switches *almost instantaneously* from an inactive state to an active state.

The modelling study by [41] took the abstraction of the phenomenon of cell cycle control to its highest level at that time. A phase is represented there, from a phenomenological perspective, by a self-sustained stable state. The integrity of the phase is ensured by a *checkpoint* that prevents overlap with the next phase by transiently maintaining the stability of the current phase. The authors structurally represent a checkpoint by a bistable switch that generates the stability of a state characteristic of a phase. The state remains stable as long as a certain set of regulatory events, *i.e.* abrupt activation or inhibition of a regulator's concentration, has not occurred. Only when all the required events have occurred can the transition to a phase take place. We mentioned earlier that the G1 phase cannot begin until the cyclinE/cdk2 complex has activated all the pre-replication complexes. Following the example provided by [41] in Figure 3.3, the G1 phase has a checkpoint if it remains stable precisely as long as activation of the cyclinE/cdk2 complex (represented here by the variable SK , the *starting kinases*) has not occurred. Here, the G1 phase is characterised by inactivation of the cyclinA/cdk1-2 and cyclinB/cdk1 complexes, both abstracted by the variable CDK , which the authors contrast with the S/G2/M block of phases, characterised by an active CDK variable.

The lower part of Figure 3.3 shows that the cell cannot switch from a G1 state to an S state while SK is inactive, indicating that activation of SK is required before that of CDK . Only when SK is activated at the end of G1 does the system switch to the S/G2/M state. This switch from G1 to S relies on activation by SK of the bistable switch formed between CDK and the *Enemies*, a variable representing the main cyclin inhibitors (*cf.* the upper part of Figure 3.3), in favour of CDK activation. The S/G2/M state can then sustain itself once the CDK inhibit their own inhibitor, in

this case the *Enemies*.

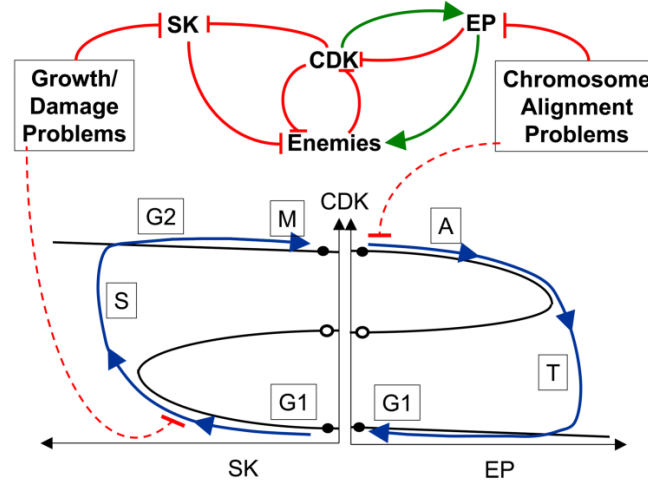


Figure 3.3: **Graph of the stable states of the model by [41], showing the G1 and S/G2/M phases and the G1/S and metaphase/anaphase transitions (bottom), and the stable switches representing checkpoint mechanisms (top).** The positive loop formed between *CDK* and the *Enemies* generates two stable states. A low *CDK* concentration is associated with G1, whereas high activity is associated with S/G2/M. Activation of *SK* allows stable activation of *CDK*. Once *CDK* is activated, it activates its own repression by activating *EP*. Activation of the latter activates mitotic exit by directly inhibiting *CDK*, returning to the state that characterises the G1 phase. When DNA damage is detected during the G1 phase, the G1/S transition is *additionally* blocked *via* inhibition of *SK* by the variable “Growth / Damage Problems”. This figure is adapted from [41].

3.2.2 Two intrinsically discrete properties of the checkpoint phenomenon

The continuous models described above have revealed purely qualitative dynamic properties of the cell cycle that emerge from coupling in series the bistable switches formed by cyclin/cdk complexes and their inhibitors. These dynamic properties, which characterise the **checkpoint phenomenon**, are as follows:

1. *The temporal separation of phases* is ensured by a bistable switch that separates two successive phases and controls the transition from one phase to the next. These phases can be characterised by states that are mutually exclusive from a logical perspective. Phase transitions are abrupt activations or inhibitions of cyclin/cdk complexes or their inhibitors, marking a clear transition from one phase to another. These abrupt activations and inhibitions can, moreover, be regarded as discrete events. Only when the phases are distinct can all the events belonging to each phase be isolated and two adjacent phases avoid overlapping. A bistable switch maintains the stability of a phase until all the events required during a phase have occurred. When this is the case, the phase initiates its own destabilisation and the transition to the next phase takes place.

2. *The unidirectionality* of the cell cycle is ensured by the hysteresis phenomenon that emerges from a bistable switch. The signal that allows a cell to switch into a phase is not the one that destabilises that same phase. In other words, the transition from one phase to the next is *irreversible* because it is triggered by a different signal from the one that initiated it. Coupling bistable switches in series through negative feedback loops (an odd number of inhibitions) generates a phenomenon of sequential stabilisation of the different phases, each of which induces its own destabilisation once all the required events have occurred. We shall illustrate this property with the first two checkpoints of the cell cycle. The first stable switch during the cell cycle is activated by growth factors and irreversibly commits the cell to G1 *via* the autonomous stabilisation of cyclinE/cdk2. This stable switch is formed by a positive loop between E2F and the cyclinE/cdk2 complex [94]. Once stabilised, this complex activates both its own inhibition and the stabilisation of the S phase by stably activating cyclinA/cdk1 [41]. Three other checkpoints are then also coupled in series: the S/G2, G2/M and metaphase/anaphase transitions.

[41] and [94] emphasise that checkpoints reflect mechanisms of interdependence between discrete events. They do not slow down a phase if a crucial event does not occur, but block it completely for as long as that event has not occurred. However, these checkpoint phenomena are not formalised, since choosing a continuous formalism does not provide the appropriate language for expressing logical dependency constraints between discrete events, or at least unnecessarily complicates the problem of formalising this constraint. Nevertheless, an initial effort to formalise the phases of the cell cycle has been made, if only because a checkpoint is a notion inseparable from a phase: the phase must be formalised beforehand.

[43] is the study that offers the finest granularity. The variables cyclinD/cdk4-6, cyclinE/cdk2, cyclinA/cdk2 and cyclinB/cdk1 allow the G1, S and M phases, respectively, to be represented by a (transiently) stable active state. The study by [41] identifies the regulatory event that allows the G1 checkpoint to be reached and hence the G1/S transition to occur. Activation of *SK* is therefore required before that of *CDK*. This constraint, expressed in logical terms, conveys a fundamental characteristic of the checkpoint phenomenon. Nevertheless, this constraint on the chronology of events is not formalised within the continuous framework, or only in terms of measured time. The model by [43] is a perfect example: the ATR/Chk1 pathway was added to the system of ODEs to constrain the duration of the G1 and S phases so that they do not overlap. Yet adding a delay should not be what makes the checkpoint phenomenon operational. The phenomenon is rooted in the very structure of the regulations. Logical formalisms such as Boolean networks or René Thomas's formalism are the most appropriate for representing a phase by a set of discrete events *and* logical constraints on the transitions between these phases. However, we emphasise in advance that, to our knowledge, there are no qualitative models dedicated to formalising phases and therefore checkpoints.

3.3 Discrete models and the formalisation of the checkpoint phenomenon

[29] were the first to prove that a logical modelling formalism can reproduce the key asymptotic properties of the cell cycle exhibited by the models in [24]. This ODE-based model was translated

into a generic Boolean network of the cell cycle that captures the mechanisms controlling entry into the cell cycle shared by yeast and mammals. Analysis of the state transition graph reveals the same attractors, transposed into a discrete world. The quiescent phase is represented by a stable state of the state transition graph, deduced from the Boolean functions associated with the variables of [24]. The self-sustained oscillation of cyclin/cdk complexes is represented by a terminal strongly connected component. This complex attractor is associated with the cell cycle itself, under a constant supply of a sufficient concentration of growth factors. This model does not verify a property inseparable from the cell cycle, namely the correct succession of the G1, S, G2 and M phases. The constraints operating at the critical transitions of the cell cycle are therefore not addressed.

3.3.1 An example of discrete modelling of the regulations involved in certain checkpoints

The model by [88] proposes extending the analysis of the complex attractor of [29] using summary graphs (Figure 3.4). The authors define k -partitions of a complex attractor, where the par-

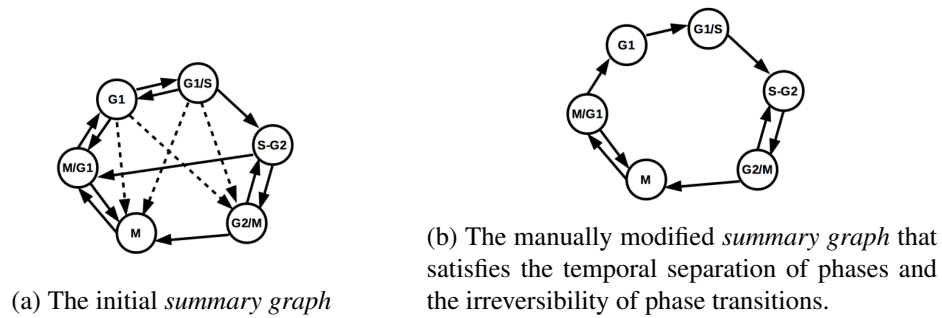


Figure 3.4: A *summary graph* of the cell cycle proposed by [88] based on the Boolean model by [29]. The markers used to define a phase are the states (active or inactive) of the variables $E2F$, cyclins E, A and B, Cdc20 and Cdh1. This figure is adapted from [88].

titions are non-empty and mutually exclusive, and k is an integer. The disjointness of the partitions ensures that the phases do not overlap, that is, the temporal separation of phases: one of the characteristics of the checkpoint phenomenon mentioned above. The states belonging to the cell cycle were classified into six partitions by defining a characteristic formula (restricted to propositional logic) for G1, G1/S, S/G2, G2/M, M and M/G1. Each encodes a particular state of $E2F$, cyclinE/cdk2, cyclinA/cdk1-2 and cyclinB/cdk1, and they are defined so as to be mutually exclusive. These partitions are actually semi-partitions, since the authors exhibit a non-empty *unknown partition* gathering all the states that have not been classified into one of the phases or portions of phases mentioned above, *i.e.* those that satisfy none of the characteristic phase formulae. From the state transition graph and the six partitions, the authors reconstruct the transitions between the different phases and aim to verify the only correct ordering, namely $G1 \rightarrow G1/S \rightarrow S/G2 \rightarrow G2/M \rightarrow M \rightarrow M/G1$, which guarantees that all the events of one phase have occurred before the next phase is reached, and that the cell cycle is unidirectional. Using the REVEAL algorithm [12], the authors infer a Boolean network from the modified summary graph that satisfies the correct sequence defined above, *cf.* Figure 3.4. However, the authors did not focus on specifying the constraints that a phase must satisfy before making its irreversible

transition to the next phase. Indeed, these constraints apply to events that must occur in a certain order, and summary graphs do not have the appropriate expressive power to identify key events (*i.e.* transitions) that cannot be permuted. In terms of events, a phase remains a black box whose domain is fixed arbitrarily.

3.3.2 An example of formalising certain chronological constraints during the cell cycle

The model by [76] uses CTL logic, presented in the next chapter, to formalise a characteristic path of the cell cycle, based on an updated version of the model by [29]. The cell cycle is represented mathematically by a CTL formula of the form:

$$E[G1 \cup S \wedge E[S \cup G2 \wedge E[G2 \cup M \wedge E[M \cup G1]]]]$$

which can be represented schematically as:

$$G1 \xrightarrow{G1/S} S \xrightarrow{S/G2} G2 \xrightarrow{G2/M} M \xrightarrow{M/G1} G1$$

where:

$G1, S, G2$ and M are characteristic formulae of the marker states of the corresponding phase,

E is the existential quantifier over paths (of a state transition graph),

$\cup$ is a binary temporal operator; for example, $\varphi \cup \psi$ (read “ φ until ψ ”) expresses, along a sequence of states, that a propositional formula φ holds until a state is reached where another propositional formula ψ holds, *cf.* Figure 3.5.

Again, the authors do not focus on defining the characteristic formulae of the phases, but *formally* investigate the existence of a *particular* sequence of states of the cyclinE/cdk2 (denoted E for simplicity), cyclinA/cdk1 (denoted A) and cyclinB/cdk1 (denoted B) complexes within the state transition graph associated with their model:

$$\begin{aligned} (E = 0 \wedge A = 0 \wedge B = 0) &\xrightarrow{E+} (E = 1 \wedge A = 0 \wedge B = 0) \xrightarrow{A+} (E = 1 \wedge A = 1 \wedge B = 0) \\ &\xrightarrow{E-} (E = 0 \wedge A = 1 \wedge B = 0) \xrightarrow{B+} (E = 0 \wedge A = 1 \wedge B = 1) \\ &\xrightarrow{A-} (E = 0 \wedge A = 0 \wedge B = 1) \xrightarrow{B-} (E = 0 \wedge A = 0 \wedge B = 0) \end{aligned}$$

where $(E = 0 \wedge A = 0 \wedge B = 0)$ specifies a cell at the beginning of G1. Some transitions exhibited by this sequence are associated with the initiation of a phase; for example, $A+$ initiates the S phase and $B+$ the M phase. The existence of this particular experimentally observed sequence is verified automatically (and very efficiently) using the Nu-SMV model checker. *In silico* loss-of-function experiments revealed that blocking the activation of certain variables prevents certain phase transitions. For example, simulating the system after loss of function of the cyclinE/cdk2 complex shows cell arrest before the S phase, that is, before the $A+$ transition. A collection of perturbations allowed the authors to identify certain regulatory events required before the transition from one phase to the next.

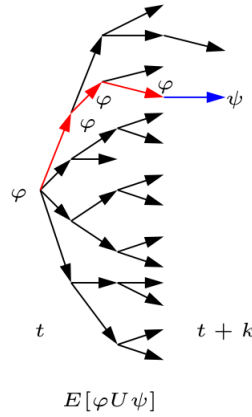


Figure 3.5: **Constraint on the order of phases expressed using a CTL formula and the tree representation of paths in a state transition graph** The formula $E[\varphi U \psi]$ encodes a constraint on the ordering of φ and ψ that ensures their temporal separation. [19] and [76] characterise a cell cycle checkpoint by the formula $\neg E[(\neg\varphi) U \psi]$, cf. the main text for further details.

Moreover, the initial choice of asynchronous semantics reveals that other transitions are possible from certain states of their cell cycle sequence. Analysis of alternative transitions along the characteristic cell cycle sequence defined above verified that certain events do not occur before others; for example, the activations of E and A cannot be permuted, allowing G1 and S to be temporally separated. Checkpoints are also studied from the perspective of the irreversibility of certain events. For example, their model validates the irreversibility of the inactivation of E . The efficiency of the model-checking algorithm used allowed the authors to verify a collection of sequences defined *a priori*, some of which are biologically admissible and others not.

This study, inspired by [19], introduces the logical formalisation of a checkpoint with the CTL formula $\neg E[(\neg\varphi) U \psi]$, which verifies that visiting states characterised by φ is obligatorily required before those characterised by ψ . It literally means that there is no path allowing states characterised by ψ to be reached without necessarily first passing through those characterised by φ . Formalising φ and ψ beforehand is essential and instead relies on a characterisation of the events required during a phase and before initiation of the next phase. This modelling study does not establish which events make up a phase, implying that the question of checkpoint satisfiability cannot be elucidated. Instead, this study aims to demonstrate the relevance of formal methods to the formalisation and automatic verification of the dynamic properties listed by [29]. The challenge now is to focus on a reasonable definition of φ and ψ , that is, of the domain of a cell cycle phase. The characteristic formula of the domain of a phase is deduced from a set of events and an initial state associated with it.

3.3.3 A first attempt to formalise the phases of the cell cycle

The discrete part of the hybrid modelling study of the cell cycle by [71] provides precisely a first example of modelling a phase by a sequence of required events from an initial state, using genetically modified Hoare logic. They use the syntax of Hoare logic to represent a discrete event by

incrementing ($v+$) or decrementing ($v-$) a biological variable. An initial state is represented by a propositional formula. The authors transposed the model by [41] into a discrete world using René Thomas’s multivalued formalism. One limitation of the interaction graph of the model by [41], with regard to formalising phases, is the choice of variables, which did not allow the S, G2 and M phases to be distinguished. The authors therefore propose splitting *CDK* into two variables: *A* represents the cyclinA/cdk1-2 complex involved in the initiation and termination of the S phase, and *B* the cyclinB/cdk1 complex involved in the initiation of the M phase. The phases are formalised by:

$$\begin{aligned} G1 &: sk+; sk+; en- \\ S &: a+; sk-; sk-; b+ \\ G2 &: a-; ep+ \\ M &: en+; b-; ep- \end{aligned}$$

where the initial state of the cell cycle (that of G1) is formalised by a formula characterising the inactivation of all cyclin/cdk complexes and of *EP*, and the activation of *En*. To our knowledge, this is the first study to make explicit a set of events required during a phase.

Again, in this study the existence of this path is verified formally, but this time using genetically modified Hoare logic. This formal method automatically establishes the proof of correctness of a Hoare triple of the form $\{precondition\} \text{ path } \{postcondition\}$, where, in the example by [71], the precondition encodes a single initial state, the path is a sequence of discrete events, and the postcondition corresponds to the state reached after executing the path. Given a state transition graph, a correct Hoare triple verifies that the state characterised by the *postcondition* is reachable from a state satisfying the *precondition* by following the *path*. We shall not elaborate further, since the next chapter is dedicated to a formal presentation of this method. A set of events required per phase is made explicit and validated here. However, the constraints of temporal separation of phases and irreversibility are not defined, even though the notion of events required during a phase is introduced.

Temporal separation actually requires that *the event that closes a phase cannot occur before the one that initiates the next phase*. Following on from this study, this thesis will focus on formalising this constraint through an *incremental approach*. An essential preliminary step is to refine the formalisation of phases initially proposed by [71]. We carried out this refinement in the article [97]. This preliminary work contributed to formulating the main question of the thesis, which is why we shall describe it in this section.

The article starts from the model by [71] and proceeds to the question of event orderings within a phase. This study originated in an observation by the authors of [76]: biological knowledge admits several orderings within their model with asynchronous semantics. In the same vein, introducing the parameters $K_{v\omega}$ can further relax the constraint on event ordering within a phase, in the sense that no logical activation function for the variables is presupposed, implying that no model is chosen arbitrarily, *cf.* Chapter 2, Section 2.3. Only after moving beyond an arbitrary choice of a single model and a single ordering can biological knowledge be formally queried to exhibit *all* event orderings within a phase that are *biologically admissible*, and thus which events can close a phase and initiate the next one (an example is given in Table 3.2).

The biological knowledge of the model by [71] was formally queried using the TotemBioNet tool, described in the next chapter. Very briefly, TotemBioNet is a software platform that assists modellers in designing and parameterising multivalued regulatory graphs *à la Thomas*, by efficiently combining genetically modified Hoare logic and model checking of CTL formulae. It efficiently enumerates all the models associated with a given regulatory graph. The main technique lies in the ability of genetically modified Hoare logic to narrow the bounds of the domains of the parameters $K_{v\omega}$, using a constraint obtained from the automatic calculation of the weakest precondition correctness proof, *cf.* Chapter 4. It is thanks to this tool and to the path quantifiers of Hoare logic that we identify the sequences of events required during a phase that are admitted by a given regulatory graph. Of the 100,800 models of the regulatory graph of [71] that satisfy Snoussi's condition (Definition 8), 676 models satisfy the cell cycle represented by the triple:

$$\{G1_{init}\} \underbrace{sk+; sk+; en-}_{G1}; \underbrace{a+; sk-; sk-; b+}_{S}; \underbrace{a-; ep+}_{G2}; \underbrace{en+; b-; ep-}_{M} \{G1_{init}\}$$

such that $G1_{init} \equiv (sk = 0 \wedge a = 0 \wedge b = 0 \wedge ep = 0 \wedge en = 1)$. The study concludes that the G1 and G2 phases, as defined, admit all orderings of their associated events, suggesting that a phase should be considered a multiset of events rather than a sequence of events. The initiation of the S and M phases, meanwhile, is fully constrained by biological knowledge, *cf.* Table 3.2. The key events of the cell cycle constrained by checkpoints are identified. However, the challenge remains to reasonably define the constraint itself that ensures *no permutation of these key events*.

3.4 Formalising phases with checkpoints: the question remains open

Table 3.1 lists the models mentioned in the two preceding sections. The first generic models of cell cycle regulation identified the structure necessary for the emergence of a sequence of phases with checkpoints. This structure is a coupling of stable switches in series through negative feedback loops. Since the choice of a continuous formalism does not allow an exhaustive set of stable states generated by these structures to be identified, neither the temporal separation of phases nor the irreversibility of phase transitions can be proved.

Phase	Event that can initiate it	Event that can close it
G1	$sk+$ or $en-$	$sk+$ or $en-$
S	$a+$	$b+$
G2	$ep+$ or $a-$	$a-$ or $ep+$
M	$en+$	$ep-$

Table 3.2: **Events that can initiate and close the phases initially defined by [71].** A checkpoint prevents the permutation of any event that can close a phase with one that can initiate the next phase. For example, $sk+$ or $en-$ can close G1, and $a+$ is the only event that can initiate the S phase. $sk+; sk+; a+$ or $sk+; en-; a+$ must necessarily be a path that cannot be deduced from the regulatory graph of [71] starting from the initial state $G1_{init}$. The study by [97] makes explicit the events within a phase that cannot be permuted, with a view to validating the G1/S, S/G2 and G2/M checkpoints.

3.4.1 Connecting the checkpoint mechanism to its two dynamic properties

Recently, the work of [87] proposed addressing this through an in-depth qualitative study of the dynamic properties of stable switches. They exhaustively identify the set of stable states associated with a regulatory graph that includes the stable switches characterised by [24, 41, 43]. They show that these states are conditionally stable and explain that the logical condition that makes these states stable is a checkpoint. Starting from a Boolean network, they enumerate *conditionally*

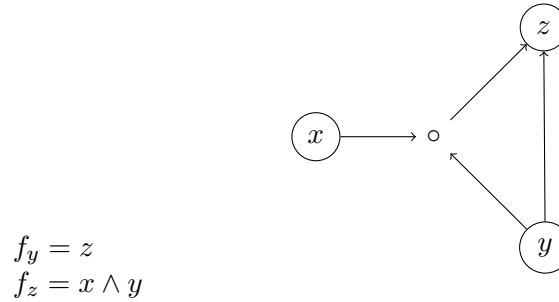


Figure 3.6: **A toy example of an expanded graph that makes a conditionally stable motif explicit.** $y \longleftrightarrow z$ is a stable motif because the reciprocal inhibition of y and z is sufficient to maintain the state $y = 1 \wedge z = 1$, but conditionally on the presence of x , which represents the condition that a motif must satisfy to become stable.

stable motifs using an expanded graph. The latter encodes both the regulatory structure and the logical rules for activating the variables of the Boolean network. Each state of a variable ($v = 1$, denoted here by v , and $v = 0$, denoted by $\neg v$) potentially forms a node of the expanded graph if the state of the variable appears in a logical function of the Boolean network. A composite node, denoted $\circ$, formalises the $\wedge$ operator, and its predecessors are its operands. An edge and its predecessors represent the sufficient condition for observing the targets of the edge. For example, in Figure 3.6, the composite node and its predecessors represent the sufficient condition for the stability of the state $x = 1 \wedge y = 1$. This type of expanded graph makes it possible to identify the exhaustive set of conditionally stable motifs, that is, a set of nodes of an expanded graph

that stably maintain their state if a condition is satisfied. A phase is then associated *a posteriori* with one of these conditionally stable motifs, and the authors make explicit the logical condition that characterises its “checkpoint”. When this condition is satisfied, the system switches stably into a phase that remains stable until the negation of the checkpoint condition is satisfied and triggers destabilisation of the motif [30]. Moreover, these results echo the work of [76], in the sense that loss of function of the events required during a phase makes the condition of the conditionally stable motif true and implies phase arrest. Using their model, the authors were thus able to make explicit the conditions for passing the restriction point in G1, the S/G2 transition and the metaphase/anaphase transition. This model offers a clearer view of the mechanisms that determine the emergence of a checkpoint.

The study by [87], coupled with that by [88], reflects the typical tendency to first calculate the (stable or conditionally stable) attractors of an (arbitrarily chosen) model and then associate them with the phase to which they are closest. [76] and [71] treat formalisation of the cell cycle, its phases and its checkpoints as a preliminary question, but study it only partially. Only a convincing formalisation of these dynamic phenomena will allow convincing verification that a regulatory structure (*e.g.* a stable switch) can generate these dynamic phenomena (*e.g.* a checkpoint). Yet, to date, we have found no modelling study dedicated to a reasonable formalisation of the phenomena *themselves* of temporal separation of phases and irreversibility of phase transitions. However, this review of the literature offers some avenues. A phase with a checkpoint is first formalised by a CTL formula of the form $E[\varphi U \psi]$ by [76] and [19]. φ is the characteristic formula of a phase and ψ the characteristic formula of the “checkpoint” states, *i.e.* the states through which it is necessary to pass before making the transition (or transitions) to the next phase. Only from this state can the transition from one phase to another take place, since it verifies that all the events required during a phase have occurred. [71] propose using the event syntax of genetically modified Hoare logic to represent this set of required events. From the sets of events characteristic of a phase, [97] identify all key events that cannot be permuted, by relaxing all arbitrarily defined constraints.

3.4.2 Intrinsic and extrinsic checkpoints

So far, all the work presented assumes that replication and mitosis proceed correctly once activated. Some models, typically that of [64], focus on modelling pathways activated in response to DNA damage during the different phases of the cell cycle. In this thesis, we shall distinguish two types of checkpoints and focus solely on modelling the first. We shall call the mechanisms that generate a correct ordering of events throughout the cell cycle *intrinsic checkpoints*. We shall call the mechanisms that supplement intrinsic checkpoints and are activated specifically when DNA damage is detected *extrinsic checkpoints*. Extrinsic checkpoints activate intrinsic checkpoints for as long as the DNA has not been repaired. Figure 3.3, page 45, provides two examples of extrinsic checkpoints. The first is the pathway involved in detecting DNA damage during replication. This pathway is named “*Growth / Damage Problems*” in the figure and corresponds to the ATR/Chk1 pathway, which inhibits the G1/S transition until the DNA is repaired. The second example is the activation of the SAC when a problem with chromosome alignment along the metaphase plate is detected, which consists in inhibiting the metaphase/anaphase transition. For an extrinsic checkpoint to be verified by a model, an intrinsic checkpoint must necessarily be formalised and verified beforehand.

3.5 Overview of checkpoint modelling

In light of this chapter, we summarise our understanding of checkpoints as follows: intrinsic checkpoints create structural dependencies between regulatory events of the cell cycle. Overall, these structures take the form of a series of bistable switches linked by negative feedback loops. These dependencies allow the start of a phase to be *blocked* until the preceding phase is complete. If an event in a phase is blocked by DNA damage sensor proteins, the cell cycle must arrest. Checkpoint phenomena inherently rely on *discrete concepts*. The main aim of this thesis is to model generically the two phenomena generated by intrinsic checkpoints: the temporal separation of phases and the irreversibility of phase transitions. The phases must therefore be formalised beforehand. The cell cycle modelling study by [71], which inspires our regulatory graph for studying phases (Chapter 5), proves that the cell cycle can be represented generically through discrete concepts. We shall carry out this proof of concept on the checkpoints of the cell cycle phases. To do so, we focus on checkpoints between phases, which are prototypical. They control the initiation or termination of a phase.

The introduction tells us that cell cycle checkpoints are not necessarily located between two adjacent phases. There are two checkpoints within phases: the metaphase/anaphase transition within the M phase and passage through the restriction point. The first represents a research challenge, laborious and still largely unresolved, which cannot coexist with the proof of concept that is the subject of this thesis. The second is the restriction checkpoint, canonically located within the G1 phase. It controls the G0/G1 transition according to the concentration of growth factors. An insufficient concentration arrests a cell in G0. A sufficient concentration allows the cell to enter G1, but an early portion remains dependent on growth factors. In this thesis, representing growth factors mathematically by an environmental variable greatly facilitates verification of the restriction checkpoint. An environmental variable is entirely under the modeller's control: its concentration is fixed. This restriction checkpoint phenomenon is fundamental and can be modelled in CTL, which is why we consider it a classical property of the cell cycle (verified in Chapter 5). *Ultimately*, all these arguments explain why we shall focus on the G1/S, S/G2 and G2/M checkpoints for the proof of concept of the thesis (Chapter 7). This consists in identifying a discrete mathematical model of the cell cycle that satisfies our specifications of the two properties of intrinsic checkpoints identified by the review of the literature.

Formal Methods Used

CHAPTER 4

Contribution of two formal methods for discrete regulatory networks

4.1	An epistemological preamble: conjunctions and refutations . .	59
4.2	Model checking of CTL formulae	60
4.2.1	CTL syntax	61
4.2.2	CTL semantics	61
4.2.3	Principle of model checking	64
4.2.4	CTL and fair paths	65
4.2.5	Some examples of properties formalised in (fair) CTL . .	66
4.3	Genetically modified Hoare logic	67
4.3.1	Syntax	68
4.3.2	Semantics	69
4.3.3	Inference rules	70
4.3.4	Computing the weakest precondition	72
4.4	Managing model exhaustiveness with TotemBioNet	73
4.4.1	Inputs and outputs	73
4.4.2	Workflow	74

4.1 An epistemological preamble: conjunctions and refutations

René Thomas’s formalism allows biological knowledge to be represented in a logical model. Chapter 2 describes the modelling approach chosen for this thesis, which consists in formalising a set of biological regulations as a biological regulatory graph with multiplexes (GRBM). The dynamics of a GRBM denoted by $\mathcal{G}$ are represented by an asynchronous state transition graph derived from a parameterisation σ of G . An instance $(\mathcal{G}, \sigma)$ is called a model in the context of this thesis. The size of GRBMs has continued to increase since the earliest modelling work in biology; the critical step therefore lies primarily in identifying parameterisations that reproduce biological properties.

Tools adapted to Boolean formalisms can compute, for a given model, the sets of states belonging to a terminal strongly connected component [79] or to a basin of attraction [82]. However, the strategy of analysing all models one by one with these tools is tedious, given the combinatorial explosion in the number of parameterisations.

The strategy of arbitrarily choosing a model without supporting biological knowledge is ruled out. Although some parameters $K_{v,\omega}$ can be identified by carrying out a thought experiment with the assistance of an expert biochemist, it is very rare to succeed in identifying a unique parameterisation at the end of this experiment. As emphasised in Chapter 2, Section 2.4.2.4, experimental biology is a science that studies living systems as a whole, whereas the parameter $K_{v,\omega}$ formalises a local context: the regulation of v given a fixed set of resources ω . The philosopher Willard Van Orman Quine expresses this as follows: “*Our statements about the external world face the tribunal of experience not individually but collectively*” [6].

The strategy selected for this thesis gives preference to automatically learning parameterisations that satisfy formally encoded properties. Two classical formal methods from theoretical computer science were selected: *model checking* and *Hoare logic*. Model checking verifies that a model satisfies a biological property formalised using a temporal logic: CTL. This formal method has already proved its worth in modelling biological regulatory networks of the cell cycle [23, 76].

This method selects, from all possible models, those that satisfy dynamic biological properties characterised by CTL formulae. *Genetically modified Hoare logic* more specifically formalises a particular time series, observed experimentally or deduced indirectly from a review of the literature. Extending the set of inference rules of classical Hoare logic makes it possible to construct a constraint satisfaction problem over the parameters belonging to the family $\mathcal{K}$. Solving this problem reduces the set of possible parameterisations to only those that satisfy the existence of the specified trace. In particular, this logic has made it possible to identify a set of models that satisfy an experimentally observed cell cycle trace [71].

The TotemBioNet tool automatically identifies the values of the parameters $K_{v,\omega}$ by efficiently combining these two formal methods, which have been adapted to René Thomas’s formalism. This tool selects the set of *biologically consistent* models, that is, those that satisfy the conjunction of the formalised dynamic properties. When new biological properties must be verified, all the other properties tested thus far must also be verified. Ultimately, the formal verification of biological properties over an exhaustive set of models aims to refute a GRBM, which occurs when the conjunction of the dynamic properties tested is unsatisfiable. This recalls Karl Popper’s criterion of scientific status: “*A theory [here, a model] that cannot be refuted by any conceivable event*”

[here, a logically formalised biological property] has no scientific character ” [1].

The choice of these two formal methods, among others such as abstract interpretation methods, to name but one, addresses the need to formalise dynamic properties in biology. CTL allows properties to be defined over the states of a transition system and has proved effective in representing and verifying asymptotic behaviours, *i.e.* the presence of a cyclic attractor or of one or more steady states, characteristic of the behaviour of living systems, *i.e.* homeostasis [76, 77, 71] or cell differentiation processes [10, 13, 21]. Genetically modified Hoare logic, for its part, formalises particular paths belonging to cyclic attractors whose presence has been verified using a model checking algorithm.

4.2 Model checking of CTL formulae

According to René Thomas’s semantics, several transitions may occur from a state of an asynchronous state transition graph (abbreviated GTA and formalised in Definition 9 on page 35). By definition, several paths are possible from a given state. CTL (*Computational Tree Logic*) is a temporal logic that allows reasoning about the future of a given initial state, taking into account the nondeterminism admitted in René Thomas’s formalism.

The set of states of a GTA is finite, but the future of an initial state is represented by a tree of infinite executions that enables the verification of asymptotic behaviours. CTL introduces syntactic rules that formalise these behaviours. This is the subject of Subsection 4.2.1. The semantics of CTL are given by the satisfaction relation $\eta \models \varphi$, where η is a state of a GTA and φ is a well-formed CTL property. The central idea of model checking consists in identifying the states that satisfy a CTL formula φ . If all the states of a GTA satisfy φ , then a parameterisation σ of a GRBM that satisfies the formalised behaviour has been identified.

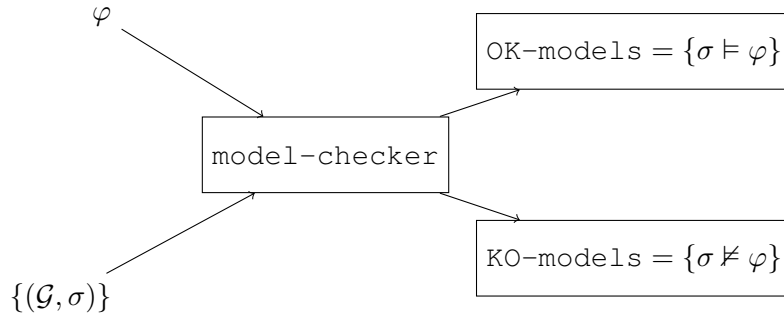


Figure 4.1: **Checking a Thomas model.** Selection (OK-models) and refutation (KO-models) of models (G, σ) using a model checker and a CTL formula φ .

The subset of parameterisations σ leading to models that do not satisfy φ is characterised by $\sigma \not\models \varphi$. If the set $\{\sigma \models \varphi\}$ is empty and it is concluded that φ correctly expresses the observed biological behaviour, then the GRBM is refuted. The following definitions of the syntax and semantics of CTL, as well as the principle of model checking, are largely inspired by [80].

4.2.1 CTL syntax

Definition 11: CTL formula

Given a GRBM $\mathcal{G} = (V, M, E)$, a well-formed CTL formula is defined inductively as follows:

- an atomic formula has the form $\top$, $\perp$ or $v \geq n$, where $v \in V$ and $n \in \llbracket 0, b_v \rrbracket$,
- if φ and ψ are formulae, then $\neg\varphi$, $\varphi \wedge \psi$, $\varphi \vee \psi$, $\varphi \Rightarrow \psi$ are formulae, and $\text{AX}(\varphi)$, $\text{EX}(\varphi)$, $\text{AF}(\varphi)$, $\text{EF}(\varphi)$, $\text{AG}(\varphi)$, $\text{EG}(\varphi)$, $\text{A}[\varphi \text{ U } \psi]$ and $\text{E}[\varphi \text{ U } \psi]$ are also formulae.

CTL temporal operators (G, F, X, U) are always preceded by a path quantifier (A, E). A path quantifier immediately followed by a temporal operator is called a temporal modality. Without temporal modalities, the CTL language reduces to the language of multiplexes (Definition 3 on page 25).

4.2.2 CTL semantics

According to René Thomas’s semantics, several paths can unfold from a given state denoted by η_0 . If we restrict ourselves to the first possible successors of η_0 , there are as many paths as variables that can be updated from η_0 . In Table 4.1, consider a single path p unfolding from η_0 . A CTL temporal operator specifies which states along the path p must satisfy the formula.

G	Globally	φ is satisfied in every state along the path p .
F	Future	φ is satisfied at an unspecified point in the future, that is, in any state directly or indirectly succeeding the initial state, or in the initial state itself (the future includes the present).
X	NeXt	φ is satisfied in the state directly succeeding the initial state.
U	Until	φ is satisfied from the initial state until ψ is satisfied. The future includes the present: ψ may already be satisfied in the initial state, so φ may never be satisfied.

Table 4.1: Semantics of the second letter of CTL temporal modalities.

Assume that φ and ψ are two well-formed CTL formulae. Intuitively, the formula $\text{AF}(\varphi)$ thus means that, from the current state, every path will lead to a state in which φ is true. Similarly, $\text{E}(\varphi \text{ U } \psi)$ means that there exists a path that will reach a state in which ψ is true, and before reaching this state, every state will satisfy φ . Readers unfamiliar with CTL may refer to [80].

A well-formed CTL formula is interpreted using a Kripke structure. This is a generic representation of a transition system, *e.g.* a GTA, from each state of which a finite set of infinite paths is unfolded.

Definition 12: Kripke structure

Let AP be a nonempty set of atomic propositions. A Kripke structure is a triple of the form $K = (\zeta, R, L)$, where:

ζ is a finite set of states,

$R \subseteq \zeta \times \zeta$ is a total transition relation such that
 $\forall \eta \in \zeta, \exists \eta' \in \zeta \mid (\eta, \eta') \in R$,

$L : \zeta \rightarrow 2^{AP}$ is a labelling function that assigns to each state the subset of atomic formulae that this state satisfies.

A Kripke structure is equipped with a labelling function L that assigns to each state the subset of atomic formulae that it satisfies, Definition 12. In the context of biological regulatory networks, we consider a set of atomic formulae that specify the state of each variable. The chosen set is $AP = \{v \geq 0 \mid v \in V \text{ and } n \in \llbracket 0, b_v \rrbracket\}$. The Kripke structure used in what follows is built on the GTA, to which we add the labelling function L defined in Definition 12.

Definition 13: Labelling function

Given a GRBM $\mathcal{G} = (V, E, M)$, the subset of atomic propositions satisfied by a state η of $\mathcal{G}$ is given by the labelling function L defined by:

$$L : \eta \rightarrow \{v \geq n \mid v \in V \text{ and } n \in \llbracket 0, \eta(v) \rrbracket \text{ and } \eta(v) \geq n\}$$

The set of atomic formulae is finite because the set of states of a variable is finite. Once the set of atomic formulae and a CTL formula φ have been defined, we introduce the satisfaction relation $\models$ relating η_0 and φ , Definition 14.

Definition 14: Semantics of a CTL formula using a Kripke structure

Let a model be represented by a Kripke structure $K = (\zeta, R, L)$, and let η_0 be an initial state. The semantics of CTL are defined inductively by:

- $\eta_0 \models \varphi$ if $\varphi \in L(\eta_0)$,
- $\eta_0 \models \neg\varphi$ if $\eta_0 \not\models \varphi$,
- $\eta_0 \models (\varphi \wedge \psi)$ if $\eta_0 \models \varphi$ and $\eta_0 \models \psi$,
- $\eta_0 \models \text{AX}(\varphi)$ if, for every successor η_1 of η_0 , $\eta_1 \models \varphi$,
- $\eta_0 \models \text{EX}(\varphi)$ if there exists a successor η_1 of η_0 such that $\eta_1 \models \varphi$,
- $\eta_0 \models \text{AF}(\varphi)$ if, for every path $\eta_0; \eta_1; \dots; \eta_i; \dots$, there exists a j such that $\eta_j \models \varphi$,
- $\eta_0 \models \text{EF}(\varphi)$ if there exists $\eta_0; \eta_1; \dots; \eta_i; \dots$, and there exists a j such that $\eta_j \models \varphi$,
- $\eta_0 \models \text{AG}(\varphi)$ if, for every path $\eta_0; \eta_1; \dots; \eta_i; \dots$, for every $j \geq 0$ such that $\eta_j \models \varphi$,
- $\eta_0 \models \text{EG}(\varphi)$ if there exists a path $\eta_0; \eta_1; \dots; \eta_i; \dots$, and there exists $j \in \mathbb{N}$ such that $\eta_j \models \varphi$,
- $\eta_0 \models \text{A}[\varphi \cup \psi]$ if, for every path $\eta_0; \eta_1; \dots; \eta_i; \dots$, there exists a j such that $\eta_j \models \psi$ and, for every $i < j$, $\eta_i \models \varphi$
- $\eta_0 \models \text{E}[\varphi \cup \psi]$ if there exists a path $\eta_0; \eta_1; \dots; \eta_i; \dots$, and there exists a j such that $\eta_j \models \psi$ and, for every $i < j$, $\eta_i \models \varphi$

All that remains is to introduce the satisfaction relation relating a model $\mathcal{R} = (\mathcal{G}, \sigma)$ and a CTL formula φ .

Definition 15: Satisfaction relation for a CTL formula and a model

Given an RRBm $\mathcal{R} = (\mathcal{G}, \sigma)$ represented by a Kripke structure $K = (\zeta, R, L)$ and a CTL formula φ , a parameterisation σ satisfies φ if every state η of $\mathcal{G}$ satisfies it:

$$\frac{\forall \eta \in \mathcal{G} : \eta \models \varphi}{\sigma \models \varphi}$$

We then write $\sigma \models \varphi$. By an abuse of notation, we write $\mathcal{R} \models \varphi$ if $\sigma \models \varphi$.

The regulatory mechanisms encoded in a graph $\mathcal{G}$ are sufficient to deduce a dynamic behaviour characterised by φ if and only if φ is satisfiable. We then say that $\mathcal{G}$ is biologically consistent with respect to φ and write $\mathcal{G} \models \varphi$. We thus have the following equivalence:

$$\mathcal{G} \models \varphi \equiv \exists(\mathcal{G}, \sigma) \mid \sigma \models \varphi$$

The properties conventionally verified are safety properties (specifying behaviours that must never occur) and liveness properties (specifying behaviours that will always eventually occur). In biol-

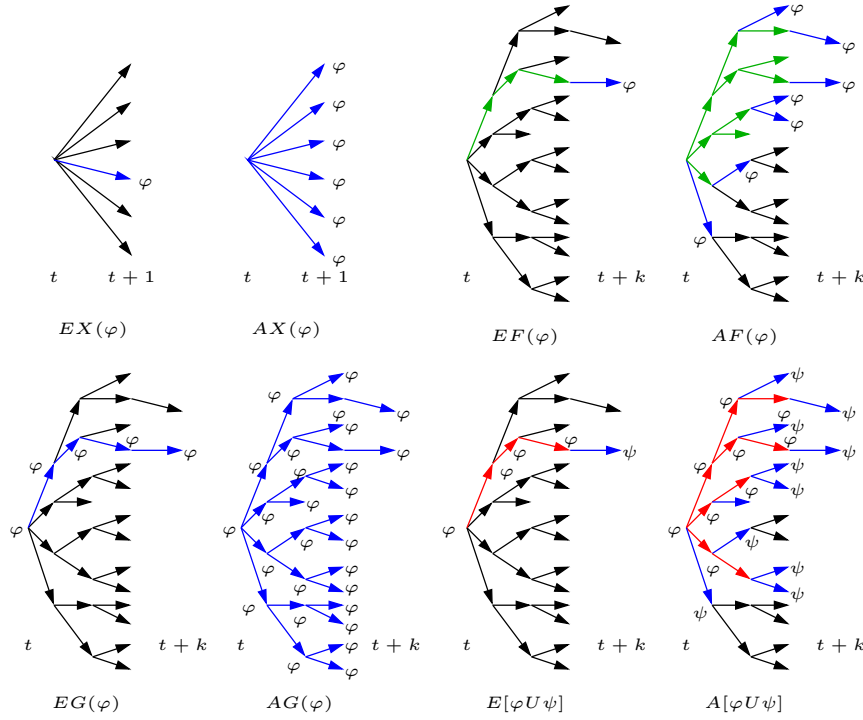


Figure 4.2: **Illustration of the CTL semantics introduced in Definition 14.** A computation tree unfolds from an initial state of a Kripke structure $K = (\zeta, R, L)$ (Definition 12). Each unfolded path is an infinite sequence of states $\eta_0; \eta_1; \dots$ such that $\eta_0 \in \zeta$ and $(\eta_i, \eta_{i+1}) \in R, \forall i \geq 0$. For this example, only one initial state is considered.

ogy, the challenge is to define a collection of properties assumed to be sufficient to qualify a GRBM as *biologically consistent*, given the variables of $\mathcal{G}$ chosen *a priori* and properties of the system under study. These properties must always be linked to experimental observations when verifying the biological consistency of $\mathcal{G}$, but hypotheses may subsequently be tested.

4.2.3 Principle of model checking

Model checking is an algorithm that determines whether a CTL formula φ is satisfiable by computing the set of states of a Kripke structure K that satisfy φ . Without loss of generality, CTL syntax is reduced to the operators $\neg, \wedge, EX, EU, AF$, since the following equivalences hold:

$$\begin{aligned}
 AG(\varphi) &\equiv \neg EF(\neg\varphi) \\
 AX(\varphi) &\equiv \neg EX(\neg\varphi) \\
 EG(\varphi) &\equiv \neg AF(\neg\varphi) \\
 EF(\varphi) &\equiv E[\top U \varphi] \\
 A[\varphi U \psi] &\equiv \neg(E[\neg\psi U (\neg\varphi \wedge \neg\psi)] \vee EG(\neg\psi))
 \end{aligned}$$

Initially, every state of K is labelled with the subset of atomic propositions of φ that it satisfies. A state is labelled with a formula of the form $\neg\varphi$ if it is not labelled with φ . A formula of the form $\varphi \wedge \psi$ is assigned to every state labelled with both φ and ψ . A formula of the form $EX(\varphi)$ is

assigned to every predecessor of every state labelled with φ .

To label states with a formula of the form $\text{AF}(\varphi)$, we proceed as follows:

1. every state already labelled with φ is labelled with $\text{AF}(\varphi)$,
2. every state all of whose successors are already labelled with $\text{AF}(\varphi)$ is labelled accordingly,
3. step 2 is repeated until the state labelling converges.

To label states with a formula of the form $\text{E}[\varphi \cup \psi]$:

1. every state already labelled with ψ is labelled with $\text{E}[\varphi \cup \psi]$,
2. every state already labelled with ϕ and having at least one successor labelled with $\text{E}[\varphi \cup \psi]$ is also labelled with $\text{E}[\varphi \cup \psi]$,
3. step 2 is repeated until the state labelling converges.

The final step returns the set of states satisfying φ . The NuSMV model checker implements an efficient symbolic model checking algorithm for CTL, designed by [16], which relies on a symbolic representation (rather than an explicit representation as in the naive algorithm presented above) of the Kripke structure, represented more compactly using an OBDD (*Ordered Binary Decision Diagrams*).

4.2.4 CTL and fair paths

The Nu-SMV model checker considers CTL semantics that admit *unfair* paths. A path p is said to be unfair if it passes infinitely often through a state with several successors but passes only finitely often through some of its successors. Paths are fair if they explore all possibilities. Consequently, liveness formulae of the form $\text{AF}(\varphi)$, where φ specifies an attractor, are systematically evaluated as false by Nu-SMV whenever there is a strongly connected component distinct from the attractor but from which the attractor can be reached.

For example, consider an asynchronous state transition graph that generates an oscillation of the variables x and y around the point $(x = 1, y = 1)$ and a steady state characterised by $x = 2 \wedge y = 1$. It is immediately clear from Figure 4.3 that this steady state is reachable regardless of the initial state considered. Suppose we wish to verify by model checking that the system always eventually stabilises with x saturated, using the formula $\text{AF}[\text{AG}(x = 2)]$. Nu-SMV concludes that this formula is false because the cycle $\eta_4 \rightarrow \eta_3 \rightarrow \eta_2 \rightarrow \eta_1 \rightarrow \eta_4$ can be followed infinitely often, even though two paths are possible from state η_4 . The path that passes infinitely often through the states $\eta_4 \rightarrow \eta_3 \rightarrow \eta_2 \rightarrow \eta_1 \rightarrow \eta_4$ is therefore unfair. To be able to verify that $\eta_4 \rightarrow \eta_5 \rightarrow \eta_6$ is indeed eventually followed, unfair paths must be ignored.

Two new path quantifiers are introduced in CTL': A' means *for every fair path*, and E' means *there exists a fair path*. Intuitively, considering only fair paths amounts to focusing on terminal strongly connected components. In other words, CTL' is well suited to expressing the asymptotic properties of systems. Adrien Richard showed in an internal report [39] that, for every CTL' formula denoted by ψ , there exists a semantic equivalence with a CTL formula φ , that is:

$$\forall \psi \in \text{CTL}', \\ \llbracket \exists \varphi \in \text{CTL} \mid (\eta \models_{\text{CTL}} \varphi) \equiv (\eta \models_{\text{CTL}'} \psi) \rrbracket$$

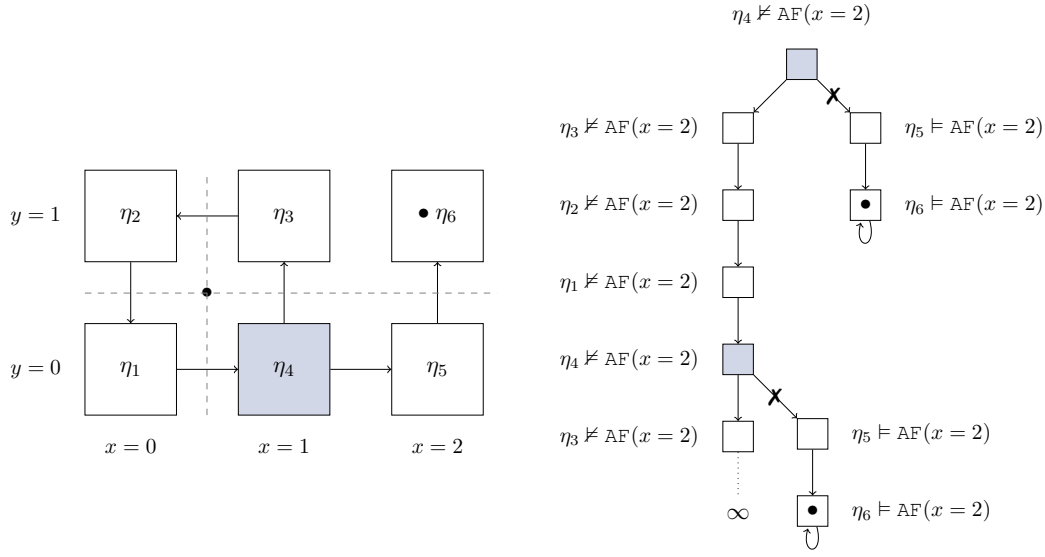


Figure 4.3: **Example of a GTA for which the formula $\mathbf{AF}(x = 2)$ is satisfied only in CTL'.** If η_3 and η_5 can be visited infinitely often, then the paths leading to them are fair. In CTL', all the states of the GTA are labelled with $\mathbf{AF}(x = 2)$, whereas in CTL the existence of an unfair cyclic path none of whose states satisfies $x = 2$ implies $\sigma \not\models \mathbf{AF}(x = 2)$.

Nu-SMV remains the model checker of choice, since there exists a translation from CTL' to CTL that preserves the semantics of formulae. Only CTL' can verify that an attractor will always eventually be reached ($\mathbf{AF}[\mathbf{AG}(\varphi)]$) along fair paths.

4.2.5 Some examples of properties formalised in (fair) CTL

We continue with some examples of dynamic properties conventionally studied in the modelling of biological regulatory networks. Given two CTL formulae φ and ψ , an asynchronous state transition graph $\mathcal{T}$ and three particular states of $\mathcal{T}$ denoted by η_1 , η_2 and η_3 , characterised by φ_1 , φ_2 , and φ_3 :

1. $\varphi_1 \Rightarrow \mathbf{EF}(\varphi_2)$ is a reachability property that verifies the existence of at least one path connecting η_1 and η_2 .
2. $\varphi_1 \Rightarrow \mathbf{EX}(\varphi_2 \wedge \mathbf{EX}(\varphi_3))$ specifies the existence of a particular path of $\mathcal{T}$: $\eta_1 \rightarrow \eta_2 \rightarrow \eta_3$.
3. $[\neg \mathbf{EF}(\varphi)] \equiv [\mathbf{AG}(\neg \varphi)]$ is a safety property that verifies that φ is not satisfied by any state of $\mathcal{T}$. For example, if φ expresses the activation of a gene, then this *safety* property verifies that the gene will never be activated.
4. $\mathbf{AF}[\mathbf{AG}(\varphi_2)]$ is a *liveness* property concerning the reachability of a steady state characterised by φ_2 . A liveness property represents a behaviour that will always eventually occur, regardless of the initial state.
5. $\mathbf{AF}[\mathbf{AG}[\mathbf{AF}(v = 1) \wedge \mathbf{AF}(v = 2)]]$ is a *liveness* property concerning the reachability of a stable oscillation of v ; more specifically, v is multivalued here and can take the values 1 and 2 infinitely often.

6. $\text{AG}(v = n) \Rightarrow \varphi$ is understood in two stages. The premise specifies a fixed environment. An environment is a variable stabilised at a value n such that $0 \leq n \leq b_v$ throughout the state space of $\mathcal{T}$. The property φ is a CTL formula satisfied in this environment.
7. $\text{A}[c \cup \varphi_1]$: the constraint c is respected before reaching state η_1 characterised by φ_1 , for every path of $\mathcal{T}$ [19].
8. $\varphi_1 \rightarrow \text{A}[\neg\varphi \cup \psi]$ is the same property as the preceding one, but formulated as a safety property. From state η_1 , the property φ will never be true before reaching a state that satisfies ψ .

The asymptotic properties expressed by formulae 4 and 5 must be interpreted in CTL'. Identifying an attractor requires unfair paths to be ignored, *cf.* Figure 4.3. For example, the subformula φ_2 in formula 4 represents, when satisfied, a set of states belonging to an attractor. In other words, this formula detects an attractor.

Checkpoints can be formalised by safety properties provided that a phase is correctly formalised. For example, if in formula 8 we take φ_1 and φ_2 to be respectively the initial and final states of a phase π_i , and φ to be the union of the domains of all the other cell cycle phases except π_i , then the safety property verifies that, once a phase has started, no other phase is visited while the current phase is under way. The interpretation of the safety property that could define a checkpoint therefore depends on the prior definition of the domains of the different cell cycle phases.

4.3 Genetically modified Hoare logic

Genetically modified Hoare logic also verifies the existence of certain paths within an asynchronous state transition graph. This logic verifies the consistency of three elements: a precondition, a sequence of instructions and a postcondition. It is inspired by the Hoare logic introduced by Tony Hoare to reason about the correctness of an imperative program [2]. A Hoare triple of the form $H \equiv \{P\} p \{Q\}$ formalises an imperative program: P and Q characterise the input and output conditions of the program respectively, and p is a sequence of instructions (increment, decrement, *while*, *if*).

Edsger Dijkstra defined an algorithm for computing the weakest precondition, which provides a unique strategy for proving the correctness of H . This algorithm takes a precondition Q and an imperative program p as input and returns the weakest precondition P_0 . The program p can execute from P and reach the postcondition Q if and only if $P \models P_0$.

This logic has been adapted to René Thomas's modelling framework for biological regulatory networks with multiplexes to determine whether an experimentally observed biological trace is correct. This section aims to explain this notion of *correctness* as transposed to RRBm modelling. Essentially, *genetically modified* Hoare logic constrains the parameterisations of a biological regulatory graph with multiplexes (GRBM). The increment and decrement instructions for a variable, *while* and *if*, allow gene expression data or experimental conditions such as *knock-out* or *knock-in* to be formalised with subtlety; for further details, we refer the reader to [86]. We will focus exclusively on increment and decrement instructions, since they suffice for the formalisation of cell cycle phases presented in this thesis.

4.3.1 Syntax

The precondition P and postcondition Q of a Hoare triple $H \equiv \{P\} p \{Q\}$ are defined using a language that extends the language of multiplexes: the assertion language. The parameters $K_{v,\omega}$ are now permitted as terms for expressing constraints on their values.

Definition 16: Assertion language

Given a GRBM $\mathcal{G} = (V, M, E)$ and the family of parameters $\mathcal{K}$ associated with $\mathcal{G}$, the assertion language $\mathcal{L}_A$ is defined inductively by:

atoms have the form $(t \geq t')$, where t and t' are well-formed terms built from V , $\mathcal{K}$ and $\mathbb{N}$ and the binary operations $+$ and $-$,

if $\varphi \in \mathcal{L}_A$ and $\psi \in \mathcal{L}_A$, then $\neg\varphi \in \mathcal{L}_A$, $(\varphi \wedge \psi) \in \mathcal{L}_A$ and $(\varphi \vee \psi) \in \mathcal{L}_A$.

For example, consider a GRBM and its associated parameter family $\mathcal{K}$ in Figure 4.4. The expressions $(K_{x,\{m_1\}} \geq K_{x,\emptyset}) \wedge (K_{x,\{m_3\}} \geq K_{x,\emptyset})$ and $(K_{x,\{m_1\}} \geq 0) \wedge (K_{x,\{m_1\}} \leq b_x)$ are two examples of assertions in $\mathcal{L}_A$.

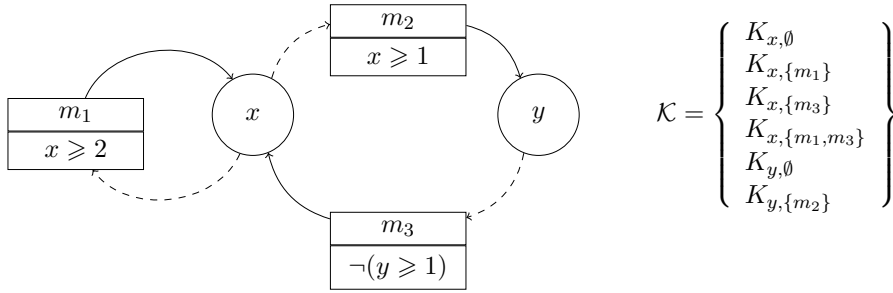


Figure 4.4: An example of a GRBM with its associated parameter family $\mathcal{K}$, illustrating the enrichment of the assertion language relative to the language of multiplexes.

In the context of this thesis, a path is restricted to a sequence of increments ($v+$) or decrements ($v-$) of a variable v . The path language represents a sequence of discrete events extracted from the literature, directly from laboratory experiments [77, 71], or after the analysis of a time series (Figure 4.5).

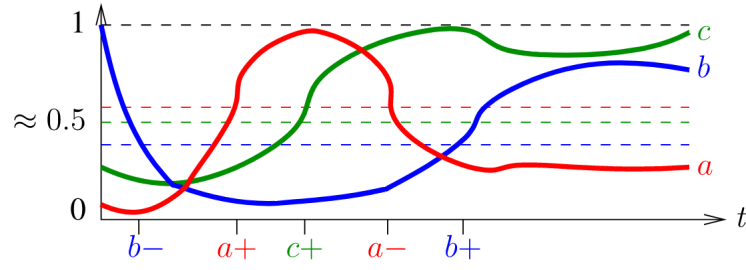


Figure 4.5: **An example of an experimentally observed biological trace.** A trace is deduced from changes in the activity of biological variables over time, where each variable is equipped with at least one activation threshold. The vertical axis represents changes in the normalised activity of a variable, and the value 0.5 chosen for this example corresponds to half the maximum activity of a variable. Figure taken from [86].

Definition 17: Path language

Given a GRBM $\mathcal{G} = (V, M, E)$, the path language $\mathcal{L}_C$ is defined inductively by:

$\forall v \in V$, the expressions $v+$ and $v-$ belong to $\mathcal{L}_C$ and are the atoms of the path language,

if $p_1 \in \mathcal{L}_C$ and $p_2 \in \mathcal{L}_C$, then $(p_1; p_2) \in \mathcal{L}_C$,

in $\mathcal{L}_C$, “;” is associative.

The elements of $\mathcal{L}_C$ are called the paths of $\mathcal{G}$. The atoms of $\mathcal{L}_C$ are called *events*.

Remark 1 – We deliberately restrict the language $\mathcal{L}_c$ initially defined in [86] to the minimum sufficient for this thesis. For example, $\mathcal{L}_c$ includes neither path quantifiers, nor the conditional branch `if`, nor the `while` loop.

Definition 18: Hoare triple

A Hoare triple associated with a GRBM $\mathcal{G}$ is an expression of the form $H \equiv \{P\} p \{Q\}$ such that $P \in \mathcal{L}_A$, $Q \in \mathcal{L}_A$ and $p \in \mathcal{L}_C$.

P and Q are respectively called the precondition and postcondition of H .

4.3.2 Semantics

The semantics of a Hoare triple are given by an asynchronous state transition graph. An experimentally observed trace that is well defined syntactically does not necessarily have a semantic equivalent. A triple of the form $\{P\} v + \{Q\}$ or $\{P\} v - \{Q\}$ has a semantic equivalent if and only if it refers to a transition belonging to an asynchronous state transition graph.

Definition 19: Semantics of a path

Given an RRBm $\mathcal{R} = (G, \sigma)$, the family of binary relations $\overset{p}{\rightsquigarrow}$ over the set of states ζ , where p ranges over $\mathcal{L}_C$, is defined inductively by:

- if p is the empty path, then trivially $\eta \overset{p}{\rightsquigarrow} \eta$ for every $\eta \in \zeta$,
- if $p \equiv v+$: if $\eta \rightarrow \eta'$ is a transition of $\mathcal{R}$ with $\eta' = \eta[v \leftarrow (\eta(v) + 1)]$, then $\eta \overset{p}{\rightsquigarrow} \eta'$,
- if $p \equiv v-$: if $\eta \rightarrow \eta'$ is a transition of $\mathcal{R}$ with $\eta' = \eta[v \leftarrow (\eta(v) - 1)]$, then $\eta \overset{p}{\rightsquigarrow} \eta'$,
and
- if $p \equiv (p_1; p_2)$: if $\eta \overset{p_1}{\rightsquigarrow} \eta'$ and $\eta' \overset{p_2}{\rightsquigarrow} \eta''$, then $\eta \overset{p}{\rightsquigarrow} \eta''$.

The semantics of a path defined by the relation $\eta \overset{p}{\rightsquigarrow} \eta'$ can be read as follows: in a transition system defined by an asynchronous state transition graph, state η' is reachable from state η along the path specified by p .

Definition 20: Semantics of a Hoare triple

Given an RRBm $\mathcal{R} = (\mathcal{G}, \sigma)$ and ζ , the set of states of its asynchronous state transition graph, a Hoare triple of the form $\{P\} p \{Q\}$ is satisfied if and only if:

$$\forall \eta \in \zeta, (\eta \models P \implies \exists \eta' \mid (\eta \overset{p}{\rightsquigarrow} \eta' \text{ and } \eta' \models Q))$$

4.3.3 Inference rules

To verify that a Hoare triple defined syntactically by $\{P\} p \{Q\}$ also has a semantic definition, each elementary instruction $v+$ or $v-$ is associated with an inference rule defining a constraint which, when satisfied, ensures that the transition defined syntactically by $v+$ or $v-$ belongs to the GTA considered (Definition 21).

Definition 21: Increment and decrement axioms

For every variable v , the increment axioms are defined by:

$$\frac{\{\Phi_v^+ \wedge Q[v \leftarrow v + 1]\}}{v + \{Q\}}$$

where $\Phi_v^+ = \bigwedge_{\omega \in E^{-1}(v)} (\Phi_v^\omega \Rightarrow K_{v,\omega} > v)$ and $\Phi_v^\omega \equiv (\bigwedge_{m \in \omega} \varphi_m) \wedge (\bigwedge_{m \in (E^{-1}(v) \setminus \omega)} \neg \varphi_m)$

and the decrement axioms are defined by:

$$\frac{\{\Phi_v^- \wedge Q[v \leftarrow v - 1]\}}{v - \{Q\}}$$

where $\Phi_v^- = \bigwedge_{\omega \in E^{-1}(v)} (\Phi_v^\omega \Rightarrow K_{v,\omega} < v)$

Φ_v^ω is the formula expressing that the set ω is the set of resources of v . This is necessary because GTA transitions are controlled by the parameters $K_{v,\omega}$, in which resources are a crucial concept. We can observe that there is a unique set ω in a state η such that $\eta \models \Phi_v^\omega$; cf. also the definition of an applicable parameter in Chapter 2, page 29. Similarly, Φ_v^+ (resp. Φ_v^-) formalises the conditions under which v can increase (resp. decrease). Indeed, among all the formulae in the conjunction, only one has a true premise (the one involving the effective parameter). Thus, the conjunction will be true if and only if the effective parameter makes the value of variable v increase (resp. decrease), cf. Figure 4.6.

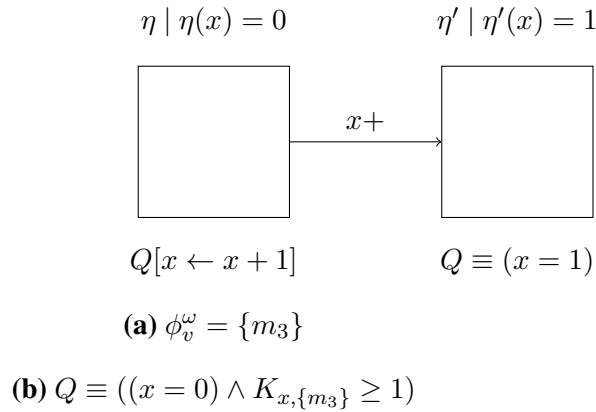


Figure 4.6: **Constraint that verifies the existence of the transition $\eta \rightarrow \eta'$ in a GTA denoted by $\mathcal{T}$ associated with the GRBM in Figure 4.4.** We have $\eta \models \phi_v^+$ if $\eta \rightarrow \eta' \in \mathcal{T}$. In other words, if the value of the parameter for x applicable in η (here $K_{x,\{m_3\}}$) exceeds its current value (here $\eta(x) = 0$), then x can be activated ($x+$).

Other rules are introduced to decompose a sequence of elementary instructions (Definition 22) or to deduce the weakest precondition (Definition 23).

Definition 22: Inference rule for sequential composition

$$\frac{\{P_1\} p_1 \{P_2\} \quad \{P_2\} p_2 \{Q\}}{\{P_1\} p_1; p_2 \{Q\}}$$

Definition 23: Inference rule for the empty path

$$\frac{P \Rightarrow Q}{\{P\} \epsilon \{Q\}}$$

Furthermore, René Thomas's formalism constrains the bounds of variables and parameters $K_{v,\omega}$ by definition. The following axiom follows:

Definition 24: Bounds axioms

For every variable v and every subset ω of its predecessors:

$$(v \geq 0) \wedge (v \leq b_v) \wedge (K_{v,\omega} \geq 0) \wedge (K_{v,\omega} \leq b_v)$$

4.3.4 Computing the weakest precondition

Edsger Dijkstra proposes a proof strategy that recursively decomposes the path p into elementary instructions, working from the postcondition Q back to the precondition P , until reaching the base case of the empty path, which returns the weakest precondition denoted by wp , [5].

Definition 25: The weakest precondition wp

Given a GRBM $\mathcal{G} = (V, M, E)$, its associated parameter family $\mathcal{K}$ and a Hoare triple $\{P\} p \{Q\}$, the *weakest precondition*, denoted by wp , is a function of p and Q defined inductively by:

- if $p \equiv \epsilon$, then $wp(\epsilon, Q) = Q$,
- if $p \equiv v+$, then $wp(v+, Q) = Q[v \leftarrow v + 1] \wedge \phi_v^+$,
- if $p \equiv v-$, then $wp(v-, Q) = Q[v \leftarrow v - 1] \wedge \phi_v^-$, and
- if $p \equiv (p_1; e)$, then $wp((p_1; e), Q) = wp(p_1, wp(e, Q))$,

where v is a variable belonging to V and e is an event of the form $v+$ or $v-$.

Remark 2 – The second and third conditions define a constraint on the domain of the parameters $K_{v,\omega}$, cf. the increment and decrement axioms, respectively, in Definition 21. The final condition defines the propagation of the constraint on the value of $K_{v,\omega}$ from the postcondition Q to the precondition P , based on the sequential composition axiom, cf. Definition 22.

Remark 3 – The bounds axiom is not mentioned in Definition 25, since it is used to simplify the weakest precondition on the fly. Indeed, knowledge of the domain bounds of the parameters $K_{v,\omega}$ and variables v must be taken into account at each rewriting step of the weakest precondition.

Dijkstra’s strategy is described as *backward* and offers two advantages. On the one hand, it entails a unique correctness proof for H and, on the other, it allows a simple rewriting process to be applied to the postcondition. A Hoare triple is satisfied if there exists a parameterisation σ of a GRBM $\mathcal{G}$ that satisfies wp . Once solved, this constraint satisfaction problem reduces the set of possible parameterisations. It is decidable, since the completeness of genetically modified Hoare logic was proved in the foundational paper [86].

4.4 Managing model exhaustiveness with TotemBioNet

This section was the subject of a *tool paper* at the international CMSB (*Computational Methods in Systems Biology*) conference in 2020 [97].

TotemBioNet is a software platform that integrates two tools, SMBioNet and HoareFol, providing efficient support for the parameterisation of a GRBM. SMBioNet is a tool developed by Adrien Richard that takes a GRBM $\mathcal{G}$ and a set of CTL formulae Φ as input, enumerates all parameterisations of $\mathcal{G}$, and selects those satisfying the conjunction of the elements of Φ . HoareFol is a tool developed by Maxime Folschette [90] that implements in OCaml the computation of the weakest precondition deduced from a Hoare triple H . It generates an Answer Set Programming (ASP) program that is sent to the Clingo solver [75] to compute the parameterisations satisfying H . TotemBioNet distinguishes itself by combining these two approaches so that modellers quickly converge on the exhaustive set of parameterisations satisfying all formalised biological knowledge.

4.4.1 Inputs and outputs

The inputs to TotemBioNet are (shown in grey in Figure 4.7):

- a GRBM $\mathcal{G}$,
- any *a priori* knowledge about the parameter values of $\mathcal{G}$,
- and properties of the system dynamics expressed in CTL, in CTL’, or by a Hoare triple.

TotemBioNet returns *all* parameterisations σ of $\mathcal{G}$ that satisfy the input properties. Let us call the set of these parameterisations Ψ .

The aim of a modelling study is often to answer a question posed by a biologist. But how can we methodically check that this question is well posed? To paraphrase a quotation from Jean-Marc Jancovici: “life comes down to ill-posed problems that admit several solutions”.

The question primarily addressed in René Thomas’s modelling framework is *whether the set of regulations formalised by a GRBM is consistent with certain dynamic properties*. The GRBM and the dynamic properties encoded in CTL or using Hoare logic may include hypotheses to be tested. If the set of biologically consistent parameterisations Ψ is empty, then the hypotheses are false, or

other knowledge with stronger support in the literature is called into question. If Ψ is too large, then the tested hypotheses are of little interest, since they provide little additional knowledge. Conversely, if the subset of parameterisations satisfying the hypotheses is relatively smaller than the original set, then:

1. the hypotheses formalised by the biologist deserve to be tested experimentally, since they have demonstrated their power of refutation in a formal framework,
2. learning parameter values informs the biologist about the logical activation rules for a set of variables involved in the formally tested hypotheses.

Ultimately, TotemBioNet helps refine questions to be tested experimentally, so as to converge more quickly towards revealing new knowledge about the system of interest to a team of biologists, and hence towards biologically relevant questions.

4.4.2 Workflow

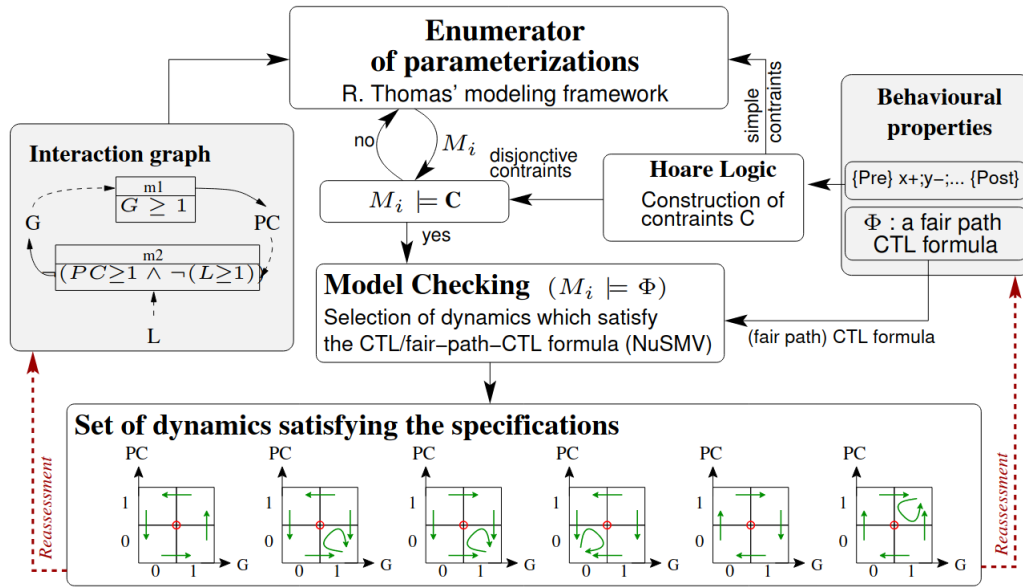


Figure 4.7: **Operation of TotemBioNet.** Inputs are shown in grey. The outputs of TotemBioNet are all the dynamics satisfying the input specifications. They lead to the formalisation of the inputs—the GRBM or the behavioural specifications—being called into question (red dashed arrows).

If a Hoare triple is defined, TotemBioNet calls HoareFol to compute the weakest precondition. Genetically modified Hoare logic introduces existential quantifiers when experimental conditions are not precise enough to determine which variable crosses its threshold first. For example, $\exists[(v_1+; v_2-), (v_2-; v_1+)]$ means that v_1 is activated at a given threshold and v_2 is deactivated at a given threshold, but the order of these events is unknown. Furthermore, universal quantifiers provide an abstraction of a collection of experimental observations that must all be satisfied (e.g. $\forall[(v_1+; v_2-), (v_2-; v_1+)]$). For existential quantifiers, the computed weakest

Contributions of the Thesis

The following three chapters are the subject of a paper in preparation (to be submitted in 2022):

“A purely discrete formalisation and formal verification of intrinsic cell cycle checkpoints: a proof of concept. ”.

-

Déborah Boyenval, Gilles Bernot, Jean-Paul Comet and Franck Delaunay.

CHAPTER 5

Our set of models for studying the phases of the cell cycle

5.1	Abstraction and modelling question	81
5.2	The variables and their discrete thresholds	81
5.3	Our initial conditions	84
5.4	Formalisation of the regulations involved in cell cycle progression	85
5.4.1	Our influence graph designed to define a canonical cell cycle	86
5.4.2	A canonical sequence of regulatory events during the phases of the cell cycle	91
5.4.2.1	The events required in G1	91
5.4.2.2	The events required in S	92
5.4.2.3	The events required in G2	93
5.4.2.4	The events required in M	93
5.4.2.5	The resulting canonical sequence	94
5.4.3	Definition of multiplexes	95
5.4.3.1	Multiplexes and mechanistic abstraction	95
5.4.3.2	Multiplexes and phenomenological abstraction	96
5.5	Initial identification of parameter values	99
5.5.1	Verification of the stability of the quiescent phase	100
5.5.2	Formalisation of a canonical cell cycle as a cyclic attractor	101
5.5.3	Asking the right questions about checkpoints using TotemBioNet	101
5.5.4	Manual identification of certain parameters	103

5.1 Abstraction and modelling question

A preliminary modelling step is necessary to validate a formalisation of intrinsic cell cycle checkpoints as a concept. The proof of concept is carried out on a non-empty set of models that formalise the main regulatory pathways involved in cell cycle control. A GRBM is first constructed to represent these pathways (sections 5.2 to 5.4). Several classical properties of the cell cycle are then defined to identify biologically consistent parameterisations of this GRBM, a qualification formally defined in chapter 4. In section 5.5, we focus on two properties: a particular ordering in the updating of the GRBM variables and control of the cell cycle by growth factors. If the set of models (where a model is a pair consisting of a GRBM and one of its parameterisations) that satisfy these properties is non-empty, then the proof of concept is successful. This set of models is identified by appropriately combining CTL and genetically modified Hoare logic, as explained at the end of the previous chapter.

The key components of cell cycle regulation are the *cyclin/cyclin-dependent kinase* complexes (abbreviated *cyc/cdk*) and their main inhibitors. A modelling question must be defined so that our GRBM is minimal, in the sense that no variable or threshold is defined unnecessarily. *Which events are required during each phase, and which regulations allow these events to occur in a particular order?* If a variable or threshold plays no part in characterising these events, then it is unnecessary.

In 2008, John Tyson and Bela Novak proposed a generic model capable of reproducing the temporal organisation of the cell cycle [41]. Jonathan Behaegel extended this model in 2016 and defined a Hoare triple characteristic of the cell cycle, of the form $\{precondition\} path \{postcondition\}$ [71]. The *path* is a sequence of events in genetically modified Hoare logic (abbreviated LHGM). Further information is provided in chapter 3, page 50. This sequence is divided into subsequences specific to each phase that highlight the events required during a given phase.

In this chapter, we make a substantial update to the model in [71], both in light of the knowledge and perspective gained since 2016 and to specialise it for mammals. Five variables, three multivalued and two Boolean, suffice to represent a sequence of events during the cycle. Once this minimal set of variables and their thresholds has been defined, the exhaustive set of regulatory pathways between these variables is inventoried and formalised as a GRBM. The reader can refer to figure 5.18 at any time for an overview that places the various details presented throughout the chapter in context.

48 parameters govern its dynamics, and our GRBM admits a set of more than 10^{20} models, which currently puts our proofs of concept in chapters 6 and 7 beyond computational reach. In the previous chapter, we demonstrated that TotemBioNet has the ability to reduce the combinatorial explosion in the number of models by exploiting *a priori* knowledge of the overall dynamics of the system under study. Such knowledge is abundant, given the accumulating studies on the cell cycle and its regulation; section 5.5 concludes that only 16 models are of interest for studying checkpoints.

5.2 The variables and their discrete thresholds

The six variables of our GRBM represent a molecule, a group of molecules that act in the same way on their targets, or entire molecular pathways.

Variable	Group of molecules	Biological entity	Domain
<i>sk</i>	“Starting kinase ”	cycE/cdk2, CAK	$\llbracket 0, 2 \rrbracket$
<i>a</i>	-	cycA/cdk1, cycA/cdk2, CAK	$\llbracket 0, 2 \rrbracket$
<i>b</i>	-	cycB/cdk1, CAK	$\llbracket 0, 2 \rrbracket$
<i>en</i>	“Enemies ”	APC-cdh1, P21, P27, PP1/2A	$\llbracket 0, 1 \rrbracket$
<i>ep</i>	“Exit protein ”	APC-cdc20	$\llbracket 0, 1 \rrbracket$
<i>gf</i>	Growth factors	EGF, FGF, PDGF, IGF, TGF β	$\llbracket 0, 1 \rrbracket$

Table 5.1: **The variables of our GRBM and their domains.** All our variables, except *gf*, draw on the notation in [41].

The cell cycle is a succession of cellular events, *e.g.*, at a coarse level: passage through the restriction point, activation of the replication machinery, or the proteins involved in the separation of sister chromatids during mitosis. Decades of experimental work [14] show that these events are controlled by *cyc/cdk*. Cdks are enzymes that phosphorylate substrates decisive for the occurrence of the various events of replication and mitosis. According to [81], cdk2 and cdk1 initiate the processes of S and M phase, respectively. Cdk4 and cdk6 regulate a cell’s entry into the cell cycle from quiescence in response to growth factors in the extracellular environment.

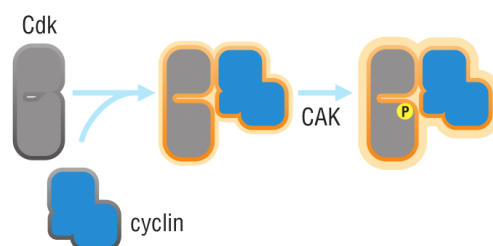


Figure 5.1: **Full activation of *cyc/cdk* complexes requires complex formation with CAK.** The variables *sk*, *a* and *b* therefore include CAK. Figure taken from [33].

Cdk1, 2, 4 and 6 are active only when bound to two activating subunits: *cyclins* and *cyclin-activating kinases*, abbreviated CAK; see figure 5.1. CAK corresponds to the cycH/cdk7 complexes and phosphorylates cdk4, 6, 2 and 1, allowing them to adopt a conformation that improves binding to their substrates. Cyclins are not simply activators of their associated cdk: they help direct it towards specific substrates, either by binding the substrate directly or by bringing it into a subcellular compartment containing the substrate.

The cycD/cdk4 and cycD/cdk6 complexes are involved in the transition between quiescence and G1 phase; they are not explicitly represented in the original model. Making these complexes explicit would imply that the modelling question focuses, at least in part, on the molecular determinants of entry into the cell cycle, which is not our case.

The cycE/cdk2 complex is a determinant of passage through the restriction point and is involved in initiating the replication process (S phase). It is activated in G1 phase and shows peak activity just before the G1/S transition [102]. This complex is aptly named *starting kinase* (variable *sk*) in

[41], and we retain this nomenclature throughout the thesis.

The *cycA/cdk2* and *cycA/cdk1* complexes are activated in S and G2 phase, respectively. They are grouped under the variable name “*a*”. The *cycB/cdk1* complex (“*b*” in table 5.1) is initially activated in late G2, and its activity increases upon entry into M phase.

The Boolean variable *gf* stands apart: we consider it an *environment* variable under the modeller’s complete control, just as an experimentalist controls the conditions of an experiment. Growth factors will make it possible to verify control over entry into the cell cycle: a cell must be sensitive to growth factors if it is to be arrestable in quiescence.

The *cyc/cdk* complexes are kept inactive by a ubiquitin ligase (APC-cdh1), phosphatases (PP1/2A), or proteins that form inhibitory complexes (p21 and p27). These inhibitors are grouped under the term *enemies* in table 5.1. APC-cdc20 is also a ubiquitin ligase that marks certain *cyc/cdk* complexes for degradation, but is defined as a separate variable because of its prominent role in mitotic exit (*exit protein* in table 5.1). *en* and *ep* are Boolean variables.

The variable *sk* has two thresholds: $sk = 0$ means that *sk* acts on no target. $sk = 1$ abstracts the concentration interval of *sk* corresponding to an initial accumulation of the *cycE/cdk2* complex, which becomes capable of indirectly activating its own transcription ($sk \rightarrow sk$) and inhibiting APC-cdh1, p21 and p27 ($sk \rightarrow en$). When $sk \geq 2$, *cycE/cdk2* is fully activated and marks the G1/S transition. At this threshold, *sk* also targets the *cycA/cdk1-2* complex ($sk \rightarrow a$). The variable *a* abstracts both the *cycA/cdk2* and *cycA/cdk1* complexes: $a = 0$ is the trivial case in which these complexes act on no target. $a = 1$ specifically represents the action of the *cycA/cdk2* complex in S phase. At this threshold, *a* activates *sk* and inhibits APC-cdh1, p21 and p27 (*en*). $a = 2$ additionally represents the action of the *cycA/cdk1* complex, which is activated later than the *cycA/cdk2* complex, in G2 phase. It also represses the activity of *en*, but additionally activates the *cycB/cdk1* complex (*b*). Since $a = 2$ represents the presence of both *cycA/cdk2* and *cycA/cdk1* complexes together, this abstraction cannot describe the action of *cycA/cdk1* without simultaneously describing the action of *cycA/cdk2*, whereas $a = 1$ describes only the action of *cycA/cdk2* (figure 5.2). Our abstraction is accompanied by the following modelling assumption:

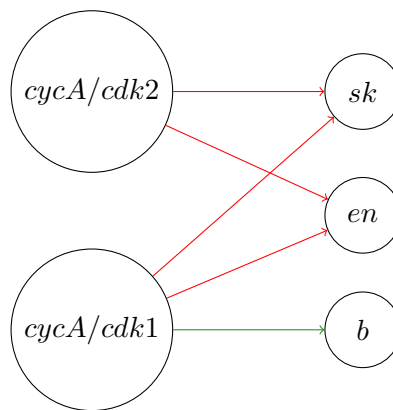


Figure 5.2: **Abstraction of *cycA/cdk2* and *cycA/cdk1* within a single variable.**

the way *a* is defined implies that *cycA/cdk1* activates *sk*, although, to our knowledge, no direct or indirect interaction between *cycA/cdk1* and *cycE/cdk2* has been reported. Whatever the reasons, if the variable *a* were to be split to distinguish the actions of *cycA/cdk2* and *cycA/cdk1*, the edge

$cycA/cdk1 \rightarrow sk$ must be treated with the utmost caution.

The variable b is associated with three concentration levels. When $b = 1$, the $cycB/cdk1$ complex is partially activated and can act indirectly only on its own activation ($b \rightarrow b$). At this threshold, b can take over from sk and a in inhibiting APC-cdh1, p21 and p27 ($b \rightarrow en$). The second threshold of b corresponds to an accumulation of cyclin B sufficient to activate the initiation of mitosis and APC-cdc20 ($b \rightarrow ep$), a protein involved in mitotic exit.

In section 5.4.2, we demonstrate that all these thresholds make it possible to define the action of each of these variables on events characteristic of the phases of the cell cycle and, by extension, the regulatory events required in G1, S, G2 and M.

5.3 Our initial conditions

The definitions of the variables and thresholds of our GRBM suffice to characterise the initial conditions of our system. Even before defining the regulations between our variables, defining these initial conditions already shows that all the expected properties will be observable:

1. A property of stable quiescence when growth factors are persistently removed from the extracellular environment.
2. Perpetual commitment to a cell cycle when growth factors are constantly active.

A cell in quiescence (denoted G0) is canonically described by inhibition, by APC-cdh1 (en), of the molecular complexes required to initiate replication and mitosis (namely sk , a and b) [100]. The APC-cdc20 protein (ep), crucial for mitotic exit, is inactive in G0. We obtain a first initial condition characterised by:

$$sk = 0 \wedge a = 0 \wedge b = 0 \wedge en = 1 \wedge ep = 0 \quad (5.1)$$

Our set of variables includes an *environment variable*: growth factors. gf is a Boolean variable under the modeller's complete control, so two initial conditions of the cell cycle must be considered. Either growth factors are inactive ($gf = 0$) and the cell is in quiescence:

$$G0 \equiv sk = 0 \wedge a = 0 \wedge b = 0 \wedge en = 1 \wedge ep = 0 \wedge gf = 0 \quad (5.2)$$

Or the growth factor concentration is sufficient to potentially activate the $cycE/cdk2$ complex (sk) through the MEK/ERK pathway ($gf = 1$), and the cell is in G1:

$$G1_{init} \equiv sk = 0 \wedge a = 0 \wedge b = 0 \wedge en = 1 \wedge ep = 0 \wedge gf = 1 \quad (5.3)$$

More precisely, $gf = 1$ means that the $cycD/cdk4-6$ complexes can activate sk by repressing the inhibitory effects of members of the pRB family, while also providing a potential resource for the genes required for replication and mitosis. Further details will be given in section 5.4.2. Since these formulas describe the state of each of our six variables, they characterise a state at the systemic level (definition 5, page 29). $G1_{init}$ describes the earliest state of G1 that we are able to define.

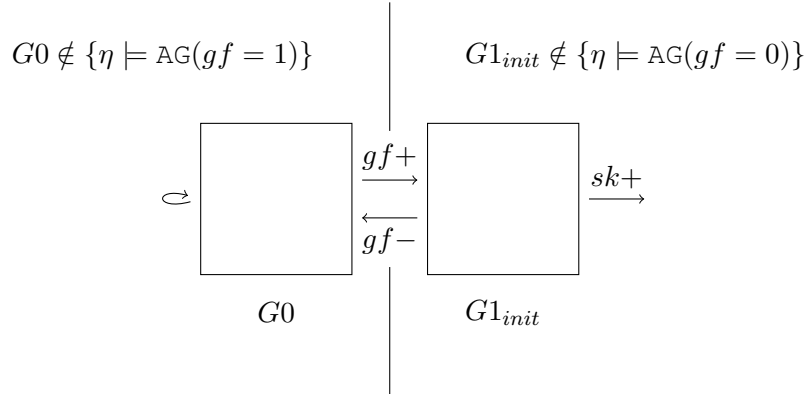


Figure 5.3: **Two environmental contexts, hence two initial states.** Each initial state is assumed to belong to a unique attractor.

Thanks to gf , the state space of our GRBM is partitioned into two parts, each of which admits a unique attractor. The first part comprises the states that satisfy the CTL formula $\text{AG}(gf = 0)$, which literally means that gf is constantly inactive. This part must admit a unique attractor: the stable state $G0$. Indeed, as long as the extracellular environment is not exposed to a sufficient level of growth factors, the cell must remain arrested in quiescence. Moreover, in practice, cells in *in vitro* culture can be synchronised in $G0$ (as we define it) by deprivation of foetal bovine serum, a solution highly concentrated in growth factors (see the list in table 5.1). The second part comprises the states that satisfy the CTL formula $\text{AG}(gf = 1)$. $G1_{init}$ is the initial state of the cell cycle, which is formally considered a complex attractor, as in [88, 87]. It is even considered the unique attractor of this partition in which gf is constitutively active.

A cell that is not quiescent is a proliferating cell: it commits to carrying out all the events required during the cell cycle. In figure 5.3, $sk+$ marks the first cell cycle event that we can formalise and, above all, marks a quiescence/proliferation transition, commonly called the *restriction point*. In section 5.4.2, we explain what enables sk to become activated in $G1_{init}$ and why $sk+$ is annotated as passage through the restriction point. [24, 81] describe the restriction point as a point of no return in the cell cycle that depends on the growth factor concentration. If growth factors are removed from the extracellular environment before the restriction point, the cell stops immediately in $G0$ or early $G1$. If they are removed after the restriction point, the cell proceeds through $G1$, S , $G2$ and M , and stops in $G0$ or early $G1$ of the next cell cycle. Formula 5.2 differs from formula 5.3 only in the value of gf . The transition between $G0/G1_{init}$ depends on gf and is completely reversible (no hysteresis), unlike the transition from quiescence to proliferation, which is irreversible (hysteresis).

5.4 Formalisation of the regulations involved in cell cycle progression

This section is organised as follows:

The influences that our variables exert on one another are first described at the highest level of abstraction using an influence graph, in which a node is a variable and an edge represents

a positive (labelled $+$) or negative (labelled $-$) action on the synthesis rate of the target variable (subsection 5.4.1).

A canonical sequence of regulatory events involving our variables is then defined using the initial conditions and the influence graph (subsection 5.4.2).

A regulatory graph with multiplexes is then derived from the influence graph and phenomenological multiplexes (subsection 5.4.3).

5.4.1 Our influence graph designed to define a canonical cell cycle

The regulation of cell cycle progression is described by a complex network of multiple positive and negative feedback loops. In chapter 3, we highlighted several studies whose abstraction approach is remarkable and which successfully identified modules characteristic of particular cell cycle phases or phase transitions [81, 83, 74]. These networks have a particular structure: they contain a positive feedback loop that acts as a bistable switch. In chapter 3, we described the crucial role of multistable switches in the temporal separation of phases and the unidirectionality of the cell cycle. We will describe our influence graph of cell cycle progression in a modular way (introduced in figure 5.12). Either the module is involved in a given phase and therefore represents a positive loop, or it is involved in a phase transition, in which case it is structurally represented by a negative feedback loop.



Figure 5.4: **Positive loop involved in passage through the restriction point, canonically located in G1.** The loop on sk allows the activation signal from gf to persist even if growth factors are removed from the extracellular environment.

The activating edge $gf \rightarrow sk$ abstracts the activating chain from growth factors (called *mitogens* in figure 5.5) to the $cycE/cdk2$ complex. The formula $\varphi_0 \equiv gf \geq 1$ is a literal in the multiplex language that represents much more than the presence of gf : it also implies activation of the MEK/ERK pathway, the presence of the $cycD/cdk4-6$ complex, the absence of Rb, and the presence of E2F. Through the MEK/ERK pathway, growth factors activate transcription of cyclin D,

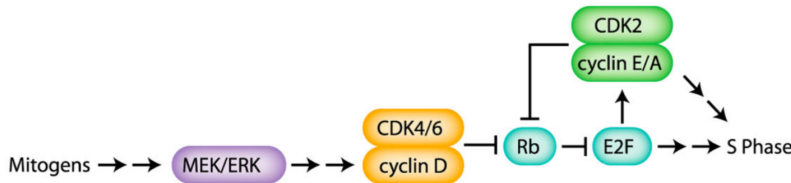
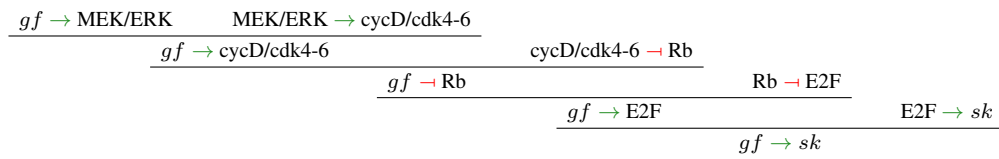


Figure 5.5: **Simplified diagram of the dominant paradigm for the regulation of passage through the restriction point.** Figure adapted from [63].

which, once associated with $cdk4-6$, indirectly relieves inhibition of the E2F transcription factor by the Rb protein. The $cycD/cdk4-6$ complex phosphorylates the Rb protein, partially dissociating

E2F from the complex it formed with Rb. E2F activates transcription of the gene encoding Cdc25, a phosphatase that activates the *cycE/cdk2* complex, and of cyclin E. In this configuration, E2F is present in sufficient quantities to activate the *cycE/cdk2* complex, which in turn hyperphosphorylates the Rb protein, thereby creating the *cycE/cdk2* activation loop in figure 5.4. When the positive loop $cycE/cdk2 \rightarrow Rb \rightarrow E2F \rightarrow cycE/cdk2$ is effective, a cell becomes insensitive to growth factor deprivation, since this loop suffices to maintain the *cycE/cdk2* complex in an active state (at threshold 1) [102]. Passage through the restriction point is therefore associated with activation of the *cycE/cdk2* complex at threshold 1. $gf \rightarrow sk$ is a typical example of the abstraction of a regulatory pathway based on transitivity:



This tree makes explicit rules that refer to positive (green) or negative (red) interactions, which must be corroborated by experimental work. Other pathways join those in figure 5.5; we refer in particular to [38], which provides a detailed, curated map of cell cycle regulation by RB/E2F. The consistency of our abstraction was checked as we proceeded by consulting one of the most extensively documented, cleaned and mammalian-model-oriented databases: Reactome*. The idea is to check manually that no regulation specified in the database between the variables of our model contradicts our abstraction of activating or inhibitory edges. For example, if an activating edge has been defined between x and y , we check that there is indeed an even (resp. odd) number of inhibitory edges abstracted in $x \xrightarrow{+} y$ (resp. $x \xrightarrow{-} y$).

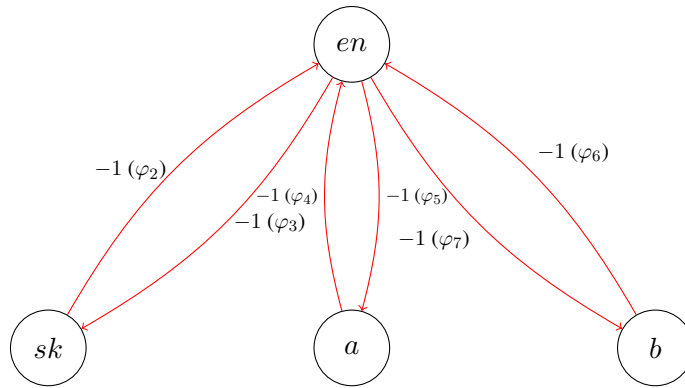


Figure 5.6: **Reciprocal inhibitions between cyclins E, A and B and the proteins APC-cdh1, p21 and p27.**

The end of G1 phase is marked by relief of the inhibition of *cycE/cdk2* (sk) by APC-cdh1, which gives us the opportunity to describe one of the mechanisms underlying the edges from en to sk , a or b , figure 5.6. Cyclins E, A and B (and, by extension, the variables sk , a and b) are ubiquitinated by APC-cdh1, leading to their degradation by the proteasome. This degradation mechanism,

*<https://reactome.org/>

which we will encounter again below, is called proteolytic destruction. The inhibitory activity of APC-cdh1 is repressed by the phosphorylation of cdh1 catalysed by sk , a and b , creating a positive feedback loop between the cyc/cdk complexes and their inhibitors en . More precisely, this phosphorylation of cdh1 causes proteolytic degradation of APC-cdh1 by the ubiquitin ligase SCF (“ubiquitinated ubiquitin”). The variable en also abstracts the *Cdk Inhibitor Proteins* (CKIs), which have the distinctive ability to inhibit the active site of cyc/cdk complexes by binding directly to them. These include the proteins p27 and p21. They are also targeted by the cyc/cdk complexes in our model, and the same degradation process as for APC-cdh1 follows. These proteins are also decisive in the process of cell cycle arrest in response to DNA lesions, a process (called an extrinsic checkpoint in chapter 3) that nevertheless lies outside our modelling question [37].

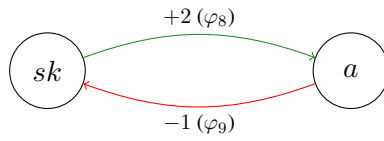


Figure 5.7: **Negative feedback loop involved in the G1/S transition.**

Once the $cycE/cdk2$ complex is fully activated ($sk = 2$), it activates its own destruction *via* the $cycA/cdk2$ complex (a). At threshold 2, sk inhibits the pRb protein and thereby releases a version of the E2F protein that specifically activates transcription of cyclin A ($sk \rightarrow a$) [20, 59] (figure 5.4). It then associates with cdk2 to form the $cycA/cdk2$ complex and can also bind to cdk1 to form the $cycA/cdk1$ complex. The $cdk2/cycA$ complex catalyses the inhibitory phosphorylation of E2F, thereby also inhibiting transcription of cyclin E and cdk2 ($a \rightarrow sk$). Inactivation of sk inhibits an event required in G1: the formation of pre-replication complexes at replication origins, described in subsection 5.4.2.

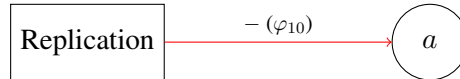


Figure 5.8: **The S/G2 transition is controlled by the repressive activity of the ETAA1-ATR-CHK1 pathway on the cyclin A-CDK1-FOXM1 axis until replication ends.** The variable a is sensitive to the endogenous stress caused by the replication process.

The replication process negatively influences a cell cycle regulator: the $cycA/cdk1$ complex. This inhibition acts through the ATR protein. It is primarily known for its decisive role in the response to DNA damage. Nevertheless, a recent study concludes that it has a role in coordinating replication and mitosis in the absence of DNA damage [85]. The authors in fact group the mechanisms involved in this coordination under the term intrinsic S/G2 checkpoint. ATR acts as a sensor of replication completion and ensures that the cell does not activate genes involved in G2 or M phase before replication ends. In section 5.4.2, we show precisely that $cycA/cdk1$ activates expression of genes involved in mitosis. Once replication begins, but as long as it remains incomplete, single-stranded DNA is coated by the RPA protein, which recruits ATR and subsequently inhibits the activity of the $cycA/cdk1$ complex through phosphorylation. This inhibition acts through activation of Chk1, an inhibitor of the $cycA/cdk1$ complex. ATR inhibits pro-mitotic factors through two

pathways. ATR inhibits FOXM1, which is to M phase what E2F is to S phase. FOXM1 activates pro-mitotic factors such as Plk1 and Aurora B, proteins that are decisive in the *Spindle Assembly Checkpoint* (SAC), and also activates transcription of cyclin B [25]. ATR activity drops abruptly at the S/G2 transition, that is, at the end of replication. This results in relief of ATR-mediated inhibition of the cycA/cdk1 complex.

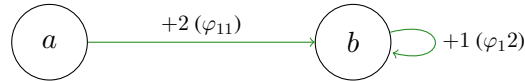


Figure 5.9: **A positive feedback loop involved in the activation of b at the very beginning of M phase.**

The cycA/cdk1 complex hyperphosphorylates the FOXM1 protein, as described previously, thereby activating transcription of cyclin B, Plk1 and the various isoforms of the Cdc25 phosphatase (Cdc25A, Cdc25B and Cdc25C). Once the cycB/cdk1 complex is formed, it reinforces its own activation ($b \rightarrow b$) through two mechanisms, concisely described in the review [44]. Briefly, cycB/cdk1 activates the Plk1 protein, which in turn activates FOXM1, causing cyclin B to accumulate at the centrosome, where the mitotic substrates of b are located. The cycB/cdk1 complex is kept partially activated by inhibitory phosphorylation by Wee1 and Myt1, two proteins not explicitly represented in our model, since we assume that the abrupt activation of b suffices to describe the activation of mitosis; see subsection 5.4.2.4. The cycB/cdk1 complex relieves its inhibition by Wee1 and Myt1 by directly phosphorylating them and by activating the Cdc25 proteins, the phosphatases antagonistic to Wee1 and Myt1.

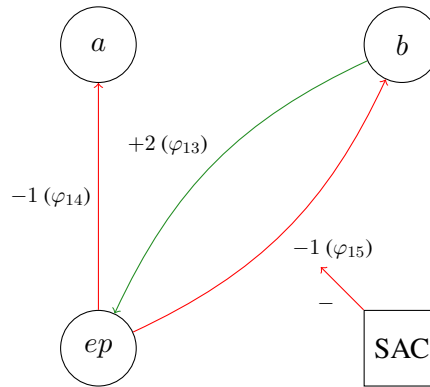


Figure 5.10: **APC-cdc20 (en) and degradation of cyclins A and B involved in the mitotic process.** Cyclin B degradation depends on the SAC. The variable ep is sensitive to the SAC.

The Cdc20 protein can bind to APC only when the latter has previously been phosphorylated by the Cdk1/CyclineB complex ($b \rightarrow ep$) [49]. APC-cdc20 activates anaphase, that is, the separation of chromosomes into sister chromatids; see figure 5.11. Once the mitotic process has begun, the *Spindle Assembly Checkpoint* (SAC) blocks APC-cdc20 activity until all chromosomes are correctly attached to the mitotic spindle at their kinetochores. The targets of APC-cdc20 include securin, a protein that inhibits the protein responsible for separating chromosomes into sister chromatids. The role of the SAC is to inhibit APC-cdc20 (and hence anaphase) as long as at least

one chromosome is not attached to the spindle. The SAC inhibits the action of *ep* on *b* and securin. Briefly, Mad2 binds to unattached kinetochores to form a complex that sequesters Cdc20. Once all

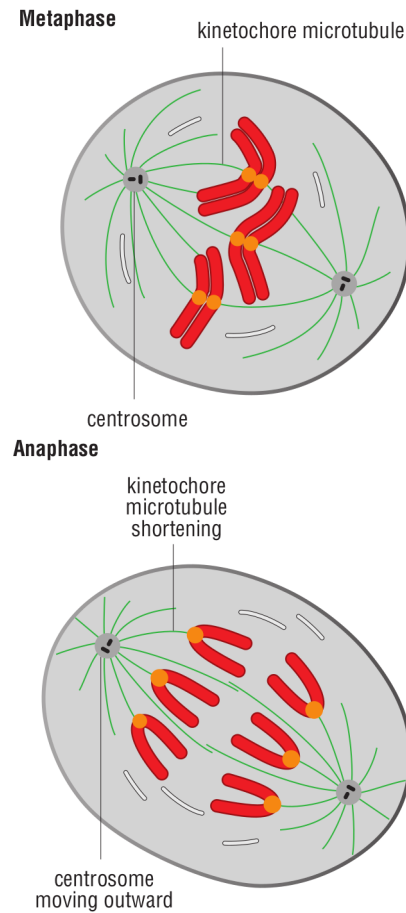


Figure 5.11: **Metaphase and anaphase.** During mitosis, the cytoskeleton of a eukaryotic cell is organised as a mitotic spindle composed of microtubules (green). During metaphase, they can bind to each sister chromatid of a chromosome (red) at a structure called a kinetochore (orange). The metaphase/anaphase transition is marked by separation of the sister chromatids, which migrate towards opposite poles of the cell at the end of anaphase. Figure adapted from [33].

kinetochores are attached to microtubules, the Mad2 protein releases Cdc20, which can associate with APC to act on cyclin B ($ep \rightarrow b$) and securin. Cyclin A, for its part, escapes the repression of its destruction by *ep*, which explains why the SAC does not target the inhibitory edge $ep \rightarrow a$. The justifications are given when defining the canonical cell cycle trace, in which destruction of *a* precedes that of *b* in M phase; see page 95.

APC-cdc20 is also a ubiquitin ligase whose substrates include cyclin A ($ep \rightarrow a$) and cyclin B ($ep \rightarrow b$), but not cyclin E [17] (absence of the edge $ep \rightarrow sk$). APC-cdh1 targets cdc20 and therefore exerts an inhibitory action on the APC-cdc20 complex ($en \rightarrow ep$).

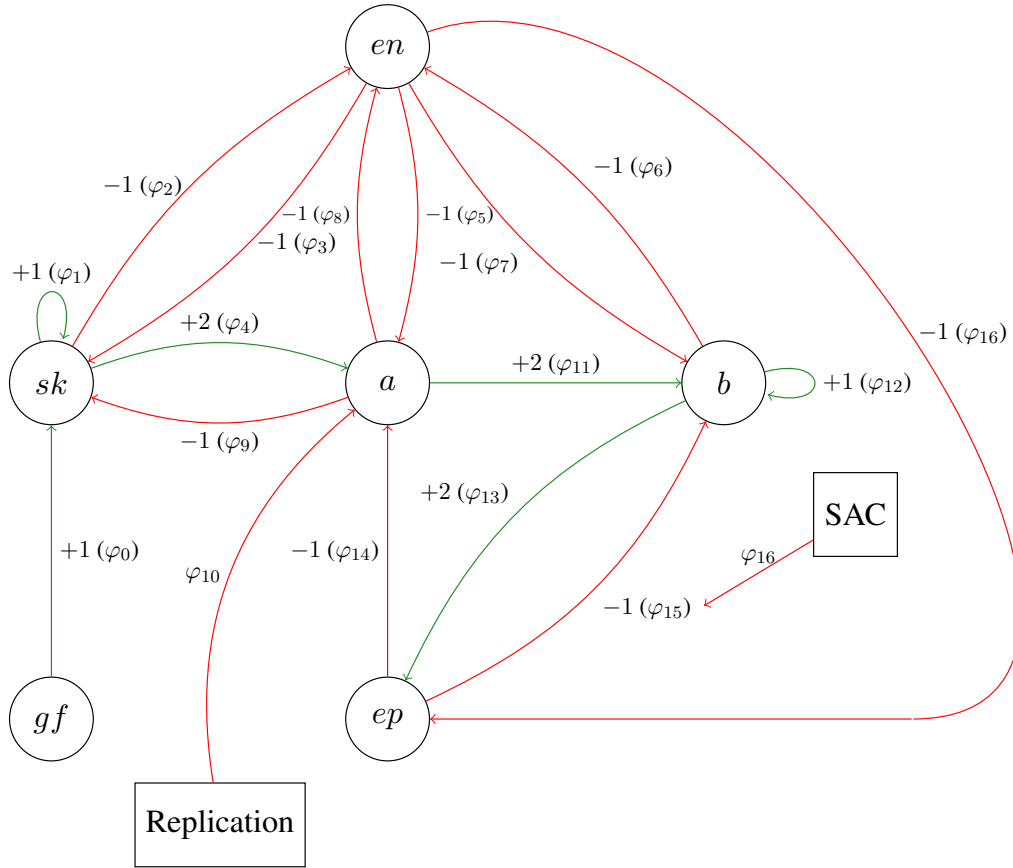


Figure 5.12: **Revision of the influence graph in [71].** The variable gf has no predecessor and is therefore called an environment variable, since it is under the modeller's complete control.

5.4.2 A canonical sequence of regulatory events during the phases of the cell cycle

Our five systemic variables regulate the processes of replication and mitosis. These influences are not represented in our graph in figure 2.3. Describing the unidirectional regulations from sk , a , b , en and ep to G1, S, G2 and M allows a set of required events to be associated with each phase. These events are represented by the atoms of the assertion language (definition 16, page 68). A canonical sequence of these events is then defined starting from our initial cell cycle state $G1_{init}$.

5.4.2.1 The events required in G1

The event $gf+$: Growth factors indirectly inhibit the Rb protein (figure 5.5), a protein that represses E2F and thus, by extension, the genes required for progression through S phase. Indeed, E2F activates transcription of genes involved in forming the pre-replication complex (*e.g.*, those encoding *cdc6* and *cdt1*) [42].

The first event $sk + (sk = 0 \rightarrow sk = 1)$: It is annotated as passage through the restriction point, based mainly on the canonical model proposed by [102] (figure 5.5). sk completes phosphorylation of Rb, initially performed by the $cycD/cdk4-6$ complex, resulting in a drastic relief of inhibition of S-phase genes under E2F control [22].

The event $en-$: The p21 protein binds to a protein capable of interacting with DNA polymerases and inhibits its activity in promoting DNA processivity [33].

The second event $sk + (sk = 1 \rightarrow sk = 2)$: The $cycE/cdk2$ complex (sk) is fully active in a hyperphosphorylated state. It contributes to activating the pre-replication complex (abbreviated CPR) at replication origins (figure 5.13). This molecular complex consists of several protein subunits, some of which are phosphorylated directly by sk . For example, phosphorylation of $cdt1$ and $cdc6$ by sk contributes to CPR assembly at replication origins [42]. Furthermore, sk has been shown to phosphorylate $Sld2$ and $Sld3$, giving the CPR its ability to unwind DNA and enabling recruitment of DNA polymerase [57]. Inhibition of $cdk2$ in human cells inhibits the initiation of S phase [27]. Our choice of variables implies the following modelling assumption: once sk has reached its second regulatory threshold, we assume that *all* replication origins are bound to an active CPR. In other words, we assume *no mutation associated with defective assembly or activation of a CPR* if all the cell cycle regulatory events occur in the expected order (see H_c , page 94). Moreover, the $cycE/cdk2$ complex indirectly activates chromatin duplication by activating the transcription factor for genes encoding the main histones, which are essential for nucleosome formation.

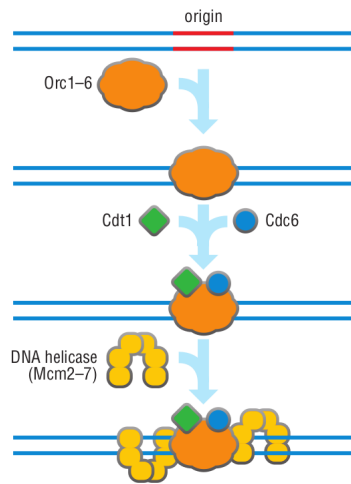


Figure 5.13: **Assembly of the pre-replication complex.** Figure taken from [35].

5.4.2.2 The events required in S

The first event $a + (a = 0 \rightarrow a = 1)$: It is associated with inactivation of the proteins that form the CPR. Indeed, the $cycA/cdk2$ complex is involved in preventing repeated replication within the same cycle. To limit genome replication to once per cell division, cells must be unable to replicate DNA in regions that have already been replicated before cell division is complete. At threshold 1,

the variable a inhibits the proteins forming the CPR through phosphorylation. As described above, a inhibits sk at threshold 1 *via* an inhibitory phosphorylation reaction involving E2F. It turns out that E2F inhibition also represses expression of the genes required for DNA replication, as well as several proteins essential for CPR assembly. Preventing the initiation of several replication cycles during the same S phase avoids replication fork stalling and DNA strand breaks [57].

The two events $sk-$: They represent inactivation of the $cycE/cdk2$ complexes following the event $a+$. We assume that they indicate effective inhibition of the pre-replication complexes.

5.4.2.3 The events required in G2

Cyclins A and B are the two major regulators of mitotic entry.

The second event $a +$ ($a = 1 \rightarrow a = 2$): The $cycA/cdk1$ complex also inhibits CPR assembly by additionally phosphorylating Orc1 and Cdc6 [57]. This complex, modelled only at threshold 2 of a , is involved in activating FOXM1, a transcription factor for genes encoding proteins essential for initiating mitosis, including Cdc25, cyclin B, Plk1 and Aurora B [25]. Plk1 and Aurora A are two proteins required for centrosome maturation and mitotic spindle formation.

The first event $b +$ ($b = 0 \rightarrow b = 1$): Cyclin B accumulates abruptly at the beginning of mitosis. This requires prior activation of the $cycB/cdk1$ complex.

5.4.2.4 The events required in M

The event $b +$: In response to the abrupt increase in cyclin B concentration, the $cycB/cdk1$ complex becomes fully activated and is then exported to the nucleus, where it begins to phosphorylate the nuclear substrates required to initiate mitosis. For example, at this threshold, b activates Plk1 and Aurora A, two proteins required for maturation and separation of the centrosome (the microtubule-organising centre) and formation of the mitotic spindle (illustrated in figure 5.11). Another example is initiation of a prophase event: nuclear envelope breakdown [48].

The event $ep +$: APC-cdc20 is involved in mitotic exit, marked by the onset of anaphase, the phase of mitosis during which sister chromatids are abruptly separated. APC-cdc20 indirectly degrades securin, a protein responsible for sister chromatid cohesion. APC-cdc20 plays an essential role in the equal partitioning of chromosomes between the two daughter cells, which is indispensable for protecting the cell against the loss of a chromosome (aneuploidy) [33].

The two events $a-$ and the two $b-$: They represent degradation of cyclins A and B following the event $ep +$. Degradation of a and b marks the withdrawal of activation of their substrates involved in chromosome condensation and nuclear envelope disassembly. [31] mention that these four events (*i.e.*, degradation of mitotic cyc/cdk) induce the termination of M phase, characterised in particular by chromosome decondensation and nuclear envelope assembly. Degradation of a represses its inhibitory action on the pre-replication complex, helping to prepare the daughter cells for replication in the next cycle.

The event $en+$: The PP2A and PP1 phosphatases dephosphorylate the mitotic substrates activated earlier in the cycle by $cycA/cdk1$ and $cycB/cdk1$. The combination of this event and the preceding four is sufficient to trigger mitotic termination in a switch-like manner [35]. [31] hypothesise that association of APC with Cdh1 at the end of mitosis maintains APC ubiquitinase activity for the next cycle.

The event $ep-$: According to our current knowledge, it has not been proved to be required in M. We assume that it occurs in M because it is necessary to return to the initial state of our G1 phase; we refer the reader to the canonical cell cycle Hoare triple in equation 5.4.

5.4.2.5 The resulting canonical sequence

Ultimately, the multiset of events required during each phase has now been defined using the key regulators of cell cycle progression:

$$\begin{aligned} G1 &: \{sk+ : 2, en- : 1\} \\ S &: \{a+ : 1, sk- : 2\} \\ G2 &: \{a+ : 1, b+ : 1\} \\ M &: \{b+ : 1, ep+ : 1, a- : 2, b- : 2, en- : 1, ep- : 1\} \end{aligned}$$

The initial condition of the canonical cell cycle is given by the formula $G1_{init}$ in section 5.3. It formalises the initial state of G1 in which gf is activated and, as a reminder, is under the modeller's control. The change of gf to 1 has therefore already taken place in our initial cell cycle state, which means that we do not include it in the multiset of events required in G1. From the literature, we have inferred a canonical cell cycle sequence starting from $G1_{init}$, divided into subsequences characteristic of G1, S, G2 and M. This sequence and preliminary division into phases represent a consensus on the dependencies between the events required during each phase of the cell cycle. The resulting canonical Hoare triple is as follows:

$$H_c \equiv \{G1_{init}\} \underbrace{sk+; en-; sk+; a+; sk-; sk-; a+; b+; b+; ep+; a-; a-; b-; b-; ep-; en+}_{p_c} \{G1_{init}\} \quad (5.4)$$

The order of required events within a phase is not a decisive question, since this thesis focuses on cell cycle checkpoints. Nevertheless, exploring the literature provides knowledge of dependencies between the events required for each phase:

In G1 phase: starting from $G1_{init}$, sk can be fully activated (second $sk+$) only if en , in the sense of APC-cdh1, has previously been inhibited by sk . The first $sk+$ precedes $en-$ [72].

In S phase: once the events required in G1 have occurred starting from $G1_{init}$, degradation of sk (two $sk-$) is induced once the $cycA/cdk2$ complex has been activated (first $a+$) [20].

In G2 phase: b is initially activated by the $cycA/cdk1$ complex. The event $a+$ occurs before $b-$ [57].

In M phase:

1. APC-cdc20 (*ep*) can act on its targets only when *b* is fully activated ($b = 2$) and APC-cdh1 (*en*) is inactive [49].
2. APC-cd20 (*ep*) targets both cyclin A and cyclin B for proteolytic degradation, and inhibition of *a* is observed to occur before that of *b*. When APC-cdc20 is activated between prophase and prometaphase, its action on *b* is inhibited as long as the SAC is active, that is, until the metaphase/anaphase transition. Its action on *a*, however, is independent of the SAC [47]. The first $a-$ therefore precedes the first $b-$. We assume that the two $a-$ occur before the two $b-$ simply in order to have a unique canonical sequence.
3. Cyclin degradation depends on APC-cdc20 in M phase [31]. The event $ep+$ occurs before $a-; a-; b-; b-$.
4. Only once cyclins A and B have been degraded ($a-; a-; b-; b-$), we assume (again in order to have a unique sequence) that a free fraction of APC can bind to its Cdh1 subunit to form APC-cdh1 ($en+$). The event $en+$ follows $a-; a-; b-; b-$.
5. We assume that en then induces degradation of Cdc20 ($ep-$) to close the loop. The event $en+$ precedes $ep-$ in the canonical sequence.

Current knowledge also allows us to explain why certain events, each belonging to a different phase, must occur in a particular order. The event that closes a phase must not take place before the event that initiates the next phase. As a reminder, this guarantees DNA integrity during the cell cycle. Here are two characteristic examples:

G1/S: the second $sk+$ is required before the first $a+$. This chronology is crucial because, if $a+$ occurs before $sk+$, CPR activation cannot take place, since $cycE/cdk2$ is an activator of CPR and $cycA/cdk2$ is one of its inhibitors [57].

S/G2: the activity of the $cycA/cdk1$ complex in G2 phase is repressed as long as replication is incomplete. Under our modelling assumptions, replication is considered complete once all the events required in S have occurred. According to René Thomas's semantics, the first event $a+$ is naturally required before the second $a+$, but both $sk-$ events are required before $a+$.

5.4.3 Definition of multiplexes

5.4.3.1 Multiplexes and mechanistic abstraction

The edges $\varphi_{[1-9]}$ and $\varphi_{[11-15]}$ of the influence graph in figure 2.3 formalise literals in the multiplex language (see figure 2.5, page 27). If the variable x activates (resp. inhibits) the variable y at threshold n , then we define $x \geq 1$ (resp. $\neg(x \geq 1)$) to formalise this regulation. Each of these formulas is associated with a label to form multiplexes. Here is an example of an atom representing the presence of sk at a concentration sufficient for self-activation:

$$\varphi_1 \equiv (sk \geq 1)$$

We associate φ_1 with the label rp (for *restriction point*) in the GRBM presented in figure 5.18. Here is an example of a negative literal formalising a concentration of sk insufficient to induce repression of the activity of en :

$$\varphi_2 \equiv \neg(sk \geq 1)$$

We associate φ_2 with the label *no_sk* in the GRBM in figure 5.18. Conceptually, a multiplex represents a potential resource. When a state satisfies the formula of a multiplex, it constitutes a resource (the set of resources of a variable is formally defined by ϕ_v^ω on page 71).

The inhibitory mechanisms characterised by the formulas φ_5 , φ_7 , φ_{14} and φ_{15} of the influence graph in figure 2.3 are grouped into a single multiplex: the one labelled *no_CI* in figure 5.18, with formula $\neg(en \geq 1 \wedge ep \geq 1)$. This is a typical case of grouping regulations of the same nature to minimise the number of parameterisations; we refer the reader to chapter 2, page 28. This formula means that the absence of the two inhibitors *en* or *ep* is necessary to constitute a potential resource for the target variables (*a* and *b*). In other words, the presence of either inhibitor cannot constitute a resource under any circumstances.

5.4.3.2 Multiplexes and phenomenological abstraction

Activation or inhibition of our variables has made it possible to formalise the various phenomena that occur during replication and mitosis. Some of these cell cycle phenomena *act back* on our variables. Two variables in our model are concerned: *a* and *b*. The incomplete replication process inhibits the *cycA/cdk2-1* complex; in other words, the end of replication is a phenomenon that represents a resource for the variable *a* (figure 5.8). A similar case is found in M phase. The action of the APC-cd20 protein on the *cycB/cdk1* complex is inhibited by the SAC (figure 5.10). Replication and the SAC would be two complex and costly processes to model at the molecular level, given the many molecular species involved and the varied nature of their regulatory mechanisms. At the time this thesis was written, 3029 genes were annotated with the term “replication” [107], and 446 with “Spindle Assembly Checkpoint” [108], for *Homo sapiens*. Still following the philosophy of Occam’s razor, we choose to use only our six variables *sk*, *a*, *b*, *en*, *ep* and *gf* to represent these processes, since they have enabled us to represent a plausible sequence of cell cycle phases (triple 5.4).

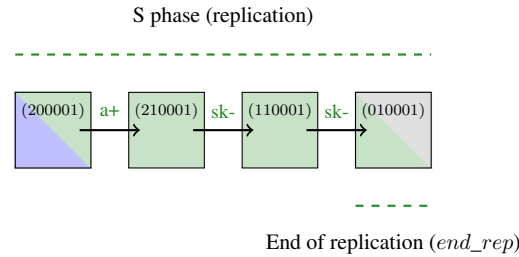


Figure 5.14: **The multiplex *end_rep* is defined using the canonical cell cycle sequence.** Shown here is the canonical S-phase sequence starting from a state characterised by the formula $sk = 2 \wedge a = 0 \wedge b = 0 \wedge en = 0 \wedge ep = 0 \wedge gf = 1$ (abbreviated 200001).

The abbreviated state 200001 is the latest marker of S phase in our canonical sequence. Only the variables *a* and *sk* are involved in the regulatory events in this representation of S phase. Consequently, only these variables will be used to define the end of replication in our model. We define the multiplex *end_rep*, whose formula is φ_{10} (figure 5.8):

$$\varphi_{10} \equiv (a \geq 1 \wedge \neg(sk = 1))$$

SAC activation is observable in our model when ep is activated but cannot yet act on cyclin B degradation ($b-$). However, even if the SAC is active, a can still begin to degrade [47]. This time window is again characterised using our canonical cell cycle Hoare triple: H_c . The abbreviated

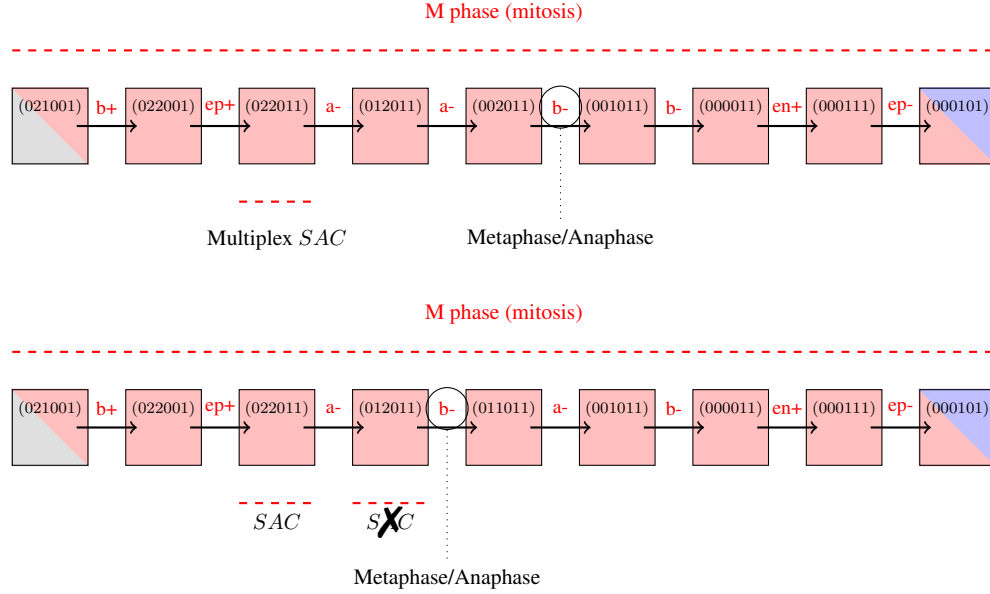


Figure 5.15: **The multiplex SAC is defined using an alternative trace of the canonical cell cycle sequence.** The canonical M sequence is shown starting from state (021001) in the upper part of the figure. No observation prevents the representation of degradation of b by the alternative trace shown in the lower part of the figure. The metaphase/anaphase transition is marked by relief of SAC-mediated repression of degradation of b (first $b-$). We will henceforth denote this event SAC_{off} .

state (022011) corresponds to the closest description of the SAC that we can achieve using our cell cycle variables (upper part of figure 5.15). The SAC is a resource of b as long as its degradation is not observed. A study of alternative traces of the path of H_c (corroborated in chapter 6) shows that, once a begins to degrade (*i.e.*, $a = 1$ after the first $a-$), degradation of b can occur ($b-$); see the lower part of figure 5.15. The state (012011) is therefore unsuitable for representing the SAC, which is assumed to be a resource of b that prevents it from degrading. Thus, the SAC is a resource of b as long as ep is active and a has not yet degraded. We define the multiplex SAC in figure 5.18 with the formula:

$$\varphi_{16} \equiv (ep \geq 1 \wedge a \geq 2)$$

Finally, $restrict_sk$ is another phenomenological multiplex, but one derived solely from observing the canonical S-phase sequence. The characteristic formula of the abbreviated state (110001) in figure 5.16 is very close to that of an abbreviated G1-phase state (100001); see the upper part of figure 5.19. Restricting ourselves to the variables required in S phase, the multiplex $restrict_sk$ must represent a resource that a must possess in this particular S-phase state. This resource prevents a from making the transition that takes the system back to G1:

$$\varphi_{17} \equiv (sk \geq 2 \wedge a \geq 1)$$

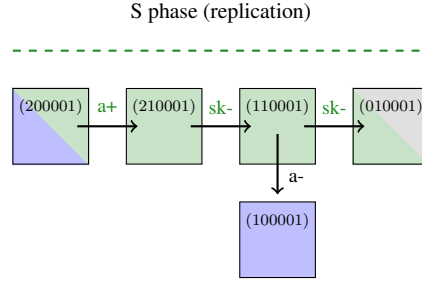


Figure 5.16: **The multiplex $restrict_sk$ is also defined using the canonical cell cycle sequence.** Introducing this multiplex prevents a return to the current G1 phase (blue) from an S state (green).

Lastly, passage through the restriction point (abbreviated PPR, as a reminder) is associated in subsection 5.4.2.1 with the time of the first $sk+$, that is, in G1 phase according to H_c . However, experimental observations show that passage through the restriction point, specifically associated with this $sk+$ event, can also occur at the end of M phase [63]. The problem is therefore to define a reasonable division of canonical M phase into an early and a late portion.

Once this problem has been resolved, it is necessary to restrict PPR to the late portion of M phase alone. The action of growth factors is shown to be decisive for passage through the restriction point in subsection 5.4.2.1. The action of gf on sk is restricted to the late portion of M, characterised using the first $b-$ event characteristic of the metaphase/anaphase transition, figure 5.17. More precisely, we add a condition necessary for gf to activate sk . Expressed in terms of resources, this gives the following multiplex $restrict_gf$:

$$\varphi_0 \equiv (gf \geq 1 \wedge \neg(b \geq 2))$$

which means that sk has gf as a resource only when b is not saturated, that is, after the SAC in M (hence in the late portion).

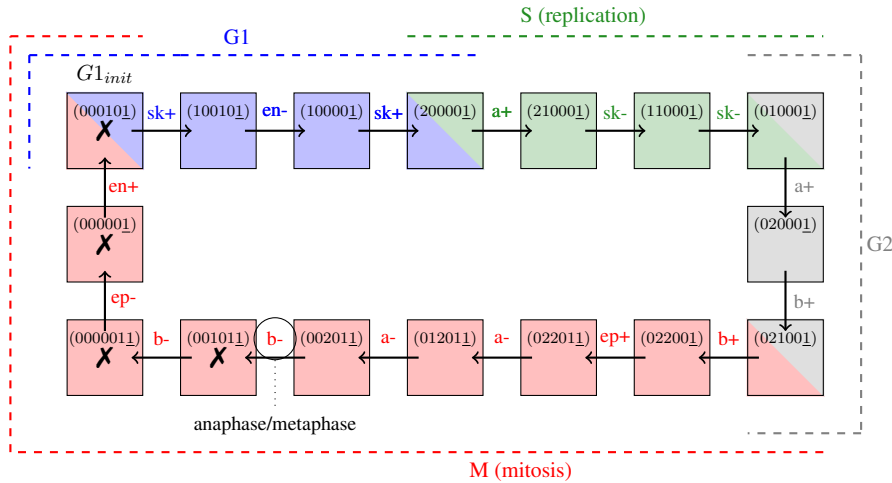


Figure 5.17: **The time window of the cell cycle during which passage through the restriction point (the first $sk+$) is permitted.** An example using our triple H_c . sk has the necessary resource ($restrict_sk$) to undergo PPR in the states marked with a cross.

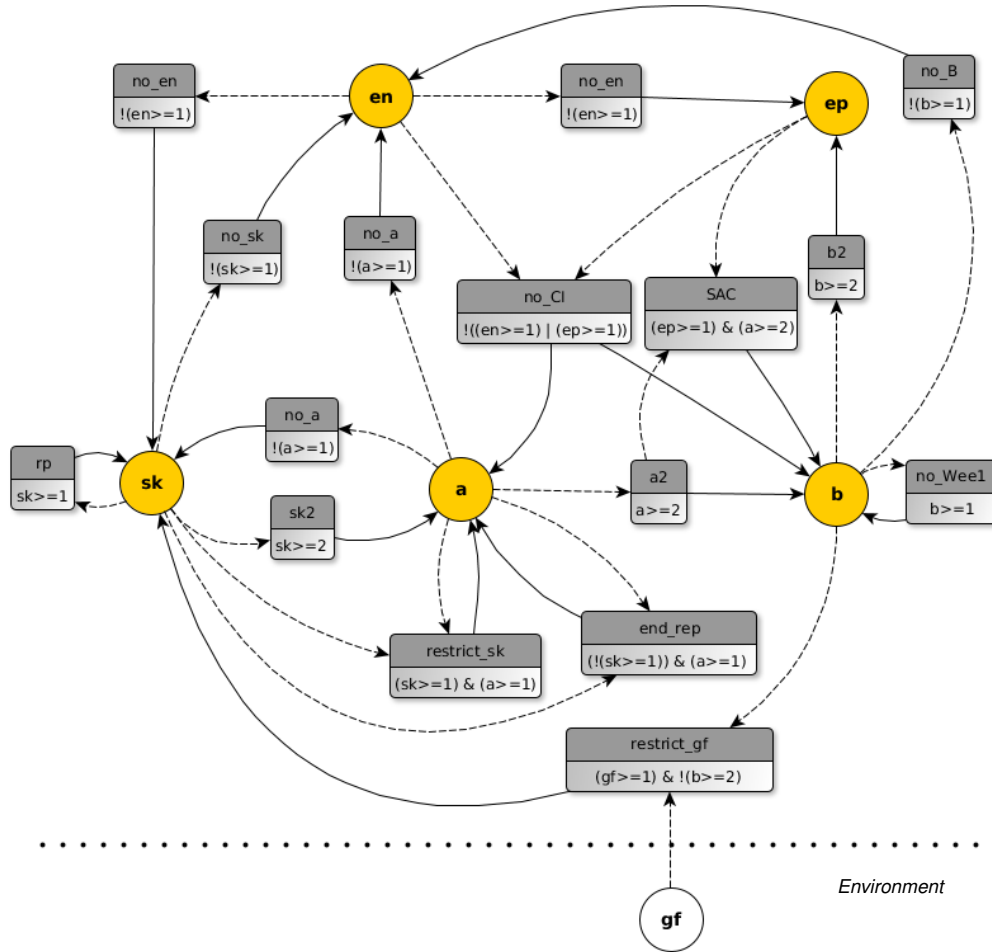


Figure 5.18: **The GRBM of the mammalian cell cycle.** Multiplexes are represented by rectangles in two parts: the label and the associated formula. Their formula characterises either a literal in the multiplex language or known cooperation between certain variables to *potentially* activate their target. The cooperation between the predecessors of a variable is not known at this stage. Instantiating all the elements of the parameter family $\mathcal{K}$ will make it possible to identify these cooperation rules. This graph was initially inspired by the abstraction proposed in the model of [41] and is drawn using the yEd software. We refer the reader to a demonstration of the use of TotemBioNet for further information [97].

5.5 Initial identification of parameter values

When the cell is not proliferating, it is in a quiescent phase. Our GRBM must generate only two basins of attraction: a cyclic attractor that includes our canonical cell cycle (H_c), and a stable state corresponding to our quiescent phase (G_0). The transition from one basin of attraction to another is controlled by growth factors in the extracellular environment, that is, by the modeller, who projects into one basin or the other by fixing the value of gf , an environment variable. We

will return to this in detail below. Figure 5.19 illustrates the two attractors formalised from the variables and thresholds of our GRBM.

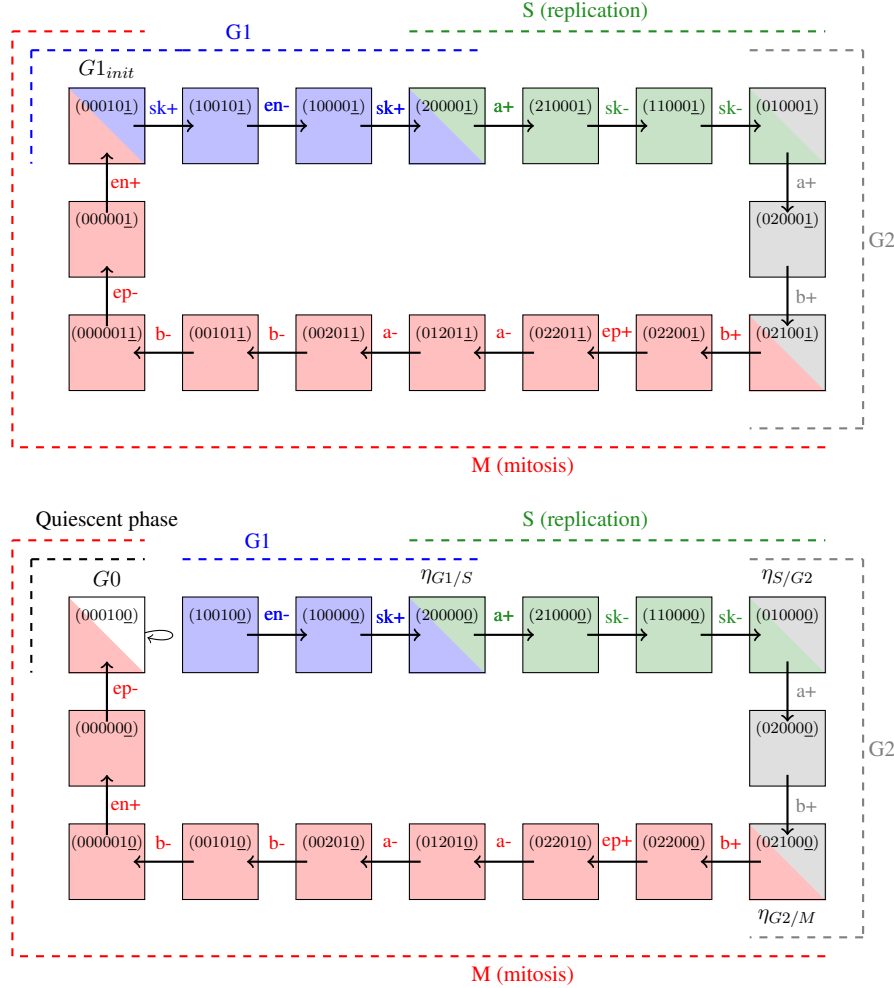


Figure 5.19: **Two attractors associated with our GRBM.** TotemBioNet will make it possible to detect the presence of a unique attractor in the context $AG(gf = 1)$ (resp. $AG(gf = 0)$) using formula 5.8 (resp. formula 5.5). The canonical cell cycle H_c belongs to the first attractor (upper subfigure). The state $G0$ is stable and characterises the second attractor (lower subfigure). Once the restriction point has been passed (first $sk+$), the G1, S, G2 and M phases occur regardless of the growth factor concentration.

5.5.1 Verification of the stability of the quiescent phase

The quiescent phase is formalised by the formula $G0$ in section 5.3, page 84. All variables are instantiated, so this formula specifies a state. The stability of this state is easily formalised in CTL; see chapter 4, page 66. The stability of $G0$ is defined using the modalities of fair CTL:

$$AG(gf = 0) \Rightarrow AF(AG(G0)) \quad (5.5)$$

This formula means that, from any state associated with our GRBM and belonging to the hyperplane $gf = 0$, all paths will lead to the quiescent phase.

At least one of these paths must satisfy the succession of G1, S, G2 and M phases. Indeed, once the restriction point has been passed, a cell deprived of growth factors carries out the replication and mitosis events of the current cycle. Passage through the restriction point is associated here with the event $sk+$ starting from $G1_{init}$. If growth factors are removed after passage through the restriction point, the initial G1 state projected onto the hyperplane $gf = 0$ becomes:

$$G1'_{init} \equiv sk = 0 \wedge a = 0 \wedge b = 0 \wedge en = 1 \wedge ep = 0 \wedge gf = 1 \quad (5.6)$$

and the cell cycle projected onto the hyperplane $gf = 0$ becomes:

$$H'_c \equiv \{G1'_{init}\} \underbrace{\begin{array}{c} \text{en-; sk+; a+; sk-; sk-; a+; b+; b+; ep+; a-; a-; b-; b-; ep-; en+} \\ \text{G1} \quad \text{S} \quad \text{G2} \quad \text{M} \end{array}}_{p'_c} \{G0\} \quad (5.7)$$

5.5.2 Formalisation of a canonical cell cycle as a cyclic attractor

Our canonical cell cycle (H_c) is assumed to form part of a unique attractor belonging to the hyperplane $gf = 1$. The states traversed along the path p_c from state $G1_{init}$ must therefore be traversed infinitely many times. We define the following formula using the modalities of fair CTL:

$$AG(gf = 1) \Rightarrow AG[AF(G1_{init})] \quad (5.8)$$

5.5.3 Asking the right questions about checkpoints using TotemBioNet

Without using any manually constrained parameter values, TotemBioNet reduces the model search space to 649,600 models by solving the following almost instantaneously:

Snoussi's condition (definition 8, page 30),

the weakest precondition computed from H_c (see $wp(H_c)$ below),

the one computed from H'_c (see $wp(H'_c)$ below).

The model-checking algorithm of TotemBioNet requires 4 h 30 min[†] to identify the parameterisations of our GRBM that satisfy the property of arrest in quiescence of a cell constantly deprived of growth factors (formula 5.5) and that of infinite division of a cell constantly infused with growth factors (formula 5.8). We also note that, in embryonic stem cells, E2F and the cycE/cdk2 complex (targets of gf) are continuously active independently of the cell cycle [28]. This theoretically allows them to perform the cell cycle indefinitely, excluding phenomena such as senescence. The efficiency of TotemBioNet allows us to focus even more rapidly on the checkpoint modelling question by additionally exploiting constraints on parameter values defined manually in the next section. All constraints on the parameters of our GRBM are listed in appendix A, page 157. These

[†]CPU times were measured on an Intel Core i7-8650U 1.90GHz $\times$ 8 processor running Ubuntu 20.04.4 LTS.

allow identification, in less than one second (858 ms, to be precise), of a set of **16 parameterisations** satisfying all the dynamic properties that have become essential in validating cell cycle models (table 5.2).

The precondition computed from the GRBM and the Hoare triple H_c is:

$$\begin{aligned}
 & wp(H_c) \\
 & \equiv \\
 & \underbrace{sk = 0 \wedge a = 0 \wedge b = 0 \wedge ep = 0 \wedge en = 1 \wedge gf = 1}_{G1_{init}} \wedge \\
 & \underbrace{K_{sk,\{no_a,restrict_gf\}} > 0}_{sk+} \wedge \underbrace{K_{en,\{no_a,no_b\}} < 1}_{en-} \wedge \underbrace{K_{sk,\{no_a,restrict_gf,rp,no_en\}} > 1}_{sk+} \wedge \\
 & \underbrace{K_{a,\{sk2,no_CI\}} > 0}_{a+} \wedge \underbrace{K_{sk,\{rp,no_en,restrict_gf\}} < 1}_{sk-; sk-} \wedge \\
 & \underbrace{K_{a,\{end_rep,no_CI\}} > 1}_{a+} \wedge \underbrace{K_{b,\{a2,no_CI\}} > 0}_{b+} \wedge \\
 & \underbrace{K_{b,\{a2,no_Wee1,no_CI\}} > 1}_{b+} \wedge \underbrace{K_{ep,\{b2,no_en\}} > 0}_{ep+} \wedge \underbrace{K_{a,\{end_rep\}} < 1}_{a-; a-} \wedge \\
 & \underbrace{K_{b,\{no_Wee1\}} < 1}_{b-; b-} \wedge \underbrace{K_{en,\{no_a,no_sk,no_b\}} > 0}_{en+} \wedge \underbrace{K_{ep,\emptyset} < 1}_{ep-}
 \end{aligned}$$

The one computed from H'_c is:

$$\begin{aligned}
 & wp(H'_c) \\
 & \equiv \\
 & \underbrace{sk = 1 \wedge a = 0 \wedge b = 0 \wedge ep = 0 \wedge en = 1 \wedge gf = 0}_{G1'_{init}} \wedge \\
 & \underbrace{K_{en,\{no_a,no_b\}} < 1}_{en-} \wedge \underbrace{K_{sk,\{no_a,rp,no_en\}} > 1}_{sk+} \wedge \\
 & \underbrace{K_{a,\{sk2,no_CI\}} > 0}_{a+} \wedge \underbrace{K_{sk,\{rp,no_en\}} < 1}_{sk-; sk-} \wedge \\
 & \underbrace{K_{a,\{end_rep,no_CI\}} > 1}_{a+} \wedge \underbrace{K_{b,\{a2,no_CI\}} > 0}_{b+} \wedge \\
 & \underbrace{K_{b,\{a2,no_Wee1,no_CI\}} > 1}_{b+} \wedge \underbrace{K_{ep,\{b2,no_en\}} > 0}_{ep+} \wedge \underbrace{K_{a,\{end_rep\}} < 1}_{a-; a-} \wedge \\
 & \underbrace{K_{b,\{no_Wee1\}} < 1}_{b-; b-} \wedge \underbrace{K_{en,\{no_a,no_sk,no_b\}} > 0}_{en+} \wedge \underbrace{K_{ep,\emptyset} < 1}_{ep-}
 \end{aligned}$$

All the results presented are available at the following address: <https://gitlab.com/deborahboyenval/temporal-separation-prototype/-/tree/master/ProofOfConceptCheckpoint>.

To be biologically plausible, our set of 16 models must necessarily satisfy preservation of DNA integrity throughout the cell cycle. This preservation property presents a formalisation challenge, since it first requires a convincing model of the cell cycle phases. Our triple H_c captures all the observations that form a consensus on the sequence of activations and inactivations of

the main regulators of replication and mitosis. However, the question of *formalising the phases* remains unresolved, since this choice of sequence makes too many *a priori* assumptions about the order of events within a phase. Nevertheless, TotemBioNet is highly effective at detecting the unsatisfiability of a canonical cell cycle trace, and this would suffice, at this modelling stage, to call into question the definition of the events required within each phase. We cannot, however, conclude that the phases have been formalised, but only that an initial verification of the biological consistency of our GRBM has been achieved.

Dynamic property	Formalisation	Logic
Validation of the canonical cell cycle	$\{G1_{init}\} p_c \{G1_{init}\}$ - (5.4)	LHGM
A cell that has passed its restriction point completes the cell cycle even when deprived of growth factors	$\{G1'_{init}\} p'_c \{G0\}$ - (5.7)	LHGM
Arrest in the quiescent phase	$AG(gf = 0) \Rightarrow AF[AG(G0)]$ - (5.5)	FAIR-CTL
The canonical cell cycle belongs to a cyclic attractor	$AG(gf = 1) \Rightarrow AG[AF(G1_{init})]$ - (5.8)	FAIR-CTL

Table 5.2: **The cell cycle properties verified before addressing the checkpoint modelling question.** LHGM stands for “genetically modified Hoare logic ” and FAIR-CTL for the fair version of CTL.

The approach to formal verification of *Thomas-style* models, presented in chapter 4 and illustrated in this chapter, assumes that we know how to formalise biologists’ research hypotheses. The formalisation of phases encounters a difficulty with this principle, as explained in chapter 3. The set of 16 models resulting from this chapter provides solid starting material for testing a formalisation of the domains of cell cycle phases from required events (chapter 6), and for testing a formalisation of intrinsic checkpoints that constrains the events that can occur within each phase domain (chapter 7).

5.5.4 Manual identification of certain parameters

The parameters $K_{v,\omega}$ define a context in which a set ω of resources of a biological variable v is fixed. We have translated the accumulated knowledge of the cell cycle into constraints on the values of certain parameters $K_{v,\omega}$. These constraints are obtained from a thought experiment (section 2.4.2.4, page 31) and may, in the best case, be corroborated by one or more mutually consistent experimental observations.

The variable en : APC-cdh1 is active only when cdk1 and cdk2 are inactive, that is, when sk , a and b are inactive [57]. Using the terminology of Boolean formalisms, we could define the following activation function for en :

$$f_{en} : sk = 0 \wedge a = 0 \wedge b = 0$$

This activation rule for en allows us to identify all the parameters of en , which is activated only when it has all three of its resources: namely, the absence of inhibition by sk (no_sk), a (no_a)

and $b(no_b)$.

$$\begin{aligned}
K_{en,\emptyset} &= 0 \\
K_{en,\{no_sk\}} &= 0 \\
K_{en,\{no_a\}} &= 0 \\
K_{en,\{no_b\}} &= 0 \\
K_{en,\{no_sk,no_a\}} &= 0 \\
K_{en,\{no_sk,no_b\}} &= 0 \\
K_{en,\{no_a,no_b\}} &= 0 \\
K_{en,\{no_sk,no_a,no_b\}} &= 1
\end{aligned}$$

This parameter set is simply the truth table of conjunction.

The variable sk : Activation of the $cycA/cdk1$ complex ($a = 2$) prevents sk from acting on the activation of pre-replication complexes ($sk = 2$) [57]. We further assume that activation of a at threshold 1 is decisive in repressing sk activity, to the extent that the absence of this resource prevents sk from acting on its targets, regardless of the value of the environment variable gf [102].

$$\forall \omega \mid \{no_a\} \notin \omega, \quad K_{sk,\omega} = 0$$

and:

$$\begin{aligned}
K_{sk,\emptyset} &= 0 \\
K_{sk,\{restrict_gf\}} &= 0 \\
K_{sk,\{rp\}} &= 0 \\
K_{sk,\{no_en\}} &= 0 \\
K_{sk,\{restrict_gf,rp\}} &= 0 \\
K_{sk,\{restrict_gf,no_en\}} &= 0 \\
K_{sk,\{rp,no_en\}} &= 0 \\
K_{sk,\{rp,restrict_gf,no_en\}} &= 0
\end{aligned}$$

There is another configuration in which sk cannot act on its targets: when none of its activators is active. sk is activated by growth factors (resource $restrict_gf$) and by itself at threshold 1 (resource rp):

$$\forall \omega \mid \{restrict_gf,rp\} \notin \omega, \quad K_{sk,\omega} = 0$$

and:

$$\begin{aligned}
K_{sk,\emptyset} &= 0 \\
K_{sk,\{no_en\}} &= 0 \\
K_{sk,\{no_a\}} &= 0 \\
K_{sk,\{no_en,no_a\}} &= 0
\end{aligned}$$

Next, sk cannot become fully activated as long as the p21 protein is active ($en = 1$):

$$\forall \omega \mid \{no_en\} \notin \omega, \quad K_{sk,\omega} < 2$$

Consequently:

$$\begin{aligned}
 K_{sk,\emptyset} &< 2 \\
 K_{sk,\{no_a\}} &< 2 \\
 K_{sk,\{rp\}} &< 2 \\
 K_{sk,\{restrict_gf\}} &< 2 \\
 K_{sk,\{no_a,rp\}} &< 2 \\
 K_{sk,\{no_a,restrict_gf\}} &< 2 \\
 K_{sk,\{rp,restrict_gf\}} &< 2 \\
 K_{sk,\{no_a,rp,restrict_gf\}} &< 2
 \end{aligned}$$

The concentration of sk cannot reach saturation when passage through the restriction point ($sk+$) has not occurred [102]:

$$\forall \omega \mid \{rp\} \notin \omega, \quad K_{sk,\omega} < 2$$

$$\begin{aligned}
 K_{sk,\emptyset} &= < 2 \\
 K_{sk,\{restrict_gf\}} &< 2 \\
 K_{sk,\{no_en\}} &< 2 \\
 K_{sk,\{no_a\}} &< 2 \\
 K_{sk,\{restrict_gf,no_en\}} &< 2 \\
 K_{sk,\{restrict_gf,no_a\}} &< 2 \\
 K_{sk,\{no_en,no_a\}} &< 2 \\
 K_{sk,\{restrict_gf,no_en,no_a\}} &< 2
 \end{aligned}$$

The variable a : The end of replication, represented by the multiplex *end_rep* (figure 5.14), relieves repression of the activity of the *cycA/cdk1* complex. The resource *end_rep* is necessary for activation of the *cycA/cdk1* complex ($a = 2$); in other words, the absence of this resource does not allow a to become activated at threshold 2:

$$\begin{aligned}
 K_{a,\emptyset} &< 2 \\
 K_{a,\{restrict_sk\}} &< 2 \\
 K_{a,\{no_CI\}} &< 2 \\
 K_{a,\{sk2\}} &< 2 \\
 K_{a,\{restrict_sk,no_CI\}} &< 2 \\
 K_{a,\{restrict_sk,sk2\}} &< 2 \\
 K_{a,\{no_CI,sk2\}} &< 2 \\
 K_{a,\{restrict_sk,no_CI,sk2\}} &< 2
 \end{aligned}$$

The variable b : Partial activation of b by a is *necessary* for self-activation of b [44], which is interpreted as attraction of b towards the value 1 in this resource configuration:

$$\forall \omega \mid a2 \notin \omega, \quad K_{b,\omega} < 1$$

Consequently:

$$\begin{aligned}
 K_{b,\emptyset} &= 0 \\
 K_{b,\{no_Wee1\}} &= 0 \\
 K_{b,\{no_CI\}} &= 0 \\
 K_{b,\{no_Wee1,no_CI\}} &= 0
 \end{aligned}$$

Partial activation of b by a is *sufficient* for self-activation of b [44] :

$$\forall \omega \mid a2 \in \omega, \quad K_{b,\omega} \geq 1$$

Thus:

$$\begin{aligned} K_{b,\{a2,no_Wee1\}} &\geq 1 \\ K_{b,\{a2,no_CI\}} &\geq 1 \\ K_{b,\{a2,SAC\}} &\geq 1 \\ K_{b,\{a2,no_Wee1,no_CI\}} &\geq 1 \\ K_{b,\{a2,no_Wee1,SAC\}} &\geq 1 \end{aligned}$$

Next, PPA and PP2A (*en*) are phosphatases that counteract the action of b on mitotic substrates, modelled when b crosses its second modelling threshold [31].

$$K_{b,\omega} < 2, \quad \forall K_{b,\omega} \setminus K_{b,\{a2,SAC,no_Wee1\}} \mid no_CI \in \omega$$

Therefore:

$$\begin{aligned} K_{b,\emptyset} &< 2 \\ K_{b,\{a2\}} &< 2 \\ K_{b,\{no_Wee1\}} &< 2 \\ K_{b,\{a2,no_Wee1\}} &< 2 \\ K_{b,\{a2,SAC\}} &< 2 \end{aligned}$$

The exception for the parameter $K_{b,\{a2,SAC,no_Wee1\}}$ arises because the SAC prevents degradation of b before that of a during mitosis. When the SAC is triggered (multiplex *SAC* satisfied), a is fully active (multiplex *a2* satisfied), and b is saturated (multiplex *no_Wee1* satisfied), degradation of b is blocked. Consequently:

$$K_{b,\{a2,SAC,no_Wee1\}} = 2;$$

The variable ep : The APC protein can bind to *cdc20* only if it has previously been phosphorylated by the *cycB/cdk1* complex. APC-*cdc20* (ep) can act on its targets only (a necessary condition) when b is fully activated ($b = 2$) [49]. Consequently:

$$\forall \omega \mid b2 \notin \omega, \quad K_{ep,\omega} < 1$$

This allows us to constrain the following parameters:

$$\begin{aligned} K_{ep,\emptyset} &= 0 \\ K_{ep,\{no_en\}} &= 0 \end{aligned}$$

The variables a and b : As soon as the APC-*cdc20* variable is activated (ep), the activity of the complexes a and b is completely repressed, causing the values of these two variables to tend towards zero [31]. As soon as ep becomes active, the multiplex *no_CI* becomes false and does not belong to the resources of the variables concerned:

$$\forall \omega_a \mid no_CI \notin \omega_a, \quad K_{a,\omega_a} = 0 \quad \text{and} \quad \forall K_{b,\omega_b} \mid \{no_CI,a2\} \notin \omega_b, \quad K_{b,\omega_b} = 0$$

This constraint implies that activation of *en*, that is, APC-cdh1, p21, p27, PP1 and PP2A, also causes inactivation of a and b . This assumption is accepted for the variable a because it does not

contradict the constraints already defined on its parameter values; consequently:

$$\begin{aligned} K_{a,\emptyset} &= 0 \\ K_{a,\{end_rep\}} &= 0 \\ K_{a,\{sk2\}} &= 0 \\ K_{a,\{restrict_sk\}} &= 0 \\ K_{a,\{sk2,restrict_sk\}} &= 0 \end{aligned}$$

This assumption is not accepted for parameters of b that include the resource $a2$; see the paragraph discussing constraints on the values of b , page 106. Thus:

$$\begin{aligned} K_{b,\emptyset} &= 0 \\ K_{b,\{SAC\}} &= 0 \\ K_{b,\{no_Wee1\}} &= 0 \\ K_{b,\{no_Wee1,SAC\}} &= 0 \end{aligned}$$

CHAPTER 6

Formalisation of cell cycle phases

6.1	Preamble	111
6.2	Syntactic definition of a canonical phase	112
6.3	Formalising a phase from its canonical definition	113
6.3.1	Definition of the hyperrectangle of a Hoare triple	113
6.3.2	Towards formalising a phase as a first-order predicate	115
6.4	Application to our cell cycle model	116
6.4.1	The canonical phases of our model	116
6.4.2	The hyperrectangles of our canonical phases	118
6.4.3	The phases of our model	121
6.4.3.1	Implementation of the predicate <i>estAdmis_σ</i> in Prolog	121
6.4.3.2	Fully constrained event order for G1, S and G2 phases	122
6.4.3.3	The remarkable case of M phase	123

6.1 Preamble

In Chapter 5, the satisfiability of the classical temporal properties of the cell cycle was proved. They express reachability and stability properties that can be formalised in CTL or using genetically modified Hoare logic (LHGM) and verified formally without difficulty. However, the formal verification of cell cycle checkpoints cannot follow the same approach, since it presupposes that we *already* know how to formalise the underlying biological knowledge, such as cell cycle *phases*.

In the introduction, five cell cycle checkpoints were described biologically: passage through the restriction point, control of the initiation of replication (G1/S), its completion (S/G2), entry into mitosis (G2/M), and spindle assembly during M phase (SAC). Although these five checkpoints differ in the nature of the molecular events that occur and in their timing within the cell cycle, a generic mechanism emerges: *a checkpoint prevents any event initiating a given phase from occurring until all events of the preceding phase have been properly completed*. This statement explicitly highlights the need to define what a phase is before formalising a checkpoint. For example, the cell cycle was formalised by a Hoare triple in the preceding chapter, or by a terminal strongly connected component, to cite only [29, 88]. The definition of the cell cycle here typically follows that of a given model and has not been approached from a generic viewpoint independent of its specific formalisation. This chapter is therefore devoted to an initial generic formalisation of a phase that separates the concept itself from the proof of concept *via* the instantiation of a model.

First, a phase is defined canonically by a Hoare triple of the form $\{P\} p \{Q\}$, where p is a sequence of molecular events assumed to be complete, *cf.* Section 6.2. The *single* sequence chosen corresponds either to a canonical order of events, that is, the order with the broadest consensus in the relevant literature, or to a reasonable hypothesis formulated to answer the following question: *which events can define a cell cycle phase?* The second step consists in completely relaxing the constraint on the order of events within a phase by considering all permutations of p . This yields an *over-approximate* definition of a phase in Section 6.3.1. The final step aims to identify the permutations of p admitted within an RRBM *via* the computation of the weakest precondition of genetically modified Hoare logic, *cf.* Section 6.3.2. Finally, the proof of our phase concept is carried out starting from:

the set of 16 solution models obtained at the end of Chapter 5 (Section 5.5.3, page 5.5.3)

the specification of the canonical cell cycle H_c (Chapter 5, page 94).

6.2 Syntactic definition of a canonical phase

The cell cycle is described as a sequence of cellular events subdivided into G1, S, G2 and M phases. In Chapter 5.4, we defined a GRBM dedicated to representing cell cycle phases. We then chose to view the cell cycle, and by extension a phase, as a canonical sequence of regulatory events observed from an initial state. The *a priori* assumption underlying this choice is that, once the regulatory events associated with a phase have occurred, the cellular events characteristic of that phase must necessarily have taken place. Indeed, all thresholds of our variables describe the action of cell cycle regulators on the cellular events characteristic of a phase, cf. Section 5.4.2, page 91. Also in Chapter 5, the syntax of genetically modified Hoare logic proved suitable for representing these regulatory events. Recall that an event in LHGM has the form $v+$ or $v-$ and means that a variable crosses one of its regulatory thresholds. We propose to define a canonical phase as a particular case of the syntax of genetically modified Hoare logic LHGM, introduced by [65], which we call an *elementary Hoare triple*.

Definition 26: Elementary syntactic path

Given a GRBM $\mathcal{G}$, an *elementary syntactic path* p is an expression of the form $p = e_1; \dots; e_n$, where the e_i are events of $\mathcal{G}$, and its length is denoted by $|p| = n$.

Definition 27: Elementary Hoare triple

Given a GRBM $\mathcal{G}$, an *elementary triple* $H = \{P\} p \{Q\}$ is a Hoare triple such that:

- P defines a unique state associated with $\mathcal{G}$,
- p is an elementary syntactic path for $\mathcal{G}$,
- Q is the postcondition uniquely obtained from P by following the path p .

Notation 1: Canonical phase

A *canonical phase* π is characterised syntactically by the elementary Hoare triple $H_\pi = \{P_\pi\} p_\pi \{Q_\pi\}$, where $\pi \in \{G1, S, G2, M\}$ and p_π is the canonical path of the phase, that is, the sequence of molecular events with the broadest consensus in the literature or within a hypothesis to be tested.

Remark 4 – To move from a syntactic definition of a canonical phase of the form $H_\pi \equiv \{P_\pi\} p_\pi \{Q_\pi\}$ to its semantic definition, H_π must first be satisfiable:

$$\exists(\mathcal{G}, \sigma) \mid \forall \eta_{P_\pi} \models P_\pi, \sigma, \eta_{P_\pi} \models \text{wp}(p_\pi, Q_\pi)$$

A canonical phase is then characterised semantically by the union of the states traversed by p_π from its initial state η_{P_π} to its final state. In this case, the succession of states that semantically characterises π exists in the GTA derived from the model $(\mathcal{G}, \sigma)$; see 6.4.1 for an illustration.

Notation 2: Checkpoint state

Finally, consider two successive cell cycle phases, denoted respectively by π_i and π_{i+1} , with $H_{\pi_i} \equiv \{P_{\pi_i}\} p_{\pi_i} \{Q_{\pi_i}\}$ and $H_{\pi_{i+1}} \equiv \{P_{\pi_{i+1}}\} p_{\pi_{i+1}} \{Q_{\pi_{i+1}}\}$. The final state of π_i , which satisfies Q_{π_i} , also satisfies $P_{\pi_{i+1}}$ and is denoted by $\eta_{\pi_i/\pi_{i+1}}$. It is termed a *checkpoint state*, since it satisfies the completion of all events of the path p from the state satisfying P_{π_i} , if H_{π} is satisfiable.

Remark 5 – When each phase $\pi \in \{G1, S, G2, M\}$ is defined by an elementary Hoare triple, we deduce a syntactic and semantic definition of the canonical cell cycle, denoted by H_c . Its initial state is characterised by the precondition of G1, denoted by P_{G1} . Since it is a cycle, its postcondition is equivalent to its precondition, and the canonical path is defined so that the four phases follow one another in order from P_{G1} . The resulting elementary Hoare triple has the form: $H_c \equiv \{P_{G1}\} p_{G1}; p_S; p_{G2}; p_M \{P_{G1}\}$. Proving the validity of each canonical phase suffices to prove the validity of the canonical cell cycle H_c . Indeed, if H_{G1} , H_S , H_{G2} and H_M are valid, then H_c is valid because $p_{G1}; p_S; p_{G2}; p_M$ is a sequential composition in modified Hoare logic, cf. Definition 22.

The syntactic definitions of our *canonical phases* for the graph $\mathcal{G}_c$ are presented in Section 6.4.1 and mark the beginning of our proof of concept.

6.3 Formalising a phase from its canonical definition

6.3.1 Definition of the hyperrectangle of a Hoare triple

The definition of a canonical phase exhibits a multiset of events that must occur during a phase π before the beginning of the next phase π_{i+1} . This multiset associates each possible event of the form $v+$ or $v-$ with its number of occurrences in π . When no constraint on the order of events within π is known, we propose to define an *over-approximation* of a phase by the *hyperrectangle* associated with its elementary Hoare triple.

Definition 28: Hyperrectangle of an elementary Hoare triple

Given an elementary Hoare triple $H \equiv \{P\} p \{Q\}$ in a GRBM $\mathcal{G} = (V, M, E)$ and a variable v of $\mathcal{G}$, the lower (resp. upper) bound of v for H is defined as follows:

let $\sharp_v^-$ (resp. $\sharp_v^+$) be the number of occurrences of the event $v-$ (resp. $v+$) in p ,

let η be the initial state characterised by the formula P ; the lower (resp. upper) bound of v for H is, by definition, $\min_v^H = \max(0, \eta(v) - \sharp_v^-)$ (resp. $\max_v^H = \min(b_v, \eta(v) + \sharp_v^+)$).

The hyperrectangle of H is then the subset of ζ corresponding to the Cartesian product $h(H) = \prod_{v \in V} [\min_v^H, \max_v^H]$, and its characteristic property is denoted by $\psi_H \equiv \bigwedge_{v \in V} (v \geq \min_v^H \wedge v \leq \max_v^H)$.

As an example, consider the following elementary Hoare triple:

$$H_\pi \equiv \{x = 0 \wedge y = 0 \wedge z = 0\} x+; y+; x-; y+ \{x = 0 \wedge y = 2 \wedge z = 0\}$$

which defines a phase π from the toy influence graph in Figure 6.1. Counting the events belonging

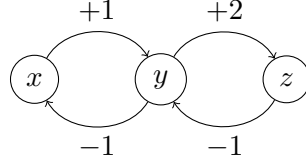
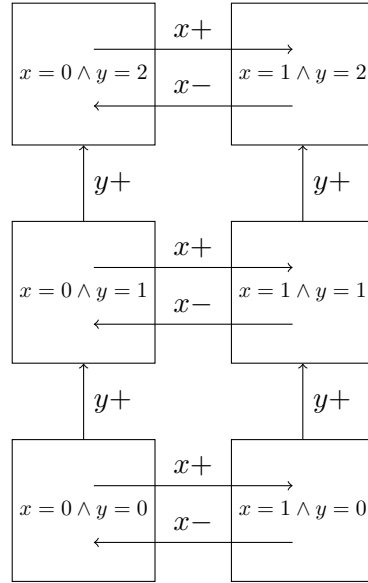


Figure 6.1: **An influence graph representing the events of a phase π .** Only the two variables x and y matter for representing π , and z is considered to matter for another phase.

to the path of H_π yields the multiset associated with π : $\{x+ : 1, x- : 1, y+ : 2\}$. The variable z does not matter in defining the domain of π , since no event involving z is observed in p_π . Thus $\min_z^{H_\pi} = \max_z^{H_\pi}$. The variables x and y range respectively from $\min_x^{H_\pi} = 0$ to $\max_x^{H_\pi} = 1$ and from $\min_y^{H_\pi} = 0$ to $\max_y^{H_\pi} = 2$, so the characteristic formula for the domain of π is $\psi_H = (x \geq 0 \wedge x \leq 1) \wedge (y \geq 0 \wedge y \leq 2) \wedge (z = 0)$. The hyperrectangle is contained here in the hyperplane ($z = 0$). The hyperrectangle $h(H)$ exhibits all permutations of p :



where all events are drawn without replacement from the multiset associated with π . The over-approximate definitions of G1, S, G2 and M phases are applied to our cell cycle model in Section 6.4.2.

To sum up: given a cell cycle phase π , it is conventionally considered that only the set of events composing it matters. Within this framework, all permutations of its events that remain within the bounds must be considered *a priori*. Consequently, $h(H_\pi)$ is the set of states that this

phase could *potentially* cover. The hyperrectangle of a phase can be represented by a GTA in which the states belonging to the set ζ are connected by transitions corresponding to the events that characterise π . To illustrate this approach, consider a toy example in which our canonical G2 phase, defined in the preceding chapter, exhibits an oscillation of the variable en :

$$\begin{aligned} &\{sk = 0 \wedge a = 1 \wedge b = 0 \wedge ep = 0 \wedge en = 1 \wedge gf = 1\} \\ &\quad a+; \mathbf{en}-; b+; \mathbf{en}+ \\ &\{sk = 0 \wedge a = 2 \wedge b = 1 \wedge ep = 0 \wedge en = 1 \wedge gf = 1\} \end{aligned}$$

In this case, we observe that some permutations of events cause en to exceed its bounds: these are all permutations in which $en+$ precedes $en-$, since $en = 1$ (its maximum value) in the precondition. This contradicts the bounds axiom of genetically modified Hoare logic. A hyperrectangle admits no departure from the bounds, since a variable must remain within its defined domain.

6.3.2 Towards formalising a phase as a first-order predicate

The hyperrectangle $h(H_\pi)$ defines the phase π with a minimum of constraints. In other words, no *a priori* biological knowledge has been used to verify the feasibility of permutations of p_π . However, we already know that some biological knowledge does not admit every possible permutation. Some of them will not allow Q_π to be reached from P_π , because certain transitions are not admitted by a parameterisation σ . For example, the preceding chapter shows that our solution set of cell cycle models admits only $sk+$ as an initiating event for G1 phase, given that its canonical phase is defined by the triple:

$$\begin{aligned} &\{sk = 0 \wedge a = 0 \wedge b = 0 \wedge ep = 0 \wedge en = 1 \wedge gf = 1\} \\ &\quad sk+; en-; sk+ \\ &\{sk = 2 \wedge a = 0 \wedge b = 0 \wedge ep = 0 \wedge en = 0 \wedge gf = 1\} \end{aligned}$$

Any permutation of the canonical path with $en-$ as its first event is biologically inadmissible. Conversely, there exist permutations of the canonical path of a phase that must necessarily be possible within our mathematical model, if only the canonical path itself.

For the formalisation of a phase to be neither too strongly nor too weakly constrained, it must be defined neither by a canonical phase $H_\pi \equiv \{P_\pi\}p_\pi\{Q_\pi\}$ nor by its hyperrectangle $h(H_\pi)$, but by the *biologically admissible* part of its hyperrectangle in light of the regulatory graph. The problem amounts to formally verifying the satisfiability of all permutations of p_π exhibited by the hyperrectangle H_π .

Definition 29: Predicate $estAdmis$

Given a model $\mathcal{R} = (\mathcal{G}, \sigma)$ and a canonical phase defined by an elementary triple H of the form $\{P\} p \{Q\}$, where P (resp. Q) is its precondition (resp. postcondition) and p its canonical path, any path p' is *admitted* by $\mathcal{R}$ if:

$$p' \in \text{permutations}(p) \quad \text{and} \quad \mathcal{I}(wp(p', Q)) \quad \text{where}$$

$\text{permutations}(p)$ is the set of permutations of p ,

$wp(p', Q)$ is the function computing the weakest precondition of genetically modified Hoare logic associated with p' and Q

$\mathcal{I}$ is the interpretation of this weakest precondition using σ and η_P :

$$\mathcal{I} : \mathcal{K} \cup \zeta \rightarrow \mathbb{N}, \quad \mathcal{I}|_{\mathcal{K}} = \sigma \quad \text{and} \quad \mathcal{I}|_{\zeta} = \eta_P$$

The predicate $estAdmis_{\sigma}(p', H)$ is introduced: it is true if p' is admitted by $\mathcal{R}$ from the state characterised by P in H .

Clearly, an RRBM $\mathcal{R} = (\mathcal{G}, \sigma)$ can be considered valid only if the canonical path of every phase belongs to the set of permutations admitted by $\mathcal{R}$. The latter abstracts a body of knowledge that we can formally *query*, using our predicate $estAdmis_{\sigma}(p', H)$, to answer the following request: *which permutations p' of the canonical path of H do not contradict the parameterisations σ ?* Recall that σ denotes any solution of a set of dynamic biological properties encoded, in the context of this thesis, either with CTL or with genetically modified Hoare logic. We consider the term *biologically admissible* to be equivalent to *admissible by a set of models* $(\mathcal{G}, \sigma)$.

Remark 6 – The two preceding definitions allow us to specify the domain of a phase within the state space more precisely. It is the union of the states reachable by permutations of p_{π} admitted by $\mathcal{R}$. Naturally, the domain of a phase depends on the parameterisation σ , and recall that the set of models solving biological properties, when nonempty, rarely reduces to a singleton. Ultimately, we can define a superset that is *a priori* more precise than the hyperrectangle by taking the union of the domains of phase π for all parameterisations to be considered. The resulting set is called the *biologically admitted part* of the hyperrectangle of π .

The parts of the hyperrectangles of our four cell cycle phases that are biologically admitted by the 16 parameterisations σ are computed in Section 6.4.3.

6.4 Application to our cell cycle model

6.4.1 The canonical phases of our model

A canonical sequence of cell cycle events was defined from the state of a cell at the beginning of G1. Recall from page 94:

$$G1_{init} \equiv (sk = 0 \wedge a = 0 \wedge b = 0 \wedge ep = 0 \wedge en = 1 \wedge gf = 1)$$

The initial state of the cell cycle is the initial state of G1, and $G1_{init}$ is renamed P_{G1} in the context of formalising *phases*. An elementary Hoare triple was used to define the canonical cell cycle, from which canonical phases are deduced.

$$H_c \equiv \{P_{G1}\} \underbrace{sk+; en-; sk+; a+; sk-; sk-; a+; b+}_{p_{G1}} \underbrace{b+; ep+; a-; a-; b-; b-; ep-; en+}_{p_M} \{P_{G1}\}$$

$\underbrace{\hspace{15em}}_{p_c}$

Given the events required in each phase, the path p_C was decomposed into four elementary paths: $p_C = p_{G1}; p_S; p_{G2}; p_M$. The expressions for p_{G1} , p_S , p_{G2} and p_M are given in Table 6.1. Each of these four paths is a subsequence of the canonical path p_c and is therefore also canonical. The initial states of S, G2 and M can be deduced from P_{G1} together with p_{G1} , p_S and p_{G2} .

P_{π_i}	$P_{G1} \equiv (sk = 0 \wedge a = 0 \wedge b = 0 \wedge ep = 0 \wedge en = 1 \wedge gf = 1)$ $P_S \equiv (sk = 2 \wedge a = 0 \wedge b = 0 \wedge ep = 0 \wedge en = 0 \wedge gf = 1)$ $P_{G2} \equiv (sk = 0 \wedge a = 1 \wedge b = 0 \wedge ep = 0 \wedge en = 0 \wedge gf = 1)$ $P_M \equiv (sk = 0 \wedge a = 2 \wedge b = 1 \wedge ep = 0 \wedge en = 0 \wedge gf = 1)$
p_{π_i}	$p_{G1} \equiv sk+; en-; sk+$ $p_S \equiv a+; sk-; sk-$ $p_{G2} \equiv a+; b+$ $p_M \equiv b+; ep+; a-; a-; b-; b-; en+; ep-$
H_{π_i}	$H_{G1} : \{P_{G1}\} p_{G1} \{P_S\}$ $H_S : \{P_S\} p_S \{P_{G2}\}$ $H_{G2} : \{P_{G2}\} p_{G2} \{P_M\}$ $H_M : \{P_M\} p_M \{P_{G1}\}$

Table 6.1: The canonical phases defined using the GRBM $\mathcal{G}_c$ in Figure 5.18 and H_c .

From now on, we denote by $\eta_{M/G1}$, $\eta_{G1/S}$, $\eta_{S/G2}$ and $\eta_{G2/M}$ the initial states of the phases characterised respectively by P_{G1} , P_S , P_{G2} and P_M . Finally, the Hoare triples H_{G1} , H_S , H_{G2} and H_M define respectively the canonical phases G1, S, G1 and M associated with our GRBM, which we denote by $\mathcal{G}_c$. Each phase $\pi_i \in \{G1, S, G2, M\}$ is currently defined by an elementary Hoare triple of the form $\{H_{\pi_i} \equiv P_{\pi_i}\} p_{\pi_i} \{Q_{\pi_i}\}$. The postcondition Q_{π_i} equals the precondition of the next phase, denoted by $P_{\pi_{i+1}}$, allowing the phases to be connected through a checkpoint state in the sense of Notation 2.

Thus far, canonical phases have been defined solely in syntactic terms. The triple H_c underlying these syntactic definitions was proved valid in the preceding chapter. To define our canonical phases semantically, the validity of H_{π_i} must be proved for every $\pi_i \in \{G1, S, G2, M\}$.

Proposition 1 – The correctness of H_C implies the correctness of the four triples H_{G1} , H_S , H_{G2} and H_M .

Proof.

If H_C is correct, then every prefix of p_C is a feasible path for $\mathcal{R}_C$ from η_{G1} , in particular p_{G1} . Thus, starting from η_{G1} and following p_{G1} necessarily leads to η_S , which by definition implies

that the triple $\{P_{G1}\} p_{G1} \{P_S\}$ is correct. Similarly, the triple $H' = \{P_S\} p_S; p_{G2}; p_M \{P_{G1}\}$ is correct. The same argument applies from H' and proves the correctness of H_S as well as that of $H'' = \{P_{G2}\} p_{G2}; p_M \{P_{G1}\}$. Finally, an immediate iteration yields the correctness of H_{G2} and H_M .

□

Remark 7 – In Section 5, we mentioned that if H_{G1} , H_S , H_{G2} and H_M are correct, then applying the sequential composition rule of Hoare logic proves H_C correct. The correctness of H_C is therefore equivalent to the correctness of the four triples H_{G1} , H_S , H_{G2} and H_M .

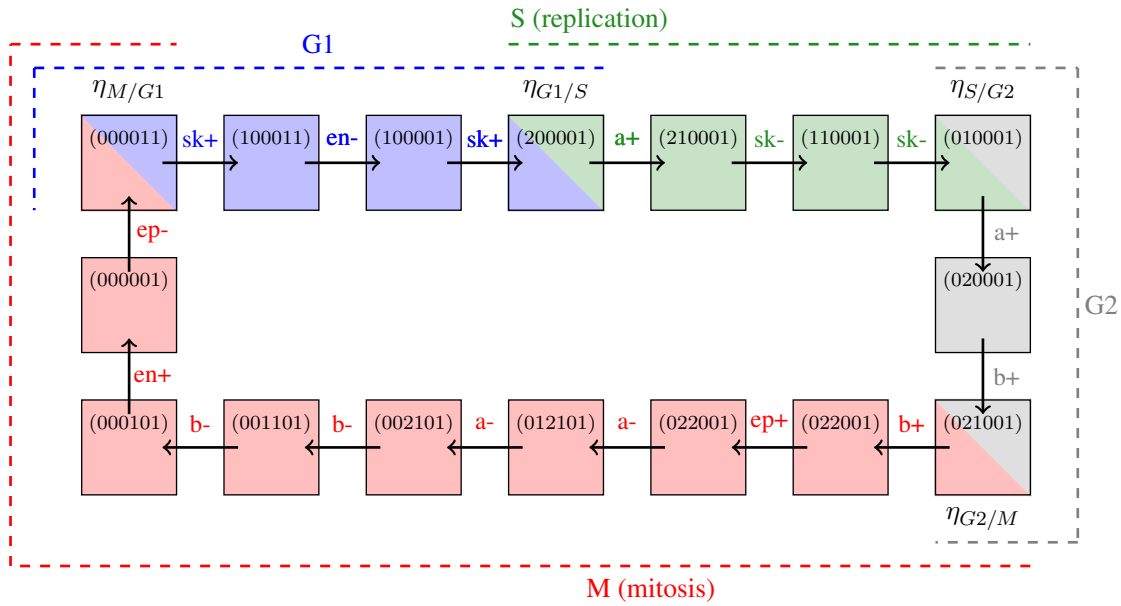


Figure 6.2: **The canonical phases H_π associated with $\mathcal{G}_C$ for every $\pi \in \{G1, S, G2, M\}$.** For clarity, a state is characterised here by a vector of the form: $(\eta(sk) \ \eta(a) \ \eta(b) \ \eta(ep) \ \eta(en) \ \eta(gf))$. For example, (000011) is equivalent to P_{G1} . States shown in two colours are the checkpoint states of our cell cycle in the sense of Notation 2.

6.4.2 The hyperrectangles of our canonical phases

The hyperrectangles of our canonical phases are as follows:

G1 phase. The hyperrectangle $h(H_{G1})$ is contained in the intersection of the hyperplanes $(a = 0 \wedge b = 0 \wedge ep = 0)$ and $(gf = 1)$, while sk ranges from 0 to 2 and en from 1 to 0. The initial state of G1 is the corner $(sk = 0 \wedge en = 1)$ of $h(H_{G1})$, and its final state $\eta_{G1/S}$ corresponds to the opposite corner $(sk = 2 \wedge en = 0)$. The characteristic property of $h(H_{G1})$ is therefore defined by:

$$\psi_{H_{G1}} \equiv (sk \geq 0 \wedge sk \leq 2) \wedge (en \geq 0 \wedge en \leq 1) \wedge (a = 0 \wedge b = 0 \wedge ep = 0 \wedge gf = 1)$$

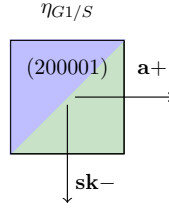


Figure 6.3: **The checkpoint state of our canonical G1 phase.** The final state of G1 phase (coloured blue) is also the initial state of S phase (coloured green), and is denoted by $\eta_{G1/S}$. This state verifies that all events of $G1$ have indeed occurred from its initial state, namely two $sk+$ events and one $en-$ event in this case. Two transitions are *a priori* possible to initiate S phase, according to $h(H_S)$.

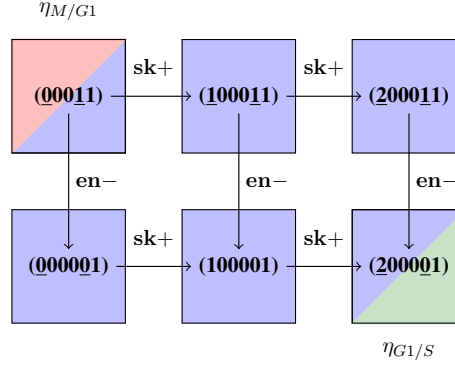


Figure 6.4: **The hyperrectangle of G1 phase, defined canonically by H_{G1} .** As in Figure 6.2, a state is represented by the vector of discrete states: $(\eta(sk) \eta(a) \eta(b) \eta(ep) \eta(en) \eta(gf))$.

All permutations of p_{G1} must be considered, since none causes a variable to exceed its bounds. The states belonging to the hyperrectangle $h(H_{G1})$ and the transitions between these states are represented graphically in Figure 6.4. The checkpoint state $\eta_{G1/S}$ is shown in Figure 6.3.

S phase. The hyperrectangle $h(H_S)$ is contained in the intersection of the hyperplanes $(b = 0 \wedge ep = 0 \wedge en = 0)$ and $(gf = 1)$, while sk now ranges from 2 to 0 and a from 0 to 1. The initial state of S is the corner $(sk = 2 \wedge a = 0)$ of $h(H_S)$, and its final state $\eta_{S/G2}$ corresponds to the opposite corner $(sk = 0 \wedge a = 1)$. All permutations of p_S are exhibited by the hyperrectangle $h(H_S)$ in Figure 6.5. The characteristic property of $h(H_S)$ is therefore defined by:

$$\psi_{H_S} \equiv (a \geq 0 \wedge a \leq 1) \wedge (sk \geq 0 \wedge sk \leq 2) \wedge (b = 0 \wedge ep = 0 \wedge en = 0 \wedge gf = 1)$$

Remark 8 – The domains of our canonical phases are not mutually exclusive, since the final state of a phase π is also the initial state of phase π_{i+1} . Excluding this case, their domains would all be disjoint, cf. Figure 6.2. Naturally, the states belonging to the intersection of the two hyperrectangles include the state characterising both the final state of G1 and the initial state of S. The intersection

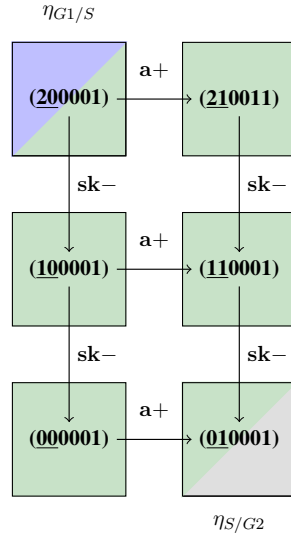


Figure 6.5: The hyperrectangle of S phase, defined canonically by the triple H_S (cf. Table 6.1).

of the two hyperrectangles also includes the state characterised by (000001) in Figures 6.4 and 6.5.

G2 phase. The hyperrectangle $h(H_{G2})$ is contained in the intersection of the hyperplanes ($sk = 0 \wedge ep = 0 \wedge en = 0$) and ($gf = 1$), while only the variables a and b vary: a from 1 to 2 and b from 0 to 1. The initial state η_{G2} is the corner ($a = 1, b = 0$) of $h(H_{G2})$, and the final state $\eta_{G2/M}$ is the opposite corner ($a = 2, b = 1$). All permutations of p_{G2} must be considered in the hyperrectangle $h(H_{G2})$ shown in Figure 6.6. The characteristic property of $h(H_{G2})$ is defined by:

$$\psi_{H_{G2}} \equiv (a \geq 1 \wedge a \leq 2) \wedge (b \geq 0 \wedge b \leq 1) \wedge (sk = 0 \wedge ep = 0 \wedge en = 0 \wedge gf = 1)$$

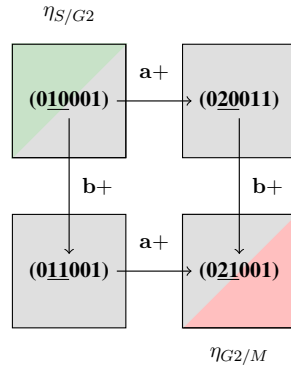


Figure 6.6: The hyperrectangle $h(H_{G2})$ of G2 phase, defined canonically by the triple H_{G2} (cf. Table 6.1).

M phase: The hyperrectangle $h(H_M)$ is contained in the intersection of the hyperplanes ($sk = 0, gf = 1$) and ($gf = 1$), while a ranges from 2 to 0, ep from 0 to 1 and then from 1 to 0, en from 0 to 1, and b potentially between 0 and 2, starting from $b = 1$. The initial state η_M , where $b = 1$, is therefore not a corner of $h(H_M)$. Depending on the permutation of events, b may vary only between 0 and 1. This is the case for every permutation including ($b-; b+; b-$), regardless of how the other events of M are interleaved. The final state is a corner of $h(H_M)$ ($a = 0, b = 0, ep = 0, en = 1$), namely η_{G1} , the initial state of G1 phase. Not every permutation of p_M must be considered. Those allowing the variables b and ep to exceed their bounds must not be taken into account. From the initial state characterised by (021001) (cf. Figure 6.2), $b+$ must occur before the second $b-$, and $ep+$ before $ep-$. The characteristic property of $h(H_M)$ is defined by:

$$\psi_{H_M} \equiv (a \leq 2 \wedge a \geq 0) \wedge (b \leq 2 \wedge b \geq 0) \wedge (en \leq 1 \wedge b \geq 0) \wedge (ep \leq 2 \wedge b \geq 0) \wedge (sk = 0 \wedge gf = 1)$$

6.4.3 The phases of our model

6.4.3.1 Implementation of the predicate $estAdmis_\sigma$ in Prolog

In practice, defining the part of $h(H_{G1})$, $h(H_S)$, $h(H_{G2})$ and $h(H_M)$ admitted by the models (G_c, σ) amounts to identifying the solutions of the predicate $estAdmis_\sigma(p'_\pi, H_\pi)$ for every $\pi \in \{G1, S, G2, M\}$. The logic programming language Prolog was chosen because $estAdmis$ relies on declaring a Horn clause that directly expresses the formula in Definition 29. Prolog provides a resolution strategy for Horn clauses within its architecture. Declaring the problem is the only concern when designing a Prolog program.

The following propositional logic formula is a Horn clause:

$$p' \in \text{permutations}(p) \text{ and } \mathcal{I}(wp(p', Q)) \implies estAdmis_\sigma(p', H)$$

where, recall, H is an elementary Hoare triple formed by P , Q and p , respectively a precondition, a postcondition and a canonical path. This formula is then rewritten directly as a Prolog predicate:

$$estAdmis(H, \mathcal{G}, \Sigma, p') : - \text{permutation}(p, p'), wp(p', Q, \Sigma).$$

This predicate takes as input H , a GRBM denoted by $\mathcal{G}^*$, and Σ , the set of parameterisations σ satisfying biological constraints ($\phi_{K_{v,\omega}}$ in Chapter 2, page 34, to give just one example). Note that σ is a term hidden within $\mathcal{I}$ (cf. Definition 29). The solutions p' are computed using the `findall` instruction of SWI-Prolog [105]:

$$\text{findall}([p'], estAdmis(H, \mathcal{G}, \Sigma, p')).$$

Solving the weakest precondition (wp) is delegated to the HoareFol tool [90], integrated into TotemBioNet [97]. The triple $\{P\} p' \{Q\}$ is proved correct for the parameterisation σ if:

$$\forall \eta_P \models P, (\sigma, \eta_P) \models wp(p', Q)$$

More precisely, the predicate wp consists of:

*Implicitly admitted by the context of Definition 29 on page 116.

a predicate making a system call to TotemBioNet, with its input file as an argument, consisting of:

1. a block in which the environment variables are instantiated[†],
2. a block in which the systemic variables and their domains are defined,
3. a block in which the multiplex formulae and regulatory edges are declared,
4. a block in which constraints on parameter values are declared (the conjunction of these constraints characterises Σ),
5. and a block in which the triple $\{P\} p' \{Q\}$ is declared.

a predicate that parses the output file of TotemBioNet to identify the number of OK-models (cf. Figure 4.7, page 74), denoted by n .

a predicate that verifies that $n > 0$.

We will not provide further details of the input data types or other implementation subtleties. For more information, readers should refer to: <https://gitlab.com/totembionet/totembionet>. In Prolog, proving the satisfiability of the predicate *estAdmis* amounts to evaluating every possible value of p' . If the canonical path p contains n events, then the time complexity is $\Theta(n!)$. It is therefore essential not to count repetitions of the same event. To address this, the predicate *permutation*(p) is replaced by a new predicate: *permutationNR*(p). The latter eliminates repetitions. The complexity becomes $\Theta(\frac{n!}{k_1! \dots k_n!})$, where the k_i represent the number of occurrences of an event. G1, S and G2 phases are associated with very few events, some of which are repeated. Consequently, results are obtained almost instantaneously (a few seconds on average). For M phase, the average time is 1 h 12 min[‡].

6.4.3.2 Fully constrained event order for G1, S and G2 phases

This section focuses exclusively on applying the predicate *estAdmis* $_{\sigma}(p'_{\pi}, H_{\pi})$ for $\pi \in \{G1, S, G2\}$. M phase yields results that differ from the others and will therefore be addressed in a dedicated section. The values of the parameter H_{π} for our canonical phases were defined above (Section 6.1). For these three phases, the biological knowledge accumulated in the 16 instances of $(\mathcal{G}_c, \sigma)$ admits only a single permutation of the canonical path: the identity permutation. The latter is necessarily admitted according to Proposition 1. No other permutation is satisfiable because at least one of its events is not feasible. These permutations therefore do not reflect a biologically credible sequence of events.

We take G1 phase as an example, but the same reasoning applies to the next two phases. The paths $p'_{G1} = en-; sk+; sk+$ and $p''_{G1} = sk+; sk+; en-$ are biologically inadmissible because $\mathcal{I}(wp(p'_{G1}, P_S)) \vee \mathcal{I}(wp(p''_{G1}, P_S))$ is unsatisfiable. The first event of p'_{G1} ($en-$) is feasible from the precondition P_{G1} if and only if $\sigma(K_{en, \{no_sk, no_a, no_b\}}) < 1$, since the variable en must be attracted towards a value strictly smaller than its current value. This constraint contradicts the parameterisation $\sigma(K_{en, \{no_sk, no_a, no_b\}}) = 1$, shared by all 16 parameterisations σ considered.

[†]Recall that they have no predecessors and are therefore influenced solely by the modeller.

[‡]CPU times were measured on an Intel Core i7-8650U 1.90GHz $\times$ 8 processor running Ubuntu 20.04.4 LTS.

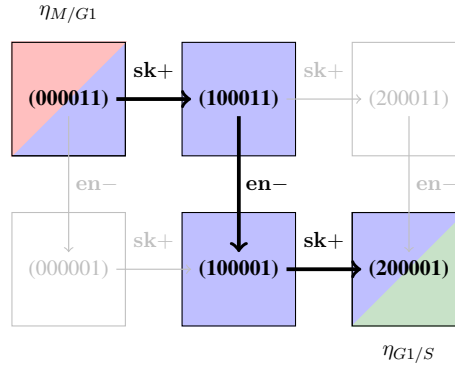


Figure 6.7: **The biologically admitted part of the hyperrectangle $h(H_{G1})$ is shown in colour.** The infeasible part is greyed out. This example shows that the definition of G1 phase reduces here to its canonical specification.

Ultimately, our canonical phases H_{G1} , H_S and H_{G2} in Section 6.4.1 also define G1, S and G2 phases in our cell cycle models. Figure 6.7 illustrates the biologically admitted part of the hyperrectangle $h(H_{G1})$. These results show that updating the discrete component of the cell cycle model of [71] made it possible to fully constrain the order of molecular events within G1, S and G2 phases. No chronometric information was needed, suggesting that the order of events within these phases does indeed come down to a matter of *chronology* and of dependencies between events. However, the same does not hold for M phase; more specifically, current biological knowledge imposes no strong constraint on the $M/G1$ transition.

6.4.3.3 The remarkable case of M phase

Our 16 models $(\mathcal{G}_c, \sigma)$ admit several permutations of the canonical path of M phase. More precisely, Figure 6.8 shows that eight orderings of p_M are solutions of the predicate $estAdmis_\sigma(p'_M, P_{G1})$.

$$p'_M \in \left\{ \begin{array}{l} b+; ep+; a-; a-; b-; b-; ep-; en+ \\ b+; ep+; a-; a-; b-; b-; en+; ep- \\ b+; ep+; a-; a-; b-; ep-; b-; en+ \\ b+; ep+; a-; b-; a-; b-; en+; ep- \\ b+; ep+; a-; b-; a-; b-; ep-; en+ \\ b+; ep+; a-; b-; a-; ep-; b-; en+ \\ b+; ep+; a-; b-; b-; a-; ep-; en+ \\ b+; ep+; a-; b-; b-; a-; en+; ep- \end{array} \right\}$$

Figure 6.8: **Set of permutations exhibited by $h(H_M)$ that are biologically admitted.**

According to Proposition 1, the canonical path of M (in red) is necessarily admitted. The knowledge accumulated in Chapter 5.4, page 5.4.2, does not identify a total order of the events of M; however, we had identified a partial order that must be satisfiable:

1. The only sequence that can initiate M phase is $b+$; $ep+$ from its initial state.
2. Cyclin A degradation begins before cyclin B degradation, meaning that $a-$ must always precede $b-$.

For a parameterisation to be validated, it must invalidate every triple $\{P_M\} p'_M \{P_{G1}\}$ such that p'_M does not satisfy these two constraints.

Every permutation of p_M that is a solution of the predicate $estAdmis_\sigma$ does indeed have the prefix $b+; ep+$. Referring to Figure 6.9, the only possible transition from $\eta_{G2/M}$ (the initial state of M) is the one that increments the variable b . The applicable parameter of b in $\eta_{G2/M}$ is $K_{b, \{a2, no_Wee1, no_CI\}} = 2$. For every other variable v apart from b , the value of the applicable parameter in $\eta_{G2/M}$ equals the current value. The same justification applies to the transition $ep+$. We then proved that a dependency between the events $a-$ and $b-$ does indeed exist. Recall that we proposed the following mechanistic explanation: destruction of cyclin A ($a-$) by APC-cdc20 (ep) begins before destruction of cyclin b ($b-$), because the former is independent of the SAC whereas the latter is *SAC-dependent*. The underlying biological regulatory mechanism is encoded by the phenomenological multiplex named SAC (Figure 5.18, page 99). As soon as APC-cdc20 acts, cyclin A is degraded ($a-$), but cyclin B is not. Within our M phase, degradation of the latter ($b-$) is postponed until $a \geq 2$ becomes false, that is, after the first $a-$.

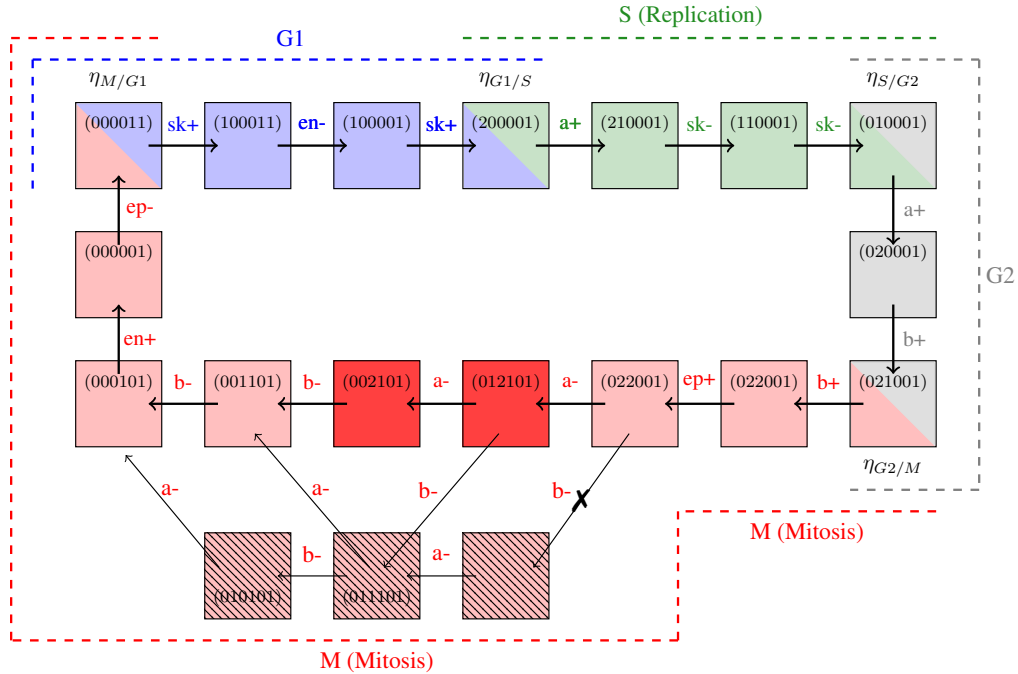


Figure 6.9: **Asynchronous cyclin degradation in M phase and SAC inactivation.** The first $a-$ event must occur before the first $b-$, so $b-$ must not be feasible from state (022001). Our GRBM verifies this property, which explains why every path in which the first $b-$ occurs before the first $a-$ was rejected. The two states shown in dark red are associated with SAC inactivation.

Furthermore, the canonical cell cycle introduces a biological annotation of an event belonging to M phase that deserves particular attention. This is the first $b-$, linked by transitivity to SAC inactivation. Only when the SAC mechanism is deactivated, that is, when all chromosomes are correctly attached to the mitotic spindle, can cyclin B degradation (the first $b-$) be initiated by APC-cdc20 (ep). Depending on the admitted path considered for M phase, the state giving rise to SAC inactivation differs (Figure 6.9).

Ultimately, the beginning of M phase is strongly constrained by the knowledge formalised in our models: only the prefix $b+; ep+; a-$ is admitted. The same does not hold for the end of M phase. The diversity of solution paths can be explained by an asynchronous variable update mode or by a set of cell cycle models that accumulates little knowledge around the $M/G1$ transition. Finally, every solution path of the predicate $estAdmis_\sigma$ is admitted by each of the 16 initial models. The valid Hoare triple[§] that establishes this result is as follows:

$$\{P_M\} \quad \begin{array}{l} \forall[(b+; ep+; a-; a-; b-; b-; en+; ep-), \\ (b+; ep+; a-; a-; b-; b-; ep-; en+), \\ (b+; ep+; a-; a-; b-; ep-; b-; en+), \\ (b+; ep+; a-; b-; a-; b-; en+; ep-), \\ (b+; ep+; a-; b-; a-; b-; ep-; en+), \\ (b+; ep+; a-; b-; a-; ep-; b-; en+), \\ (b+; ep+; a-; b-; b-; a-; ep-; en+), \\ (b+; ep+; a-; b-; b-; a-; en+; ep-)]; \end{array} \quad \{P_{G1}\}$$

All the results presented are available at: <https://gitlab.com/deborahboyenval/temporal-separation-prototype/-/tree/master/ProofOfConceptCheckpoint>.

[§]The correctness of the following triple is proved in 200 ms using TotemBioNet on an Intel Core i7-8650U 1.90GHz $\times$ 8 processor running Ubuntu 20.04.4 LTS.

CHAPTER 7

Formalisation of a checkpoint between two phases

7.1	Preamble	129
7.2	The generic predicate <i>séparation – temporelle</i>(π_i, π_{i+1})	130
7.2.1	The predicates <i>peutClore</i> and <i>peutInitier</i>	130
7.2.2	The <i>estRequis</i> predicate	131
7.3	Irreversibility of phase transitions	133
7.4	Application to our model and our cell cycle phases	136
7.4.1	Implementation of the <i>séparation – temporelle</i> predicate in <code>Prolog</code>	136
7.4.2	Our regulatory graph and its phases with checkpoints	137
7.4.3	The proof of concept	139

7.1 Preamble

The previous chapter defines a formal methodology that exhibits the permutations of a canonical cell cycle sequence admitted by a set of models. These models reproduce the classical properties of the cell cycle, but *what about checkpoints?* The canonical sequence of a phase, shown schematically in blue in Figure 7.1, exhibits a checkpoint state, shown in green. It verifies that all events associated with a phase have taken place and is therefore a mandatory point of passage to the next phase. A certain uncertainty in the order of events within a phase is taken into account and identified. The next objective is to identify whether a given regulatory graph, in this case the one presented in Chapter 5, satisfies a checkpoint at the crucial points of the cell cycle. These checkpoints are passage through the restriction point and the G1/S, S/G2, G2/M and metaphase/anaphase transitions. The constraint imposed by a checkpoint at these points must first take into account *all* events that can complete a phase or portion of a phase (denoted E below) and *all* those that can initiate the next phase or portion of a phase (denoted E' below). A checkpoint verifies *at a minimum* that every event E is *required* before every event E' . Definition 31 formalises this constraint, which is based on the *estAdmis* predicate defined in the previous chapter, and introduces the *estRequis* predicate.

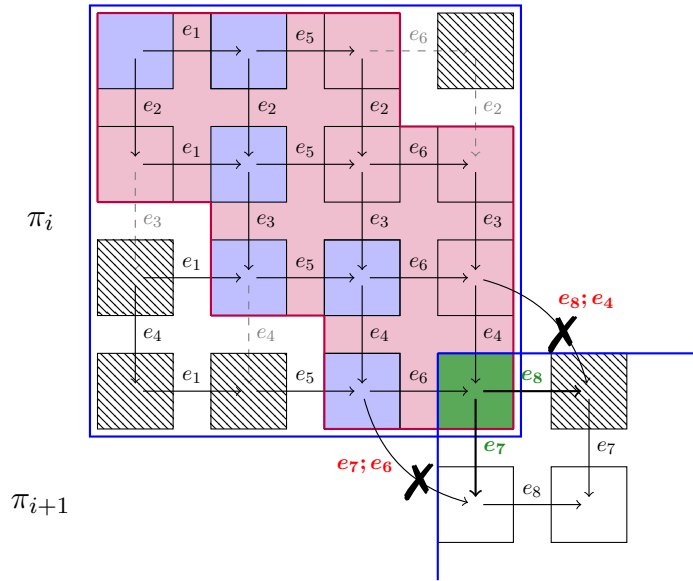


Figure 7.1: **Diagram of the minimal constraint for an intrinsic checkpoint.** Here we assume that variables v and v' matter for phase π_i , and w and w' for π_{i+1} . Events e_1 , e_5 and e_6 correspond to incrementing variable v ($v+$). Events e_2 , e_3 and e_4 correspond to decrementing variable v' ($v'-$). Finally, event e_7 corresponds to $w-$ and e_8 to $w'+$. The pink region represents the biologically admissible part of the hyperrectangle of π_i . A checkpoint between π_i and π_{i+1} verifies that the permutations $(e_7; e_6)$, $(e_8; e_4)$, $(e_7; e_4)$ and $(e_8; e_6)$ are not admitted by a set of models. In that case, all events in $\{e_1, e_2, e_3, e_4, e_5, e_6\}$ are required before those belonging to $\{e_7, e_8\}$.

The *estRequis* predicate constrains the values of certain parameters $K_{v,\omega}$ and allows us to select models satisfying an initial formalisation of checkpoints focused on the temporal separation of

phases. If the set of events of π_i is not required before those of π_{i+1} , then π_i and π_{i+1} are not temporally separated phases and cannot represent a controlled cell cycle.

Verification of the irreversibility of phase transitions follows the same approach. A formula defined in fair CTL (Definition 33) allows us to select, among the models verifying that every E is required before every E' , those also satisfying irreversibility of the π_i/π_{i+1} transitions. Unsatisfiability of the *séparation – temporelle*(π_i/π_{i+1}) predicate (Definition 31) allows the modelling question to be refocused on the canonical definition of phases π_i and π_{i+1} , or on the definition of the regulatory graph, or on the domain of parameters $K_{v\omega}$.

7.2 The generic predicate *séparation – temporelle*(π_i, π_{i+1})

7.2.1 The predicates *peutClore* and *peutInitier*

Prior definition of the biologically admitted paths of a phase using the *estAdmis*(p', H) predicate makes identifying the events that can initiate or complete a phase defined by H trivial. We need only identify the first and last events of all paths p' that are solutions of the *estAdmis* predicate.

Definition 30: Predicates *peutInitier* and *peutClore*

Given a model $\mathcal{R} = (\mathcal{G}, \sigma)$, a canonical phase π defined by an elementary Hoare triple $H \equiv \{P\} p \{Q\}$, where P (resp. Q) is its precondition (resp. postcondition) and p its canonical path, any event E' can initiate π if:

$$\exists p' \in \text{permutations}(p) \mid \text{estAdmis}_\sigma(p', H) \wedge E' = \text{premier}(p')$$

and any event E can complete π if and only if:

$$\exists p' \in \text{permutations}(p) \mid \text{estAdmis}_\sigma(p', H) \wedge E = \text{dernier}(p')$$

We denote the corresponding predicates by *peutInitier* $_\sigma(E', \pi)$ and *peutClore* $_\sigma(E, \pi)$, and the functions *premier*(p') and *dernier*(p') are trivially the first and last events of p' .

The *estAdmis* $_\sigma(p', H)$ predicate was introduced in Definition 29 (Chapter 6). According to our introduction, five checkpoints regulate the cell cycle. Thus, six phases or portions of phases must be considered instead of the classical four phases of the cell cycle.

$$G1_{\text{précocoe}} \xrightarrow[\text{PPR}]{} G1_{\text{tardif}} \xrightarrow[\text{G1/S}]{} S \xrightarrow[\text{S/G2}]{} G2 \xrightarrow[\text{G2/M}]{} M_{\text{précocoe}} \xrightarrow[\text{M/A}]{} M_{\text{tardif}} \rightarrow G1_{\text{précocoe}}$$

Passage through the restriction point (denoted *PPR*) is an event separating *G1* phase into two parts. We call *G1_{précocoe}* the part of *G1* preceding passage through the restriction point, and *G1_{tardive}* the part following it. A sufficient concentration of growth factors is required before passage through the restriction point, thus controlling (irreversible) entry into a cell cycle. Similarly, the metaphase/anaphase transition (denoted *M/A*) is an event separating *M* phase into two parts:

$M_{précocce}$ and M_{tardif} .

Definition 31: Predicate *séparation – temporelle*

Given an interaction graph $\mathcal{G}$ and a set of parameterisations Σ of $\mathcal{G}$, with canonical phases π_i ordered according to the sequence $G1_{précocce}; G1_{tardif}; S; G2; M_{précocce}; M_{tardif}$, cell cycle checkpoints are satisfiable if the *phases are temporally separated*:

$$\exists \sigma \in \Sigma \mid \forall i \in \llbracket 1, 5 \rrbracket,$$

$$\forall E, \forall E', [\text{peutClore}_\sigma(E, \pi_i) \wedge \text{peutInitier}_\sigma(E', \pi_{i+1}) \implies \text{estRequis}_\sigma(E, E')]$$

We denote the corresponding predicate by *séparation – temporelle*(π_i, π_{i+1}).

The next section is devoted to defining the $\text{estRequis}_\sigma(E, E')$ predicate, which ensures that event E' cannot take place before event E . In other words, E and E' are the key cell cycle events that must not be permutable.

Remark 9 – Only events that can initiate or complete a phase matter for the temporal separation constraint. Furthermore, any event occurring from a state that does not belong to the biologically admissible part of the hyperrectangle of a phase π_i is simply ignored.

7.2.2 The *estRequis* predicate

Now that we have identified the cell cycle events that must not be permutable within a model $\mathcal{R}$, we introduce the notation needed to characterise the estRequis_σ constraint. It verifies the non-existence of the transition semantically associated with event E' (*i.e.* the one that can initiate a phase) from the source state of the transition associated with E (*i.e.* the one that can complete the preceding phase).

Notation 3: On the *estRequis* predicate

Using the notation of Definition 30, we denote:

by η the source state of E ,

by η' the target state of E , that is, its checkpoint state deduced from the canonical definition of a phase (*cf.* Notation 2).

v' (resp. v) is the variable associated with event E' (resp. E) of the form $v' -$ or $v' +$ (resp. $v -$ or $v +$),

$K_{v', \omega}$ is the applicable parameter of v' in η (*cf.* Definition 6),

and $K_{v', \omega'}$ is the one applicable in state η' .

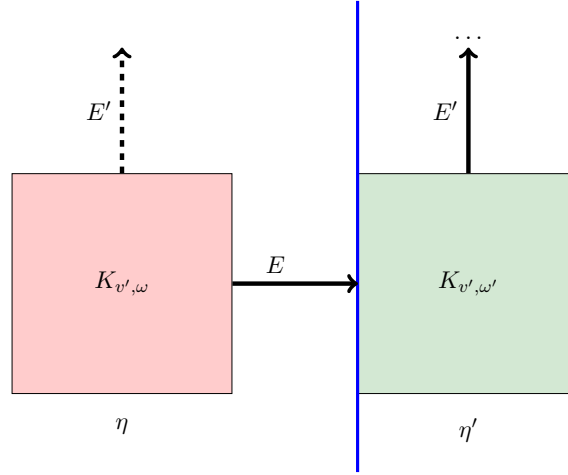


Figure 7.2: **Illustration of Notation 3.** State η belongs exclusively to a phase π_i , and η' is the state from which the transition to the next phase originates. The constraint expressed by the *estRequis* predicate focuses only on transitions that can initiate phase π_{i+1} : such a transition must not be allowed in state η . Here we illustrate the approach with a single instantiation of E and E' . The blue boundary represents the π_i/π_{i+1} transition, and η' is the checkpoint state of phase π_i , indicating that all events required during π_i have taken place.

Definition 32: Predicate *estRequis*

Given an event E that can complete a phase π_i and an event E' that can initiate the next phase π_{i+1} , E is *required* before E' in a model $\mathcal{R} = (\mathcal{G}, \sigma)$ if and only if:

$$[\sigma(K_{v',\omega}) - \eta(v')] \times [\sigma(K_{v',\omega'}) - \eta'(v')] \leq 0$$

where $K_{v',\omega}$, $K_{v',\omega'}$, η , η' , and v' are introduced in Notation 3; recall that η , η' and σ are substitutions, cf. Definitions 5 and 7 in Chapter 2.

We denote the corresponding predicate by $estRequis_\sigma(E', E)$.

Since, by definition, the event that can initiate phase π_{i+1} is a solution of the *peutInitier* predicate, the transition described by E' is necessarily admitted by at least one model $\mathcal{R} = (\mathcal{G}, \sigma)$. It follows that $\sigma(K_{v',\omega'}) - \eta'(v') \neq 0$. Model learning therefore essentially consists of identifying the substitutions σ of parameter $K_{v',\omega}$ that do not attract v' in the same direction as in state η' :

if $E' \equiv v' +$ then $\sigma(K_{v',\omega}) \leq \eta'(v')$ since $\sigma(K_{v',\omega'}) > \eta'(v')$,

if $E' \equiv v' -$ then $\sigma(K_{v',\omega}) \geq \eta'(v')$ since $\sigma(K_{v',\omega'}) < \eta'(v')$.

$$\underbrace{[\sigma(K_{v',\omega}) - \eta(v')]}_{\text{Is } E' \text{ admitted from } \eta?} \times \underbrace{[\sigma(K_{v',\omega'}) - \eta'(v')]}_{E' \text{ is admitted from } \eta'} \leq 0$$

7.3 Irreversibility of phase transitions

A cell cycle regulatory graph can be truly biologically consistent only if the transitions governed by checkpoints are *irreversible*. In other words, there must be no path connecting a phase to any other phase that does not necessarily pass, in order, through each of the canonically defined checkpoint states.

Definition 33: Predicate *irréversible*

Given an interaction graph $\mathcal{G}$ and a set of parameterisations Σ of $\mathcal{G}$, with canonical phases π_i ordered according to the sequence $G1_{précoce}; G1_{tardif}; S; G2; M_{précoce}; M_{tardif}$, irreversibility of the transitions between π_i and π_{i+1} is satisfiable if:

$$\exists \sigma \in \Sigma \mid \forall i \in \llbracket 1, 5 \rrbracket,$$

$$P_{\pi_i} \Rightarrow A \left(\psi_{H_{\pi_i}} \vee \neg \left(\bigvee_{\pi' \in \Pi \setminus \pi_i} \psi_{H_{\pi'}} \right) \cup Q_{\pi_i} \right)$$

where:

$\psi_{H_{\pi}}$ is the characteristic formula of the domain of a phase π (Definition 28 in Chapter 6),

P_{π_i} (resp. Q_{π_i}) characterises the initial (resp. final) state of π_i ,

Π is the set of π_i for all $i \in \llbracket 1, 6 \rrbracket$.

and the modality $A[\dots \cup \dots]$ is that of CTL.

We denote the predicate corresponding to this definition by $irréversible(\pi_i, \pi_{i+1})$.

This definition is illustrated in Figure 7.3 on the following page.

Notation 4: Characteristic formula for the irreversibility of the π_i/π_{i+1} transition

We denote by $\varphi_{irréversibilité}$ the CTL formula: $(P_{\pi_i} \Rightarrow A(\psi_{H_{\pi_i}} \vee \neg(\bigvee_{\pi' \in \Pi \setminus \pi_i} \psi_{H_{\pi'}}) \cup Q_{\pi_i}))$.

The formula $P_{\pi_i} \Rightarrow A(\psi_{H_{\pi_i}} \cup Q_{\pi_i})$ verifies that a phase π_i is stable until the checkpoint state characterised by Q_{π_i} is reached. This conception of a checkpoint recalls the conditionally stable motif of [87] (cf. page 52), which makes a phase transiently stable as long as the condition of the stable motif is satisfied.

The formula $P_{\pi_i} \Rightarrow A(\neg(\bigvee_{\pi' \in \Pi \setminus \pi_i} \psi_{H_{\pi'}}) \cup Q_{\pi_i})$ verifies, if the domain of a phase π_i is left, that no other phase is visited until the checkpoint state of π_i is reached. Any state that does not satisfy the characteristic formula of a phase may, however, be visited. Like the study by [88] (cf.

page 47), we take *unknown* partitions into account in characterising a checkpoint. Including these partitions avoids defining a checkpoint through an unnecessarily strong constraint.

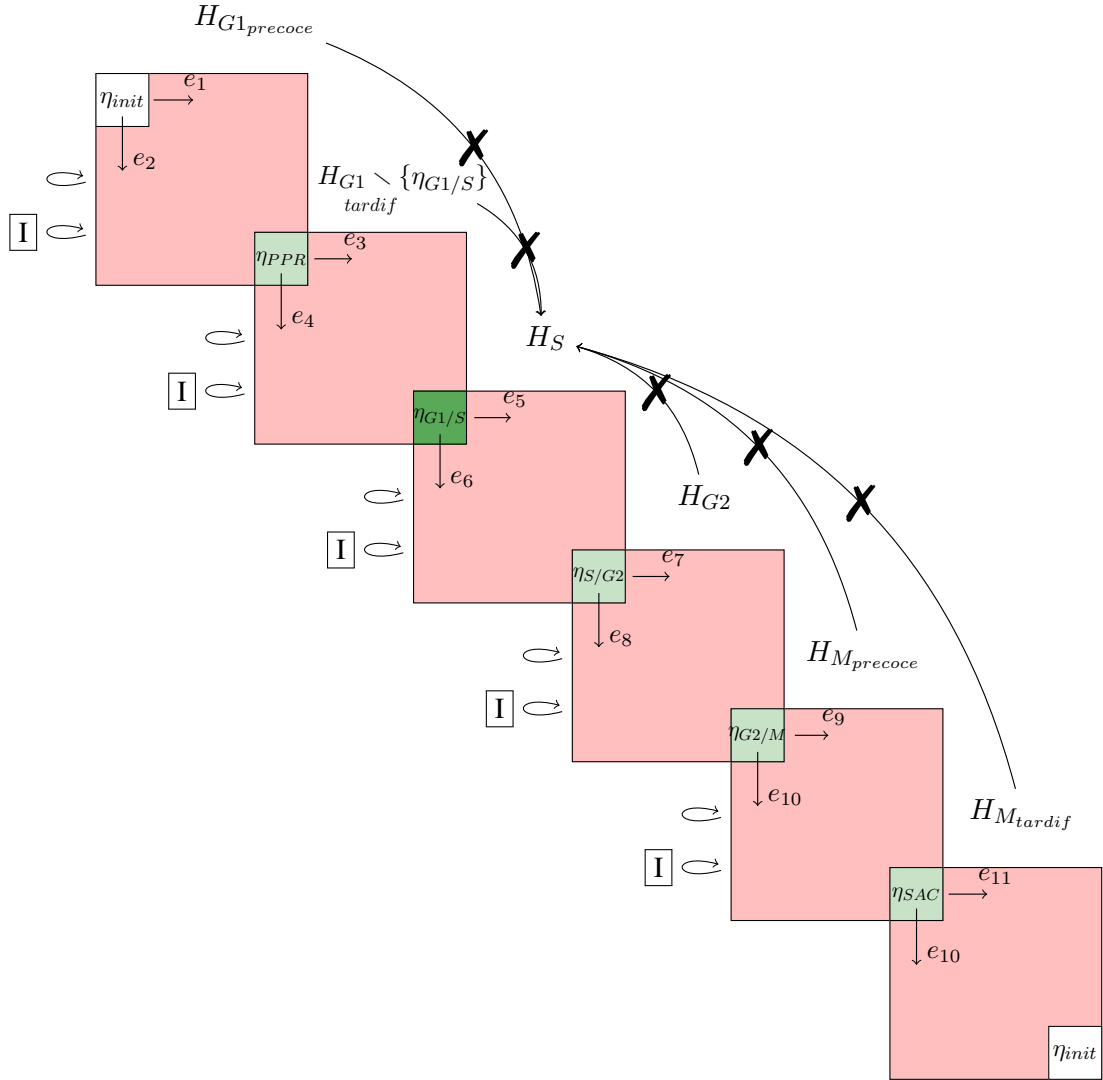


Figure 7.3: **Definition 33** illustrated with the G1/S checkpoint. The initial state of the cell cycle is represented by η_{init} (in white). Each phase or portion of a phase, with or without a checkpoint, is represented by a hyperrectangle H_π for every $\pi \in \{G1_{précoce}, G1_{tardif}, S, G2, M_{précoce}, M_{tardif}\}$. Two hyperrectangles intersect at a checkpoint state (in green), except for $H_{M_{tardif}}$ and $H_{G1_{précoce}}$, which intersect at η_{init} . The late portion of M phase has no checkpoint because the M/G1 checkpoint does not exist (*cf.* the introduction). Events that can initiate a phase from a checkpoint state represent phase transitions. For example, e_3 or e_4 represent passage through the restriction point, e_5 or e_6 represent the G1/S transition, and e_9 or e_{10} represent the metaphase/anaphase transition. The G1/S transition can occur only *via* checkpoint state $\eta_{G1/S}$ (dark green), so the part of the hyperrectangle of $G1_{tardif}$ excluding this state must be stable (loops). The loop labelled **I** means that the space of unknown states, that is, states satisfying no characteristic formula of a phase, may be visited before returning to the ongoing phase (represented here by the pink part of a hyperrectangle).

Remark 10 – The biologically admissible part of the hyperrectangle of a phase exhibits certain paths admitted by biological knowledge. Some states of the hyperrectangle may not be reachable from the initial state of the phase (illustrated by the greyed-out squares in Figure 7.1), but this does not imply that these states do not allow an exit from the phase. For this reason, the hyperrectangle of a phase (characterised by ψ_{H_π}) is considered in Definition 33, rather than only its biologically admissible part, as in Definition 31.

In fine, checkpoints between adjacent phases ensure their *temporal separation* when the sets of events required for these phases take place separately. This is what the *séparation – temporelle*(π_i, π_{i+1}) predicate of Definition 31 ensures when it is satisfiable. Ensuring that every event that can complete a phase indeed takes place before every event that can initiate the next phase is not a sufficient constraint to prove that a phase has a checkpoint. For this, it is additionally necessary to verify that no transition between a phase π_i and its neighbouring phase π_{i+1} occurs from any state other than the checkpoint state of π_i . This is what the *irréversible*(π_i, π_{i+1}) predicate of Definition 33 verifies when it is satisfiable. This predicate ensures the *unidirectionality* of the cell cycle. Temporal separation of phases *and* unidirectionality of the cell cycle are the two essential characteristics for characterising cell cycle checkpoints, according to the modelling study by [41]. Thus, a regulatory graph $\mathcal{G}$ admits phases with checkpoints when the following condition is true:

$$\exists \sigma \in \Sigma \mid \forall i \in \llbracket 1, 5 \rrbracket, \text{séparation} - \text{temporelle}(\pi_i, \pi_{i+1}) \wedge \text{irréversible}(\pi_i, \pi_{i+1}) \quad (7.1)$$

where:

Σ is the set of parameterisations associated with a GRBM $\mathcal{G}$,

π_i represents the canonical phases (or portions of phases) defined by an elementary Hoare triple (Definition 27 in Chapter 6), ordered according to the sequence $G1_{\text{précoc}}; G1_{\text{tardif}}; S; G2; M_{\text{précoc}}; M_{\text{tardif}}$.

We call the corresponding predicate *checkpoint*(π_i, π_{i+1}).

7.4 Application to our model and our cell cycle phases

7.4.1 Implementation of the *séparation – temporelle* predicate in Prolog

The *séparation – temporelle* predicate is a Horn clause that can be declared directly in Prolog.

$$\text{séparation} - \text{temporelle}(H_{\pi_i}, H_{\pi_{i+1}}, \mathcal{G}, \Sigma, E, E') : -$$

$$\text{peutClore}(H_{\pi_i}, \mathcal{G}, \Sigma, E), \text{peutInitier}(H_{\pi_{i+1}}, \mathcal{G}, \Sigma, E'), \text{estRequis}(H_{\pi_i}, H_{\pi_{i+1}}, \mathcal{G}, \Sigma, E, E').$$

with the following declarations beforehand:

$$\begin{aligned} & \text{dernier}([_|E], E). \\ & \text{premier}([E|_], E). \\ \text{peutClore}(H, \mathcal{G}, \Sigma, E) & : - \text{estAdmis}(H, \mathcal{G}, \Sigma, p'), \text{dernier}(p', E). \\ \text{peutInitier}(H, \mathcal{G}, \Sigma, E) & : - \text{estAdmis}(H, \mathcal{G}, \Sigma, p'), \text{premier}(p', E). \end{aligned}$$

The *séparation – temporelle* predicate takes as input:

H_{π_i} , the *canonical* definition of a phase π_i , recalling that the initial state, elementary syntactic path and resulting final state are defined *a priori*,

$H_{\pi_{i+1}}$, the *canonical* definition of the phase immediately following π_i ,

$\mathcal{G}$, a GRBM,

and Σ , the set of parameterisations of $\mathcal{G}$ satisfying biological constraints.

The reader may refer to Section 6.4.3.1, page 121, for the `Prolog` implementation of the *estAdmis* predicate.

The *estRequis* predicate constructs the constraint of the same name introduced in Definition 32, page 132. The implementation is at the first-prototype stage. For now, this predicate delegates the task to a shell program that:

1. identifies the $K_{v,\omega}$ terms of the constraint and instantiates the terms $\eta(v')$ and $\eta'(v')$ using H_{π_i} , $H_{\pi_{i+1}}$ and $\mathcal{G}$,
2. tests the truth value of the constraint for each instance σ of Σ .

According to Formula 7.1, temporal separation of π_i and π_{i+1} holds if there exists at least one parameterisation $\sigma \in \Sigma$ admitting that E is required before E' . The `findall` instruction identifies all solution pairs (E, E') :

$$\text{findall}([E, E'], \text{séparation} - \text{temporelle}(H_{\pi_i}, H_{\pi_{i+1}}, \mathcal{G}, \Sigma, E, E'))$$

Proving the satisfiability of the predicate amounts to evaluating all biologically admitted paths for π_i and π_{i+1} , from which E and E' are extracted. We denote by n (resp. m) the number of events of π_i (resp. π_{i+1}), and the time complexity for identifying the solutions of the predicate is:

$$\Theta\left(\frac{n!}{k_1! \dots k_n!}\right) + \Theta\left(\frac{m!}{k_1! \dots k_m!}\right)$$

where the k_i represent the number of occurrences of an event. Again, we will not burden the reader with implementation subtleties. The prototype is available at <https://gitlab.com/deborahboynval/temporal-separation-prototype>.

7.4.2 Our regulatory graph and its phases with checkpoints

Definition 31 is enough to raise the question of the relevance of a regulatory graph: *can it represent the cell cycle phases that have a checkpoint, i.e. $G1_{précoc}$, $G1_{tardif}$, S , $G2$, $M_{précoc}$, M_{tardif} ?* The variables and thresholds of our GRBM (Figure 5.18, page 99) enabled us to formalise $G1$, S , $G2$ and M phases from a canonical definition. For the reasons discussed in Chapter 3, page 54, only the interphase checkpoints will be considered for the proof of concept, that is, $G1/S$, $S/G2$ and $G2/M$. Nevertheless, this section provides guidance for modelling the checkpoint within M phase.

The previous chapter concludes with the uncertainty in our knowledge of M phase, more specifically concerning the location of its checkpoint state. It is from this state that the

metaphase/anaphase transition can occur, after inactivation of the SAC. This transition separates $M_{précocce}$ from M_{tardif} and is characterised in our study by event $b-$ ($b = 2 \rightarrow b = 1$) belonging to the path of H_M . Unlike the other phase transitions, the metaphase/anaphase transition (abbreviated M/A) does not isolate a unique set of events required before it:

$$\{P_M\} \quad b+; ep+; a-; a-; \underbrace{b-}_{M/A}; b-; en+; ep- \quad \{P_{G1}\}$$

is the canonical triple of M phase. The multiset of events of $M_{précocce}$ suggested by this triple is $\{b+ : 1, ep+ : 1, a- : 2\}$. No information in the literature allows these four events to be considered required before the metaphase/anaphase transition (represented by $b-$), as corroborated by the solutions of the $estAdmis_\sigma$ predicate for M phase. Figure 7.4 shows that two sequences are admitted before the metaphase/anaphase transition: No model can be refuted on the basis of the

$$p'_M \in \left\{ \begin{array}{l} \textcolor{red}{b+; ep+; a-; a-; \underbrace{b-}_{M/A}; b-; ep-; en+} \\ b+; ep+; a-; a-; \underbrace{b-}_{M/A}; b-; en+; ep- \\ b+; ep+; a-; a-; \underbrace{b-}_{M/A}; ep-; b-; en+ \\ b+; ep+; a-; \underbrace{b-}_{M/A}; a-; b-; en+; ep- \\ b+; ep+; a-; \underbrace{b-}_{M/A}; a-; b-; ep-; en+ \\ b+; ep+; a-; \underbrace{b-}_{M/A}; a-; ep-; b-; en+ \\ b+; ep+; a-; \underbrace{b-}_{M/A}; b-; a-; ep-; en+ \\ b+; ep+; a-; \underbrace{b-}_{M/A}; b-; a-; en+; ep- \end{array} \right\}$$

Figure 7.4: **The different positions of the metaphase/anaphase transition exhibited by the paths admitted in M phase.** The path in red is the canonical path of M as defined in Table 6.1.

multiset of events it admits for $M_{précocce}$. The additional degree of freedom admitted in M phase concerning the order of its events means that it is not possible to isolate $M_{précocce}$ and M_{tardif} , even though the metaphase/anaphase transition is unambiguous thanks to our GRBM.

7.4.3 The proof of concept

π_i/π_{i+1}	E	E'	$ \sigma \models \text{séparation} - \text{temporelle}(\pi_i, \pi_{i+1})$	Computation time*
G1/S	$\{sk+\}$	$\{a+\}$	16/16	2.1 sec
S/G2	$\{sk-\}$	$\{a+\}$	8/16	3.3 sec
G2/M	$\{b+\}$	$\{b+\}$	16/16	1h12min

Table 7.1: **Solutions of the *séparation – temporelle* predicate for 3 cell cycle checkpoints.**

Remark 11 – The event that can complete G2 is the same as the event that can initiate M, but only from a syntactic point of view. The first (resp. second) $b+$ in G2 (resp. M) is associated with monophosphorylation (resp. hyperphosphorylation) of the cyclinB/cdk1 complex. Hyperphosphorylation of this complex is associated with activation of mitosis. $E \equiv b+$ in G2 is necessarily required before $E' \equiv b+$ in M, since René Thomas’ unitary semantics implies that b first passes through its state $\eta(b) = 1$ (thus in G2) before reaching its state $\eta(b) = 2$ (thus in M).

In stricto sensu, our regulatory graph does not satisfy the *séparation-temporelle* predicate of Definition 31. It does, however, satisfy all the checkpoints it can exhibit, cf. the first column of Table 7.1. One of the clauses of the predicate aims to identify the number of models satisfying the checkpoint being tested among the 16 models satisfying the classical properties of the cell cycle (Chapter 5).

The S/G2 checkpoint is satisfied by any model $(\mathcal{G}, \sigma)$ such that $\sigma(K_{a,\{no_CI, restrict_sk\}}) \leq 1$, since the $estRequis_\sigma(E, E')$ constraint is characterised in this case by:

$$[(\sigma(K_{a,\{no_CI, restrict_sk\}}) - 1) \times (\sigma(K_{a,\{no_CI, end_Rep\}}) - 1)] \leq 0 \quad (7.2)$$

and since the constraint $\sigma(K_{a,\{no_CI, end_Rep\}}) > 1$ is deduced from computation of the weakest precondition of the canonical cell cycle. The reader is referred to page 72 for further details. $K_{a,\{no_CI, restrict_sk\}}$ is an unidentified parameter. The knowledge accumulated in Chapter 5 allows its domain to be reduced to the interval $\llbracket 0, 1 \rrbracket$, hence the 8 models selected out of the 16. This knowledge is fully consistent with the S/G2 checkpoint; more specifically, S and G2 phases are temporally separated. Following the same approach, the $estRequis_\sigma(E, E')$ constraints for the other two checkpoints in Table 7.1 are, respectively:

$$[(\sigma(K_{a,\{no_CI\}}) - 0) \times (\sigma(K_{a,\{sk2, no_CI\}}) - 0)] \leq 0 \quad (7.3)$$

and

$$[(\sigma(K_{b,\{a2, no_CI\}}) - 1) \times (\sigma(K_{b,\{a2, no_Wee1, no_CI\}}) - 1)] \leq 0 \quad (7.4)$$

The weakest precondition associated with the canonical cell cycle implies $\sigma(K_{a,\{sk2, no_CI\}}) = 1$ and $\sigma(K_{b,\{a2, no_Wee1, no_CI\}}) = 2$, and the 16 models do indeed satisfy $\sigma(K_{a,\{no_CI\}}) < 1$ and $\sigma(K_{b,\{a2, no_CI\}}) < 2$, so the G1/S and G2/M checkpoints are indeed satisfiable.

Remark 12 – According to Remark 11, it would be tempting to regard verification of the $estRequis_\sigma$ constraint for the G2/M checkpoint as pointless. Indeed, René Thomas’ unitary semantics, by definition, constrains the first $b+$ in G2 to precede the second in M, even when

*CPU times were measured on an Intel Core i7-8650U 1.90GHz $\times$ 8 processor running Ubuntu 20.04.4 LTS.

$K_{v',\omega} = K_{v',\omega'}$. However, the $estRequis_\sigma(b+, b+)$ constraint highlights that it is precisely the addition of resource no_Wee1 to the resource set of variable b , in the state preceding checkpoint state $\eta_{G2/M}$, that allows b to move from a monophosphorylated state in G2 to a hyperphosphorylated state in M, thus enabling the G2/M transition.

Irreversibility of the phases is tested using TotemBioNet and the formula $\varphi_{irréversibilité}$, cf. Notation 4. TotemBioNet enabled the satisfiability of this formula to be checked using the Nu-SMV model checker, again starting from the 16 remaining models. The results are reported in Table 7.2.

Checkpoint	$\varphi_{irréversibilité}$	$ \sigma \models \varphi_{irréversibilité}$
$G1/S$	$P_{G1} \implies \mathbb{A}[(\psi_{H_{G1}} \vee \neg(\psi_{H_S} \vee \psi_{H'_{G2}} \vee \psi_{H_M})) \cup P_S]$	16/16
$S/G2$	$P_S \implies \mathbb{A}[(\psi_{H_S} \vee \neg(\psi_{H_{G1}} \vee \psi_{H_{G2}} \vee \psi_{H_M})) \cup P_{G2}]$	16/16
$G2/M$	$P_{G2} \implies \mathbb{A}[(\psi_{H_{G2}} \vee \neg(\psi_{H_{G1}} \vee \psi_{H_S} \vee \psi_{H_M})) \cup P_M]$	16/16

Table 7.2: Checking the satisfiability of formula $\varphi_{irréversibilité}$ for all checkpoints that our regulatory graph allows us to study.

Recall that:

$$\begin{aligned}
\psi_{H_{G1}} &\equiv (sk \geq 0 \wedge sk \leq 2) \wedge (en \geq 0 \wedge en \leq 1) \wedge (a = 0 \wedge b = 0 \wedge ep = 0 \wedge gf = 1) \\
\psi_{H_S} &\equiv (a \geq 0 \wedge a \leq 1) \wedge (sk \geq 0 \wedge sk \leq 2) \wedge (b = 0 \wedge ep = 0 \wedge en = 0 \wedge gf = 1) \\
\psi_{H_{G2}} &\equiv (a \geq 1 \wedge a \leq 2) \wedge (b \geq 0 \wedge b \leq 1) \wedge (sk = 0 \wedge ep = 0 \wedge en = 0 \wedge gf = 1) \\
\psi_{H_M} &\equiv (a \leq 2 \wedge a \geq 0) \wedge (b \leq 2 \wedge b \geq 0) \wedge (en \leq 1 \wedge b \geq 0) \wedge (ep \leq 2 \wedge b \geq 0) \\
&\quad \wedge (sk = 0 \wedge gf = 1)
\end{aligned}$$

Checking the satisfiability of the conjunction of the $\varphi_{irréversibilité}$ formulae for these three checkpoints is very efficient with TotemBioNet (82 milliseconds). Thus, we have just proved that phases $G1$, S and $G2$ associated with our regulatory graph have checkpoints in the sense that they are temporally separated and that the transition to the next phase is irreversible. Our proof of concept for modelling the cell cycle checkpoint phenomenon is therefore validated in the sense that we have identified biologically relevant biological examples (our 8 remaining models are relevant to the question posed) satisfying Definitions 31 and 33. All the results presented are available at: <https://gitlab.com/deborahboyenval/temporal-separation-prototype/-/tree/master/ProofOfConceptCheckpoint>.

CHAPTER 8

Conclusion and perspectives

“An expert is a man who has made all the mistakes that could be made on a particular subject.”

-
Niels Bohr

8.1 Assessment

In the modelling of complex biological systems, the notion of a checkpoint is a concept encountered relatively often, particularly in the example of the cell cycle. This notion carries several key properties: first, we think of the need to complete a set of tasks before starting the next set; but we must also consider the prohibition on moving backwards among these identified sets of tasks, ordered relative to one another. The notion of a checkpoint necessarily includes an aspect of blocking until the current set of tasks is complete, *even if these tasks are delayed*. It follows that correct ordering of these tasks must not depend on their completion times.

We therefore assumed that it should be possible to formalise a checkpoint in a mathematical framework where time is not quantitative. This is why, in this thesis, we chose a purely discrete formalism, that of René Thomas, in which only the chronology of events matters. Our main contribution was to demonstrate that the notion of a checkpoint could indeed be formalised within such a framework. This framework enabled us to use tools from theoretical computer science, such as formal verification of models taking the form of finite-state automata and Hoare logic for the static analysis of biological traces.

The task could have been straightforward if the biological systems considered were not cyclic, mainly because the question would have amounted to a succession of checks of unreachability properties, as allowed, for example, by model-checking algorithms for LTL or CTL formulae [19, 52, 76], or by optimised automata traversal algorithms based on abstract interpretation techniques [60, 67]. Unfortunately, most physiological systems are cyclic, implying that from any reachable state it is possible to reach all other reachable states. The notion of a checkpoint therefore had to be more subtle than “simple reachability queries”.

The thesis ultimately shows that two notions are fundamental to checkpoints: the temporal separation of successive sets of events, formalised by the *séparation-temporelle* predicate in Chapter 7, page 130, and the unidirectionality of the dynamics along these sets of events, formalised by the *irréversibilité* predicate, also in Chapter 7, page 133.

8.1.1 Methodological contributions

Specifically, *temporal separation* of two successive sets of events is ensured if all the events that can complete the first set are necessary for each of the events that can initiate the second. Genetically modified Hoare logic gave us the opportunity to formalise the notion of an event explicitly, thereby facilitating the definition of this predicate. The properties of temporal separation and irreversibility complement one another to define the fundamental dynamical property of biological checkpoints.

The *temporal separation predicate* ensures that an event that can initiate a phase cannot take place within the characteristic domain of the preceding phase. The predicate for the irreversibility of phase transitions goes further by isolating an ongoing phase from all other phases, thereby forcing the succession of phases to occur only in the defined direction, for example $G1 \rightarrow S \rightarrow G2 \rightarrow M$ in the case of the cell cycle.

This isolation of phases required a certain subtlety to be taken into account, already addressed by [88]. Our irreversibility predicate allows a trace to leave the set of marker states of a phase (*i.e.* the hyperrectangle of a canonical trace), provided that it always returns, without ever visiting a state belonging to another phase. In other words, the transition from one phase to subsequent phases takes place only through the final state of the current phase, which also marks the initial state of the phase immediately following it.

Here again, the symbolic and formal nature of Thomas' theory made it possible to define irreversibility as a predicate, whereas an initial, more brute-force approach would have been to examine each transition leaving a canonical phase using semantic methods.

8.1.2 Contributions to cell cycle modelling

Our proof of concept relied on the paradigmatic example of cyclic behaviour with checkpoints: the cell cycle. In this specific case, checkpoints ensure DNA integrity during replication and mitosis. The thesis proves that our checkpoint concept is directly applicable to the cell cycle, given that a phase can be described by a succession of events taking place from an initial state.

We drew on the rich state of the art in mathematical modelling of the cell cycle to model an abstract checkpoint mechanism using a biological regulatory graph. The precise biological meaning of the variables useful for addressing the question of phase succession and checkpoints was fully reassessed. Essentially, the biological entities abstracted by the variables were matched to their actual role in the cell cycle. The abstract interaction graph of Tyson and Novák [41], updated by [71] and [83], was supplemented with multiplexes that explicitly reflect dynamical constraints imposed “collectively” on the system by these biological entities. These constraints guarantee the absence of certain traces and are crucial for checkpoints. The successful implementation of the checkpoint definition in the case of the cell cycle would not have been possible without our regulatory graph enriched with multiplexes and updated on the basis of recent biological knowledge.

We then deduced a finite set of asynchronous transition graphs using the family $\mathcal{K}$ of dynamical parameters. This method offers the advantage of avoiding an arbitrary choice of model. Indeed, experimental data rarely allow a unique activation logic to be inferred for the variables of an interaction graph, a choice liable to introduce biases into a model's predictions [98]. Global knowledge of the system's dynamics was also used to reduce the combinatorial explosion.

The definition of a phase could be biologically validated only if a canonical sequence of events was reflected by at least one transition graph. This validation was successfully performed by applying genetically modified Hoare logic to a set of multivalued models of the mammalian cell cycle. From this set, the domain of a phase is defined by the states that this phase can cover by executing, not its canonical sequence, but all the permutations of that canonical sequence allowed by a set of parameterisations of a regulatory graph.

8.2 Perspectives

8.2.1 Model inference using our predicates

Our specification of a checkpoint is defined by the conjunction of our predicates presented in Chapter 7, and a model is updated when at least one of these predicates is false. The question of temporal separation is handled symbolically, by solving a constraint on the values of the dynamical parameters of a regulatory graph, and informs the modeller about how to constrain locally the action of certain regulators on certain variables. The question of cell cycle unidirectionality is resolved by enumerating the correct values of the dynamical parameters using TotemBioNet, which can make updating a model tedious when the irreversibility predicate is not satisfied. A characteristic formula of the correct parameterisations would be easier to handle than a long enumeration, along the lines of the weakest precondition in genetically modified Hoare logic.

A method consisting of exhibiting all transitions leaving the hyperrectangle of a phase and prohibiting *all* of them by modifying the model in an *ad hoc* manner would be excessively brute-force. The method proposed with our *irréversibilité* predicate is much more appropriate in that it allows the current phase to be left provided that no other phase is encountered. In certain practical cases of model debugging, however, it may suffice to identify the outgoing transitions that cause the predicate not to be satisfied and to modify the model so that they are no longer admitted. We could exploit the ability of some model checkers to return paths constituting a counterexample when the tested property is not satisfied. It would also be possible to develop a model-checking algorithm that handles the satisfiability of the irreversibility predicate symbolically. Such a model checker would return a constraint on the dynamical parameters ensuring that the predicate is satisfied by all states of the transition system considered.

8.2.2 On our modelling of mammalian cell cycle checkpoints

Our modelling identified five crucial points during the cell cycle. Our proof of concept focused on the restriction point, the checkpoints at the entry to and exit from S phase (G1/S and S/G2), and the checkpoint at entry into mitosis (G2/M), all within a single environmental context, namely one admitting a cyclic attractor that includes our definition of the cell cycle. The modelling work needs to be continued to characterise the missing regulations that will allow the mitotic exit checkpoint (metaphase/anaphase transition) to be modelled, thereby extending our proof of concept to all intrinsic cell cycle checkpoints. With a view to model reduction or explainability, the REVEAL algorithm [12] or the method of learning from a transition system (LFIT) [84] could, in theory, infer the set of sufficient and minimal regulatory rules needed to represent a specific phase with a checkpoint.

8.2.3 Some applications in biology

The field of synthetic biology could adopt our approach to the specification and automatic verification of a checkpoint in order to equip an oscillatory system with one or more checkpoints, for example [78]. An experimental feat enabled, for example, the design of a minimal model of cell cycle progression in yeast, in which certain canonical regulations are removed but phase ordering is still respected [46]. Such an approach in more complex eukaryotic organisms would benefit from being supported by our approach.

The ability of checkpoints to arrest the cell cycle in response to DNA damage motivates extensive research in the biomedical field [104]. Some anticancer therapeutic strategies exploit the cytostatic properties of pharmaceutical molecules to block cell cycle progression (*e.g.* cdk4 and cdk6 inhibitors) [51]. By preventing the activation of certain cdks, the mechanism of action of these molecules suggests the blocking of an event associated with a phase and therefore cell cycle arrest, if we refer to our *checkpoint* predicate. Other cancer therapies rely on cytotoxic molecules that induce DNA damage and activate intrinsic checkpoints *via* extrinsic checkpoints. For example, DNA damage activates p53, which in turn activates an inhibitor of *cyc*/*cdk* complexes, potentially blocking an event in a phase and thus blocking cell cycle progression. Any “multi-system” model that includes a description of the cell cycle and satisfies our *checkpoint* predicate is a strong candidate for testing a variety of hypotheses.

Finally, we mention other avenues for modelling, such as applying our predicates to biological systems other than the cell cycle [68, 93, 96], testing cytostatic or cytotoxic molecules on cell cycle dynamics, studying synergy between mutants that could lead to cancer phenotypes (*e.g.* uncontrolled proliferation or an accumulation of mutations), or exploring coordination between the phases of the cell cycle, the circadian cycle [53, 69] and metabolism [32, 92]. This could require adding chronometric information in the form of transition delays [73, 71], or defining priority classes within a discrete model [29, 67].

Bibliography

- [1] Karl R. Popper. *Conjectures et Réfutations*. Payot, 1963.
- [2] Tony Hoare. “An axiomatic basis for computer programming”. In: *Commun. ACM* 12(10) (1969), pp. 576–580.
- [3] S.A. Kauffman. “Metabolic stability and epigenesis in randomly constructed genetic nets”. In: *Journal of Theoretical Biology* 22.3 (1969), pp. 437–467. ISSN: 0022-5193. DOI: [https://doi.org/10.1016/0022-5193\(69\)90015-0](https://doi.org/10.1016/0022-5193(69)90015-0). URL: <https://www.sciencedirect.com/science/article/pii/0022519369900150>.
- [4] René Thomas. “Boolean formalization of genetic control circuits”. In: *Journal of Theoretical Biology* 42.3 (1973), pp. 563–585. ISSN: 0022-5193. DOI: [https://doi.org/10.1016/0022-5193\(73\)90247-6](https://doi.org/10.1016/0022-5193(73)90247-6). URL: <http://www.sciencedirect.com/science/article/pii/0022519373902476>.
- [5] Edsger W. Dijkstra. “Guarded Commands, Nondeterminacy and Formal Derivation of Programs”. In: *Commun. ACM* 18.8 (Aug. 1975), pp. 453–457. ISSN: 0001-0782. DOI: [10.1145/360933.360975](https://doi.org/10.1145/360933.360975). URL: <https://doi.org/10.1145/360933.360975>.
- [6] W.V. Quine. *Les deux dogmes de l’empirisme*. Gallimard, 1980.
- [7] Leland Hartwell and Ted Weinert. “Checkpoints: Controls That Ensure the Order of Cell Cycle Events”. In: *Science (New York, N.Y.)* 246 (Dec. 1989), pp. 629–34. DOI: [10.1126/science.2683079](https://doi.org/10.1126/science.2683079).
- [8] E.H Snoussi. “Qualitative dynamics of a piecewise-linear differential equations : a discrete mapping approach”. In: *Dynamics and stability of Systems* 4 (1989), pp. 189–207.
- [9] René Thomas. “Regulatory networks seen as asynchronous automata: A logical description”. In: *Journal of Theoretical Biology* 153.1 (1991), pp. 1–23. ISSN: 0022-5193. DOI: [https://doi.org/10.1016/S0022-5193\(05\)80350-9](https://doi.org/10.1016/S0022-5193(05)80350-9). URL: <http://www.sciencedirect.com/science/article/pii/S0022519305803509>.
- [10] Denis Thieffry and René Thomas. “Dynamical behaviour of biological regulatory networks—II. Immunity control in bacteriophage lambda”. In: *Bulletin of Mathematical Biology* 57.2 (1995), pp. 277–297. ISSN: 0092-8240. DOI: [https://doi.org/10.1016/0092-8240\(94\)00037-D](https://doi.org/10.1016/0092-8240(94)00037-D). URL: <https://www.sciencedirect.com/science/article/pii/009282409400037D>.
- [11] Matthew O’Connell et al. “O’Connell M, Raleigh J, Verkade H, Nurse P. Chk1 is a weel kinase in the G2 DNA damage checkpoint inhibiting cdc2 by Y15 phosphorylation. EMBO J 16: 545-554”. In: *The EMBO journal* 16 (Mar. 1997), pp. 545–54. DOI: [10.1093/emboj/16.3.545](https://doi.org/10.1093/emboj/16.3.545).
- [12] Shoudan Liang, S. Fuhrman, and R. Somogyi. “Reveal, a general reverse engineering algorithm for inference of genetic network architectures.” In: *Pacific Symposium on Biocomputing. Pacific Symposium on Biocomputing* (1998), pp. 18–29.

- [13] L. Mendoza, D. Thieffry, and E. R. Alvarez-Buylla. “Genetic control of flower morphogenesis in *Arabidopsis thaliana*: a logical analysis.” In: *Bioinformatics* 15.7 (July 1999), pp. 593–606. ISSN: 1367-4803. DOI: [10.1093/bioinformatics/15.7.593](https://doi.org/10.1093/bioinformatics/15.7.593). eprint: <https://academic.oup.com/bioinformatics/article-pdf/15/7/593/9732060/150593.pdf>. URL: <https://doi.org/10.1093/bioinformatics/15.7.593>.
- [14] Paul Nurse. “A Long Twentieth Century of the Cell Cycle and Beyond”. In: *Cell* 100 (Feb. 2000), pp. 71–8. DOI: [10.1016/S0092-8674\(00\)81684-0](https://doi.org/10.1016/S0092-8674(00)81684-0).
- [15] Mikhail Blagosklonny and Arthur Pardee. “The Restriction Point of the Cell Cycle”. In: *Cell cycle (Georgetown, Tex.)* 1 (Mar. 2002), pp. 103–10. DOI: [10.4161/cc.1.2.108](https://doi.org/10.4161/cc.1.2.108).
- [16] A. Cimatti et al. “NuSMV 2: An OpenSource Tool for Symbolic Model Checking”. In: *CAV*. 2002.
- [17] Jerry Hsu et al. “E2F-dependent accumulation of hEmi1 regulates S phase entry by inhibiting APC(Cdh1)”. In: *Nature cell biology* 4 (June 2002), pp. 358–66. DOI: [10.1038/ncb785](https://doi.org/10.1038/ncb785).
- [18] Hidde de Jong. “Modeling and Simulation of Genetic Regulatory Systems: A Literature Review”. In: *J. Comput. Biol.* 9 (2002), pp. 67–103.
- [19] Nathalie Chabrier and François Fages. “Symbolic Model Checking of Biochemical Networks”. In: vol. 2602. Feb. 2003, pp. 149–162. ISBN: 978-3-540-00605-3. DOI: [10.1007/3-540-36481-1_13](https://doi.org/10.1007/3-540-36481-1_13).
- [20] Markus Welcker et al. “Multisite Phosphorylation by Cdk2 and GSK3 Controls Cyclin E Degradation”. In: *Molecular cell* 12 (Sept. 2003), pp. 381–92. DOI: [10.1016/S1097-2765\(03\)00287-9](https://doi.org/10.1016/S1097-2765(03)00287-9).
- [21] Gilles Bernot et al. “Application of formal methods to biological regulatory networks: Extending Thomas’ asynchronous logical approach with temporal logic”. In: *Journal of theoretical biology* 229 (Sept. 2004), pp. 339–47. DOI: [10.1016/j.jtbi.2004.04.003](https://doi.org/10.1016/j.jtbi.2004.04.003).
- [22] Adrian P. Bracken et al. “E2F target genes: unraveling the biology”. In: *Trends in Biochemical Sciences* 29.8 (2004), pp. 409–417. ISSN: 0968-0004. DOI: <https://doi.org/10.1016/j.tibs.2004.06.006>. URL: <https://www.sciencedirect.com/science/article/pii/S0968000404001513>.
- [23] Nathalie Chabrier-Rivier et al. “Modeling and querying biomolecular interaction networks”. In: *Theoretical Computer Science* 325.1 (2004). Computational Systems Biology, pp. 25–44. ISSN: 0304-3975. DOI: <https://doi.org/10.1016/j.tcs.2004.03.063>. URL: <https://www.sciencedirect.com/science/article/pii/S030439750400218X>.
- [24] Béla Novák and John J. Tyson. “A model for restriction point control of the mammalian cell cycle”. In: *Journal of Theoretical Biology* 230.4 (2004). Special Issue in honour of Arthur T. Winfree, pp. 563–579. ISSN: 0022-5193. DOI: <https://doi.org/10.1016/j.jtbi.2004.04.039>. URL: <https://www.sciencedirect.com/science/article/pii/S0022519304002449>.
- [25] Robert Costa. “FoxM1 dances with mitosis”. In: *Nature cell biology* 7 (Mar. 2005), pp. 108–10. DOI: [10.1038/ncb0205-108](https://doi.org/10.1038/ncb0205-108).

- [26] Yuichi Machida, Joyce Hamlin, and Anindya Dutta. “Right Place, Right Time, and Only Once: Replication Initiation in Metazoans”. In: *Cell* 123 (Nov. 2005), pp. 13–24. DOI: [10.1016/j.cell.2005.09.019](https://doi.org/10.1016/j.cell.2005.09.019).
- [27] Yuichi Machida, Jamie Teer, and Anindya Dutta. “Acute Reduction of an Origin Recognition Complex (ORC) Subunit in Human Cells Reveals a Requirement of ORC for Cdk2 Activation”. In: *The Journal of biological chemistry* 280 (Aug. 2005), pp. 27624–30. DOI: [10.1074/jbc.M502615200](https://doi.org/10.1074/jbc.M502615200).
- [28] Klaus Becker et al. “Self-renewal of human embryonic stem cells is supported by a shortened G1 cell cycle phase”. In: *Journal of cellular physiology* 209 (Dec. 2006), pp. 883–93. DOI: [10.1002/jcp.20776](https://doi.org/10.1002/jcp.20776).
- [29] Adrien Fauré et al. “Dynamical analysis of a generic Boolean model for the control of the mammalian cell cycle”. In: *Bioinformatics* 22 14 (2006), e124–31.
- [30] Daniele Guardavaccaro and Michele Pagano. “Stabilizers and Destabilizers Controlling Cell Cycle Oscillators”. In: *Molecular cell* 22 (May 2006), pp. 1–4. DOI: [10.1016/j.molcel.2006.03.017](https://doi.org/10.1016/j.molcel.2006.03.017).
- [31] Jonathon Pines. “Mitosis: A matter of getting rid of the right protein at the right time”. In: *Trends in cell biology* 16 (Feb. 2006), pp. 55–63. DOI: [10.1016/j.tcb.2005.11.006](https://doi.org/10.1016/j.tcb.2005.11.006).
- [32] Zheng Chen and Steven Mcknight. “A Conserved DNA Damage Response Pathway Responsible for Coupling the Cell Division Cycle to the Circadian and Metabolic Cycles”. In: *Cell cycle (Georgetown, Tex.)* 6 (Oct. 2007), pp. 2906–12. DOI: [10.4161/cc.6.23.5041](https://doi.org/10.4161/cc.6.23.5041).
- [33] Jane L. Lubischer. “The Cell Cycle, Principles of Control. David O. Morgan.” In: *Integrative and Comparative Biology* 47.5 (June 2007), pp. 794–795. ISSN: 1540-7063. DOI: [10.1093/icb/icm066](https://doi.org/10.1093/icb/icm066). eprint: <https://academic.oup.com/icb/article-pdf/47/5/794/1665242/icm066.pdf>. URL: <https://doi.org/10.1093/icb/icm066>.
- [34] A. Richard and J.-P. Comet. “Necessary conditions for multistationarity in discrete dynamical systems”. In: *Discrete Applied Mathematics* 155.18 (2007), pp. 2403–2413.
- [35] Matt Sullivan and David Morgan. “Finishing mitosis, one step at a time”. In: *Nature reviews. Molecular cell biology* 8 (Dec. 2007), pp. 894–903. DOI: [10.1038/nrm2276](https://doi.org/10.1038/nrm2276).
- [36] G. Bernot, J.-P. Comet, and Z. Khalis. “Gene regulatory networks with multiplexes”. In: *European Simulation and Modelling Conference Proceedings*. ISBN 978-90-77381-44-1. France, Oct. 2008, pp. 423–432.
- [37] Arnaud Besson, Steve Dowdy, and James Roberts. “CDK Inhibitors : Cell Cycle regulators and Beyond”. In: *Developmental cell* 14 (Mar. 2008), pp. 159–69. DOI: [10.1016/j.devcel.2008.01.013](https://doi.org/10.1016/j.devcel.2008.01.013).
- [38] Laurence Calzone et al. “A comprehensive modular map of molecular interactions in RB/E2F pathway”. In: *Molecular systems biology* 4 (Feb. 2008), p. 173. DOI: [10.1038/msb.2008.7](https://doi.org/10.1038/msb.2008.7).
- [39] Barbara Di Fiore and Jonathon Pines. “Fair paths in CTL”. In: *personnal communication* (2008). URL: <https://gitlab.com/totembionet/totembionet>.

- [40] Helfrid Hochegger, Shunichi Takeda, and Tim Hunt. “Cyclin-dependent kinases and cell-cycle transitions: Does one fit all?” In: *Nature reviews. Molecular cell biology* 9 (Oct. 2008), pp. 910–6. DOI: [10.1038/nrm2510](https://doi.org/10.1038/nrm2510).
- [41] John J Tyson and Bela Novak. “Temporal organization of the cell cycle.” In: *CB Current biology* 18 (2008), R759–R768. DOI: [10.1016/j.cub.2008.07.001](https://doi.org/10.1016/j.cub.2008.07.001).
- [42] Leroy W. Wheeler, Nathan H. Lents, and Joseph J. Baldassare. “Cyclin A-CDK activity during G1 phase impairs MCM chromatin loading and inhibits DNA synthesis in mammalian cells”. In: *Cell Cycle* 7.14 (2008). PMID: 18635963, pp. 2179–2188. DOI: [10.4161/cc.7.14.6270](https://doi.org/10.4161/cc.7.14.6270).
- [43] C. Gérard and A. Goldbeter. “Temporal self-organization of the cyclin/Cdk network driving the mammalian cell cycle”. In: *Proceedings of the National Academy of Sciences* 106 (2009), pp. 21643–21648.
- [44] Arne Lindqvist, Veronica Rodriguez-Bravo, and René Medema. “The decision to enter mitosis: Feedback and redundancy in the mitotic entry network”. In: *The Journal of cell biology* 185 (May 2009), pp. 193–202. DOI: [10.1083/jcb.200812045](https://doi.org/10.1083/jcb.200812045).
- [45] Marcos Malumbres and Mariano Barbacid. “Malumbres M, Barbacid MCell cycle, CDKs and cancer: a changing paradigm. Nat Rev Cancer 9: 153-166”. In: *Nature reviews. Cancer* 9 (Apr. 2009), pp. 153–66. DOI: [10.1038/nrc2602](https://doi.org/10.1038/nrc2602).
- [46] Damien Coudreuse and Paul Nurse. “Driving the cell cycle with a minimal CDK control network”. In: *Nature* 468 (Dec. 2010), pp. 1074–9. DOI: [10.1038/nature09543](https://doi.org/10.1038/nature09543).
- [47] Barbara Di Fiore and Jonathon Pines. “How cyclin A destruction escapes the spindle assembly checkpoint”. In: *The Journal of cell biology* 190 (Aug. 2010), pp. 501–9. DOI: [10.1083/jcb.201001083](https://doi.org/10.1083/jcb.201001083).
- [48] Olivier Gavet and Jonathon Pines. “Progressive Activation of CyclinB1-Cdk1 Coordinates Entry to Mitosis”. In: *Developmental Cell* 18.4 (2010), pp. 533–543. ISSN: 1534-5807. DOI: <https://doi.org/10.1016/j.devcel.2010.02.013>. URL: <https://www.sciencedirect.com/science/article/pii/S1534580710001115>.
- [49] Eusebio Manchado, Manuel Eguren, and Marcos Malumbres. “The anaphase-promoting complex/cyclosome (APC/C): cell-cycle-dependent and -independent functions”. In: *Biochemical Society Transactions* 38.1 (Jan. 2010), pp. 65–71. ISSN: 0300-5127. DOI: [10.1042/BST0380065](https://doi.org/10.1042/BST0380065). eprint: <https://portlandpress.com/biochemsoctrans/article-pdf/38/1/65/547226/bst0380065.pdf>. URL: <https://doi.org/10.1042/BST0380065>.
- [50] A. Richard. “Negative circuits and sustained oscillations in asynchronous automata networks”. In: *ArXiv abs/0907.5096* (2010).
- [51] Asako Sakaue-Sawano et al. “Drug-induced cell cycle modulation leading to cell-cycle arrest, nuclear mis-segregation, or endoreplication”. In: *BMC Cell Biology* 12 (2010), pp. 2–2.
- [52] Gustavo Arellano et al. ““Antelope”: A hybrid-logic model checker for branching-time Boolean GRN analysis”. In: *BMC bioinformatics* 12 (Dec. 2011), p. 490. DOI: [10.1186/1471-2105-12-490](https://doi.org/10.1186/1471-2105-12-490).

- [53] Elisabetta Maria et al. “Design, optimization and predictions of a coupled model of the cell cycle, circadian clock, DNA repair system, irinotecan metabolism and exposure control under temporal logic constraints”. In: *Theor. Comput. Sci.* 412 (May 2011), pp. 2108–2127. DOI: [10.1016/j.tcs.2010.10.036](https://doi.org/10.1016/j.tcs.2010.10.036).
- [54] Andrew Murray. “A brief history of error”. In: *Nature cell biology* 13 (Oct. 2011), pp. 1178–82. DOI: [10.1038/ncb2348](https://doi.org/10.1038/ncb2348).
- [55] Asako Sakaue-Sawano, Kenji Ohtawa, and Atsushi Miyawaki. “Drug-induced cell cycle modulation leading to cell-cycle arrest, nuclear mis-segregation, or endoreplication”. In: *BMC cell biology* 12 (Jan. 2011), p. 2. DOI: [10.1186/1471-2121-12-2](https://doi.org/10.1186/1471-2121-12-2).
- [56] Marieke Aarts et al. “Forced Mitotic Entry of S-Phase Cells as a Therapeutic Strategy Induced by Inhibition of WEE1”. In: *Cancer discovery* 2 (May 2012), pp. 524–39. DOI: [10.1158/2159-8290.CD-11-0320](https://doi.org/10.1158/2159-8290.CD-11-0320).
- [57] Melvin Depamphilis et al. ““The Octet”: Eight Protein Kinases that Control Mammalian DNA Replication”. In: *Frontiers in physiology* 3 (Sept. 2012), p. 368. DOI: [10.3389/fphys.2012.00368](https://doi.org/10.3389/fphys.2012.00368).
- [58] Brian Gabrielli, Kelly Brooks, and Sandra Pavey. “Defective Cell Cycle Checkpoints as Targets for Anti-Cancer Therapies”. In: *Frontiers in pharmacology* 3 (Feb. 2012), p. 9. DOI: [10.3389/fphar.2012.00009](https://doi.org/10.3389/fphar.2012.00009).
- [59] Cosetta Bertoli, Jan Skotheim, and Robertus Bruin. “Control of cell cycle transcription during G1 and S phases”. In: *Nature reviews. Molecular cell biology* 14 (Aug. 2013), pp. 518–28. DOI: [10.1038/nrm3629](https://doi.org/10.1038/nrm3629).
- [60] Maxime Folschette et al. “Under-approximation of Reachability in Multivalued Asynchronous Networks”. In: *Electronic Notes in Theoretical Computer Science* 299 (Dec. 2013), pp. 33–51. DOI: [10.1016/j.entcs.2013.11.004](https://doi.org/10.1016/j.entcs.2013.11.004).
- [61] Carito Guziolowski et al. “Exhaustively characterizing feasible logic models of a signaling network using Answer Set Programming”. In: *Bioinformatics (Oxford, England)* 30 (July 2013). DOI: [10.1093/bioinformatics/btt393](https://doi.org/10.1093/bioinformatics/btt393).
- [62] Tae Lee, Guang Yao, and Lingchong You. “Cell Cycle Transition, Principles of Restriction Point”. In: *Encyclopedia of Systems Biology*. Ed. by Werner Dubitzky et al. New York, NY: Springer New York, 2013, pp. 326–329. ISBN: 978-1-4419-9863-7. DOI: [10.1007/978-1-4419-9863-7_36](https://doi.org/10.1007/978-1-4419-9863-7_36). URL: https://doi.org/10.1007/978-1-4419-9863-7_36.
- [63] Sabrina Spencer et al. “The Proliferation-Quiescence Decision Is Controlled by a Bifurcation in CDK2 Activity at Mitotic Exit”. In: *Cell* 155 (Sept. 2013). DOI: [10.1016/j.cell.2013.08.062](https://doi.org/10.1016/j.cell.2013.08.062).
- [64] José Carlos Mombach, Cristhian Bugs, and Claudine Chaouiya. “Modelling the onset of senescence at the G1/S cell cycle checkpoint”. In: *BMC Genomics* 15 (Oct. 2014). DOI: [10.1186/1471-2164-15-S7-S7](https://doi.org/10.1186/1471-2164-15-S7-S7).
- [65] Gilles Bernot et al. “A Genetically Modified Hoare Logic”. In: *Theoretical Computer Science* 765 (June 2015). DOI: [10.1016/j.tcs.2018.02.003](https://doi.org/10.1016/j.tcs.2018.02.003).

- [66] Celine Feillet et al. “Coupling between the Circadian Clock and Cell Cycle Oscillators: Implication for Healthy Cells and Malignant Growth”. In: *Frontiers in Neurology* 6 (May 2015). DOI: [10.3389/fneur.2015.00096](https://doi.org/10.3389/fneur.2015.00096).
- [67] Maxime Folschette et al. “Sufficient Conditions for Reachability in Automata Networks with Priorities”. In: *Theoretical Computer Science* (2015). DOI: [10.1016/j.tcs.2015.08.040](https://doi.org/10.1016/j.tcs.2015.08.040). URL: <https://hal.archives-ouvertes.fr/hal-01202671>.
- [68] Fritz Melchers. “Checkpoints that control B cell development”. In: *The Journal of clinical investigation* 125 (May 2015), pp. 1–8. DOI: [10.1172/JCI78083](https://doi.org/10.1172/JCI78083).
- [69] Pauline Traynard, François Fages, and Sylvain Soliman. “Model-based Investigation of the Effect of the Cell Cycle on the Circadian Clock through Transcription Inhibition during Mitosis”. In: *CMSB’15 - Proceedings of the thirteenth international conference on Computational Methods in Systems Biology*. Ed. by Olivier Roux and Jérémie Bourdon. Vol. 9308. Lecture Notes in BioInformatics. Nantes, France: Springer-Verlag, 2015, pp. 208–221. DOI: [10.1007/978-3-319-23401-4_18](https://doi.org/10.1007/978-3-319-23401-4_18). URL: <https://hal.archives-ouvertes.fr/hal-01236264>.
- [70] Santiago Videla et al. “Learning Boolean logic models of signaling networks with ASP”. In: *Theoretical Computer Science* 599 (2015). Advances in Computational Methods in Systems Biology, pp. 79–101. ISSN: 0304-3975. DOI: <https://doi.org/10.1016/j.tcs.2014.06.022>. URL: <https://www.sciencedirect.com/science/article/pii/S0304397514004587>.
- [71] J. Behaegel et al. “A hybrid model of cell cycle in mammals”. In: *J. Bioinformatics and Comput. Biol.* 14.1 (2016), 1640001 [17 pp.]
- [72] Steven Cappell et al. “Irreversible APCCdh1 Inactivation Underlies the Point of No Return for Cell-Cycle Entry”. In: *Cell* 166 (June 2016), pp. 167–180. DOI: [10.1016/j.cell.2016.05.077](https://doi.org/10.1016/j.cell.2016.05.077).
- [73] Emilien Cornillon et al. “Hybrid Gene Networks: a new Framework and a Software Environment.” In: *Advances in Systems and Synthetic Biology, Modelling Complex Biological Systems in the Context of Genomics*. 2016. URL: <https://hal.archives-ouvertes.fr/hal-01345019>.
- [74] Dávid Deritei et al. “Principles of dynamical modularity in biological regulatory networks”. In: *Scientific Reports* 6 (Mar. 2016), p. 21957. DOI: [10.1038/srep21957](https://doi.org/10.1038/srep21957).
- [75] M. Gebser et al. “Theory Solving made easy with Clingo 5 (Extended Version *)”. In: 2016.
- [76] Denis Thieffry et al. “Logical model specification aided by model- checking techniques: application to the mammalian cell cycle regulation”. In: *Bioinformatics* 32 (Sept. 2016). DOI: [10.1093/bioinformatics/btw457](https://doi.org/10.1093/bioinformatics/btw457).
- [77] Emilien Cornillon. “Modèles qualitatifs de réseaux génétiques : réduction de modèles et introduction d’un temps continu”. Theses. Université Côte d’Azur, Oct. 2017. URL: <https://tel.archives-ouvertes.fr/tel-01682212>.
- [78] Lauren Andrews, Alec Nielsen, and Christopher Voigt. “Cellular checkpoint control using programmable sequential logic”. In: *Science* 361 (Sept. 2018), eaap8987. DOI: [10.1126/science.aap8987](https://doi.org/10.1126/science.aap8987).

- [79] Madalena Chaves and Laurent Tournier. “Analysis Tools for Interconnected Boolean Networks With Biological Applications”. In: *Frontiers in Physiology* 9 (2018). ISSN: 1664-042X. DOI: [10.3389/fphys.2018.00586](https://doi.org/10.3389/fphys.2018.00586). URL: <https://www.frontiersin.org/article/10.3389/fphys.2018.00586>.
- [80] Edmund M. Clarke et al. “Handbook of Model Checking”. In: *Springer International Publishing*. 2018.
- [81] Claude Gérard, Didier Gonze, and Albert Goldbeter. “Revisiting a skeleton model for the mammalian cell cycle: From bistability to Cdk oscillations and cellular heterogeneity”. In: *Journal of Theoretical Biology* 461 (Oct. 2018). DOI: [10.1016/j.jtbi.2018.10.042](https://doi.org/10.1016/j.jtbi.2018.10.042).
- [82] Hannes Klarner et al. *Basins of Attraction, Commitment Sets and Phenotypes of Boolean Networks*. July 2018.
- [83] Bela Novak, Frank Heldt, and John Tyson. “Genome Stability during Cell Proliferation: A Systems Analysis of the Molecular Mechanisms Controlling Progression through the Eukaryotic Cell Cycle”. In: *Current Opinion in Systems Biology* 9 (Feb. 2018). DOI: [10.1016/j.coisb.2018.02.004](https://doi.org/10.1016/j.coisb.2018.02.004).
- [84] Tony Ribeiro et al. “Inductive Learning from State Transitions over Continuous Domains”. In: Mar. 2018, pp. 124–139. ISBN: 978-3-319-78089-4. DOI: [10.1007/978-3-319-78090-0_9](https://doi.org/10.1007/978-3-319-78090-0_9).
- [85] Joshua Saldivar et al. “An intrinsic S/G 2 checkpoint enforced by ATR”. In: *Science* 361 (Aug. 2018), pp. 806–810. DOI: [10.1126/science.aap9346](https://doi.org/10.1126/science.aap9346).
- [86] G. Bernot et al. “A Genetically Modified Hoare Logic”. In: *Theoretical Computer Science* 765 (2019). <https://doi.org/10.1016/j.tcs.2018.02.003>, pp. 145–157.
- [87] Dávid Deritei et al. “A feedback loop of conditionally stable circuits drives the cell cycle from checkpoint to checkpoint”. In: *Scientific Reports* 9 (Nov. 2019). DOI: [10.1038/s41598-019-52725-1](https://doi.org/10.1038/s41598-019-52725-1).
- [88] Ousmane Diop, Laurent Tourniel, and Vincent Fromion. “Summarizing complex asynchronous Boolean attractors, application to the analysis of a mammalian cell cycle model”. In: June 2019, pp. 1677–1682. DOI: [10.23919/ECC.2019.8795712](https://doi.org/10.23919/ECC.2019.8795712).
- [89] Babatunde Ekundayo and Franziska Bleichert. “Origins of DNA replication”. In: *PLOS Genetics* 15 (Sept. 2019), e1008320. DOI: [10.1371/journal.pgen.1008320](https://doi.org/10.1371/journal.pgen.1008320).
- [90] Maxime Folschette. *The Hoare-fol Tool*. Technical Report. Univ. Lille, CNRS, Centrale Lille, UMR 9189 - CRISTAL - Centre de Recherche en Informatique Signal et Automatique de Lille, F-59000 Lille, France, Dec. 2019. URL: <https://hal.archives-ouvertes.fr/hal-02409801>.
- [91] Bennie Lemmens and Arne Lindqvist. “DNA replication and mitotic entry: A brake model for cell cycle progression”. In: *The Journal of Cell Biology* 218 (Dec. 2019), jcb.201909032. DOI: [10.1083/jcb.201909032](https://doi.org/10.1083/jcb.201909032).

- [92] Cecile Moulin, Laurent Tournier, and Sabine Pérès. “Using a hybrid approach to model central carbon metabolism across the cell cycle”. In: *Hybrid Systems Biology*. Vol. 11705. Lecture Notes in Computer Science Chapter 9. Češka M., Paoletti N. (eds), 2019. DOI: [10.1007/978-3-030-28042-0_9](https://doi.org/10.1007/978-3-030-28042-0_9). URL: <https://hal.inrae.fr/hal-02791386>.
- [93] Suyasha Roy, Zaigham Rizvi, and Amit Awasthi. “Metabolic Checkpoints in Differentiation of Helper T Cells in Tissue Inflammation”. In: *Frontiers in Immunology* 9 (Jan. 2019), p. 3036. DOI: [10.3389/fimmu.2018.03036](https://doi.org/10.3389/fimmu.2018.03036).
- [94] Wayne Stallaert et al. “Bistable switches as integrators and actuators during cell cycle progression”. In: *FEBS Letters* 593.20 (2019), pp. 2805–2816. DOI: <https://doi.org/10.1002/1873-3468.13628>. eprint: <https://febs.onlinelibrary.wiley.com/doi/pdf/10.1002/1873-3468.13628>. URL: <https://febs.onlinelibrary.wiley.com/doi/abs/10.1002/1873-3468.13628>.
- [95] Gabriele Sulli, Michael Lam, and Satchidananda Panda. “Interplay between Circadian Clock and Cancer: New Frontiers for Cancer Treatment”. In: *Trends in Cancer* 5 (Aug. 2019). DOI: [10.1016/j.trecan.2019.07.002](https://doi.org/10.1016/j.trecan.2019.07.002).
- [96] Zhihao Sun and Jeffrey Brodsky. “Protein quality control in the secretory pathway”. In: *The Journal of Cell Biology* 218 (Oct. 2019), jcb.201906047. DOI: [10.1083/jcb.201906047](https://doi.org/10.1083/jcb.201906047).
- [97] Déborah Boyenval et al. “What is a cell cycle checkpoint? The ToTemBioNet answer.” In: *18th International Conference on Computational Methods in Systems Biology (CMSB 2020)*. 2020.
- [98] Stéphanie Chevalier et al. “Synthesis and Simulation of Ensembles of Boolean Networks for Cell Fate Decision”. In: Aug. 2020.
- [99] Camilla Elaek, Valdemaras Petrosius, and Claus Sorensen. “WEE1 kinase limits CDK activities to safeguard DNA replication and mitotic entry”. In: *Mutation Research/Fundamental and Molecular Mechanisms of Mutagenesis* 819-820 (Feb. 2020), p. 111694. DOI: [10.1016/j.mrfmmm.2020.111694](https://doi.org/10.1016/j.mrfmmm.2020.111694).
- [100] Océane Marescal and Iain Cheeseman. “Cellular Mechanisms and Regulation of Quiescence”. In: *Developmental Cell* 55 (Nov. 2020), pp. 259–271. DOI: [10.1016/j.devcel.2020.09.029](https://doi.org/10.1016/j.devcel.2020.09.029).
- [101] Loïc Paulevé et al. “Reconciling Qualitative, Abstract, and Scalable Modeling of Biological Networks”. In: *bioRxiv* (2020). DOI: [10.1101/2020.03.22.998377](https://doi.org/10.1101/2020.03.22.998377). eprint: <https://www.biorxiv.org/content/early/2020/07/08/2020.03.22.998377.full.pdf>. URL: <https://www.biorxiv.org/content/early/2020/07/08/2020.03.22.998377>.
- [102] Seth M. Rubin, Julien Sage, and Jan M. Skotheim. “Integrating Old and New Paradigms of G1/S Control”. In: *Molecular Cell* 80.2 (2020), pp. 183–192. ISSN: 1097-2765. DOI: <https://doi.org/10.1016/j.molcel.2020.08.020>. URL: <https://www.sciencedirect.com/science/article/pii/S1097276520306055>.
- [103] Xianlin Zou et al. “A Systems Biology Approach Identifies Hidden Regulatory Connections Between the Circadian and Cell-Cycle Checkpoints”. In: *Frontiers in Physiology* 11 (Apr. 2020). DOI: [10.3389/fphys.2020.00327](https://doi.org/10.3389/fphys.2020.00327).

- [104] Helen Matthews, Cosetta Bertoli, and Robertus Bruin. “Cell cycle control in cancer”. In: *Nature Reviews Molecular Cell Biology* 23 (Sept. 2021). DOI: [10.1038/s41580-021-00404-3](https://doi.org/10.1038/s41580-021-00404-3).
- [105] SWIProlog. 2021. URL: <https://www.swi-prolog.org/pldoc/man?predicate=permutation/2>.
- [106] AmiGO2 Checkpoint. 2022. URL: <http://amigo.geneontology.org/amigo/search/bioentity?q=checkpoint>.
- [107] AmiGO2 Replication. 2022. URL: <http://amigo.geneontology.org/amigo/search/annotation?q=replication>.
- [108] AmiGO2 SAC. 2022. URL: <http://amigo.geneontology.org/amigo/search/annotation?q=Spindle%5C%20Assembly%5C%20Checkpoint>.
- [109] Reactome Checkpoint. 2022. URL: <https://reactome.org/PathwayBrowser/#/R-HSA-69620&DTAB=MT>.

Appendices

APPENDICES A

Parameterisation of our cell cycle GRBM

$\phi_{K_{v,\omega}}$	ϕ_{LHGM}	Conjunction of constraints
Parameters of sk interpreted over the domain $\llbracket 0, 2 \rrbracket$		
$K_{sk,\emptyset} = 0$	-	$K_{sk,\emptyset} = 0$
$K_{sk,\{restrict_gf\}} = 0$	-	$K_{sk,\{restrict_gf\}} = 0$
$K_{sk,\{rp\}} = 0$	-	$K_{sk,\{rp\}} = 0$
$K_{sk,\{no_en\}} = 0$	-	$K_{sk,\{no_en\}} = 0$
$K_{sk,\{no_a\}} = 0$	-	$K_{sk,\{no_a\}} = 0$
$K_{sk,\{restrict_gf,rp\}} = 0$	-	$K_{sk,\{restrict_gf,rp\}} = 0$
$K_{sk,\{restrict_gf,no_en\}} = 0$	-	$K_{sk,\{restrict_gf,no_en\}} = 0$
$K_{sk,\{restrict_gf,no_a\}} < 2$	$K_{sk,\{restrict_gf,no_a\}} > 0$	$K_{sk,\{restrict_gf,no_a\}} = 1$
$K_{sk,\{rp,no_en\}} = 0$	$K_{sk,\{rp,no_en\}} < 1$	$K_{sk,\{rp,no_en\}} = 0$
$K_{sk,\{rp,no_a\}} < 2$	-	$K_{sk,\{rp,no_a\}} < 2$
$K_{sk,\{no_en,no_a\}} = 0$	-	$K_{sk,\{no_en,no_a\}} = 0$
$K_{sk,\{restrict_gf,rp,no_en\}} = 0$	$K_{sk,\{restrict_gf,rp,no_en\}} < 1$	$K_{sk,\{restrict_gf,rp,no_en\}} = 0$
$K_{sk,\{restrict_gf,rp,no_a\}} < 2$	-	$K_{sk,\{restrict_gf,rp,no_a\}} = 1$ (CS)
$K_{sk,\{restrict_gf,no_en,no_a\}} < 2$	-	$K_{sk,\{restrict_gf,no_en,no_a\}} = 1$ (CS)
-	$K_{sk,\{rp,no_en,no_a\}} > 1$	$K_{sk,\{rp,no_en,no_a\}} = 2$
-	$K_{sk,\{restrict_gf,rp,no_en,no_a\}} > 1$	$K_{sk,\{restrict_gf,rp,no_en,no_a\}} = 2$
Parameters of a interpreted over the domain $\llbracket 0, 2 \rrbracket$		
$K_{a,\emptyset} = 0$	-	$K_{a,\emptyset} = 0$
$K_{a,\{sk2\}} = 0$	-	$K_{a,\{sk2\}} = 0$
$K_{a,\{no_CI\}} = 0$	-	$K_{a,\{no_CI\}} = 0$
$K_{a,\{restrict_sk\}} = 0$	-	$K_{a,\{restrict_sk\}} = 0$
$K_{a,\{end_rep\}} = 0$	$K_{a,\{end_rep\}} < 1$	$K_{a,\{end_rep\}} = 0$
$K_{a,\{sk2,no_CI\}} < 2$	$K_{a,\{sk2,no_CI\}} > 0$	$K_{a,\{sk2,no_CI\}} = 1$
$K_{a,\{sk2,restrict_sk\}} = 0$	-	$K_{a,\{sk2,restrict_sk\}} = 0$
$K_{a,\{no_CI,restrict_sk\}} < 2$	-	$K_{a,\{no_CI,restrict_sk\}} < 2$
-	$K_{a,\{end_rep,no_CI\}} > 1$	$K_{a,\{end_rep,no_CI\}} = 2$
$K_{a,\{sk2,no_CI,restrict_sk\}} < 2$	-	$K_{a,\{sk2,no_CI,restrict_sk\}} = 1$ (CS)

Parameters of b interpreted over the domain $\llbracket 0, 2 \rrbracket$

$K_{b,\emptyset} = 0$	-	$K_{b,\emptyset} = 0$
$K_{b,\{a2\}} < 2$	-	$K_{b,\{a2\}} < 2$
$K_{b,\{no_Wee1\}} = 0$	$K_{b,\{no_Wee1\}} < 1$	$K_{b,\{no_Wee1\}} = 0$
$K_{b,\{no_CI\}} = 0$	-	$K_{b,\{no_CI\}} = 0$
$K_{b,\{a2,SAC\}} = 1$	-	$K_{b,\{a2,SAC\}} = 1$
$K_{b,\{a2,no_Wee1\}} > 0$	$K_{b,\{a2,no_Wee1\}} < 2$	$K_{b,\{a2,no_Wee1\}} = 1$
-	$K_{b,\{a2,no_CI\}} > 0$	$K_{b,\{a2,no_CI\}} = 1$
$K_{b,\{no_Wee1,no_CI\}} = 0$	-	$K_{b,\{no_Wee1,no_CI\}} = 0$
-	$K_{b,\{a2,no_Wee1,no_CI\}} > 1$	$K_{b,\{a2,no_Wee1,no_CI\}} = 2$
$K_{b,\{a2,no_Wee1,SAC\}} = 2$	-	$K_{b,\{a2,no_Wee1,SAC\}} = 2$

Parameters of en interpreted over the domain $\llbracket 0, 1 \rrbracket$

$K_{en,\emptyset} = 0$	-	$K_{en,\emptyset} = 0$
$K_{en,\{no_sk\}} = 0$	-	$K_{en,\{no_sk\}} = 0$
$K_{en,\{no_a\}} = 0$	-	$K_{en,\{no_a\}} = 0$
$K_{en,\{no_b\}} = 0$	-	$K_{en,\{no_b\}} = 0$
$K_{en,\{no_sk,no_a\}} = 0$	-	$K_{en,\{no_sk,no_a\}} = 0$
$K_{en,\{no_sk,no_b\}} = 0$	-	$K_{en,\{no_sk,no_b\}} = 0$
$K_{en,\{no_a,no_b\}} = 0$	$K_{en,\{no_a,no_b\}} < 1$	$K_{en,\{no_a,no_b\}} = 0$
$K_{en,\{no_sk,no_a,no_b\}} = 1$	$K_{en,\{no_sk,no_a,no_b\}} > 0$	$K_{en,\{no_sk,no_a,no_b\}} = 1$

Parameters of ep interpreted over the domain $\llbracket 0, 1 \rrbracket$

$K_{ep,\emptyset} = 0$	$K_{ep,\emptyset} < 1$	$K_{ep,\emptyset} = 0$
-	-	$K_{ep,\{b2\}} \text{ unconstrained}$
$K_{ep,\{no_en\}} = 0$	-	$K_{ep,\{no_en\}} = 0$
-	$K_{ep,\{b2,no_en\}} > 0$	$K_{ep,\{b2,no_en\}} = 1$

Table A.1: **List of parameters of our cell cycle GRBM.** The conjunction of all parameters in the last column characterises 16 biologically consistent models. We denote this constraint in conjunctive form by ψ_{bio} .

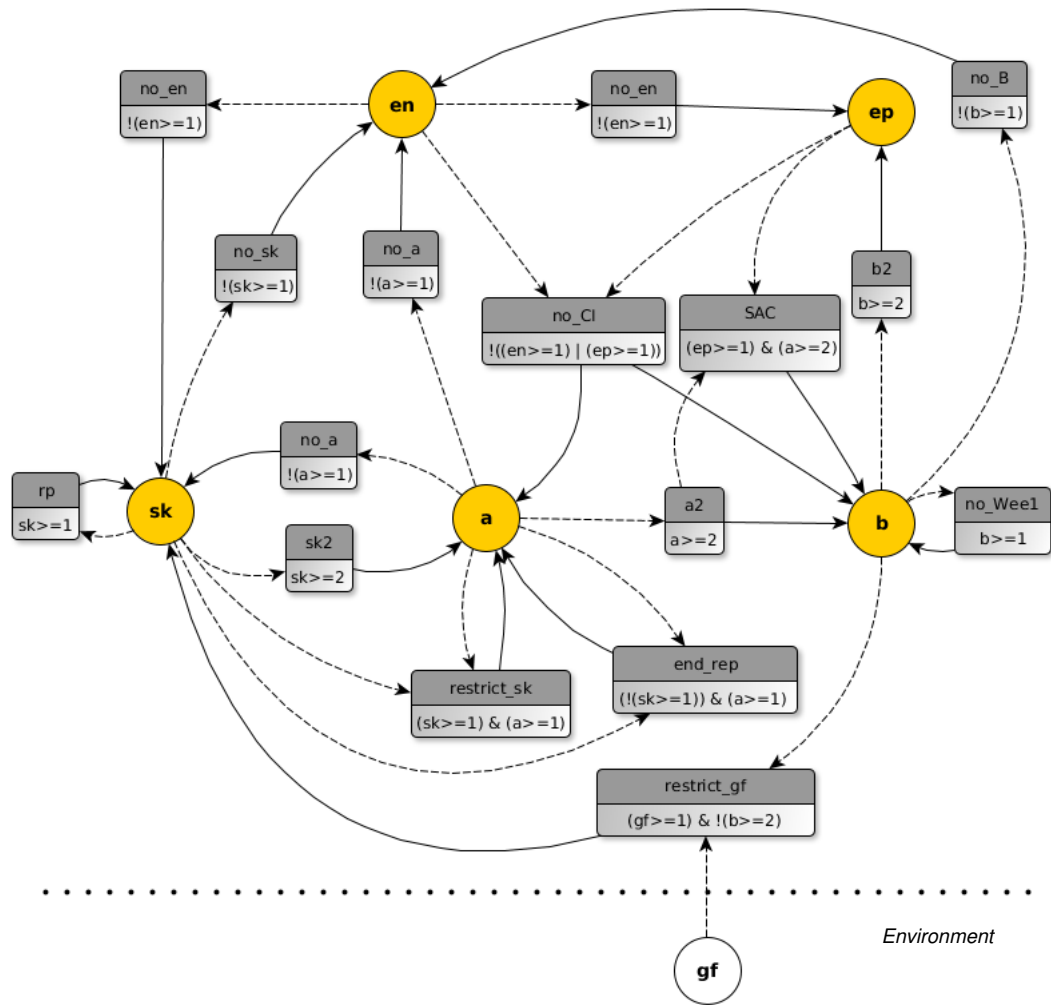


Figure A.1: The cell cycle GRBM used for the proof of concept in Chapters 6 and 7

Abstract

The cell cycle is a sequence of events *through* which a cell reproduces while preserving the integrity of its genome. This preservation property is ensured by a set of regulatory mechanisms called *checkpoints*. Their ability to arrest the cell cycle in response to DNA damage motivates extensive research in the biomedical field. This thesis argues that this arrest phenomenon, observed in a context of perturbation, is a corollary of more fundamental properties such as the temporal separation of phases and the irreversibility of phase transitions. In this context, a phase describes a particular aspect of a proliferating cell, for example the genome replication phase (S phase) or the cell division phase (M phase). The fundamental properties of checkpoints have never been formalised, essentially because phase modelling itself had never previously been addressed. The thesis presents a discrete model of cell cycle regulation and proposes a methodology for formalising a phase. A phase is initially described by a finite multiset of discrete events that occur from an initial state. By choosing René Thomas' modelling framework, we can apply genetically modified Hoare logic to compute the set of event sequences, exhibited by the initial multiset, that are compatible with a certain set of parameterisations of a regulatory graph. The set of event sequences characterising a phase temporally separates that phase from the others if there exists a parameterisation such that any event that can complete a phase is required before any event that can initiate the next phase. This problem has been specified using a Prolog predicate. The irreversibility of phase transitions is formalised using a temporal logic, and model checking identifies the parameterisations of a cell cycle model that satisfy this property. We show that this methodology for formalising and verifying the fundamental properties of checkpoints makes it possible to identify contradictions in a regulatory graph while avoiding certain arbitrary choices found in most studies of cell cycle modelling.

Abstract

The cell cycle is a series of events by which a cell reproduces itself while maintaining the integrity of its genome. This property is ensured by a set of regulatory mechanisms called *checkpoints*. Their ability to stop the cell cycle in response to DNA damage has motivated numerous studies in the field of biomedicine. This thesis argues that the phenomenon of cell cycle arrest, which is observed in the context of perturbation, is a consequence of more fundamental properties such as the temporal separation of phases and the irreversibility of phase transitions. In this context, a phase describes a particular aspect of a proliferating cell, for example the phase of genome replication (S-phase) or that of cell division (M-phase). The fundamental properties of checkpoints have never been formalized, mainly due to the fact that phase modelling itself has never been tackled before. The thesis presents a discrete model of the regulation of the cell cycle and proposes a methodology for formalizing a phase. The latter is firstly characterized by a finite multi-set of discrete events which occur from an initial state. By opting for René Thomas' modelling framework, the genetically modified Hoare logic can be applied to compute the set of sequences of events, exhibited by the initial multiset, that are consistent with a certain set of parameterizations of a regulation graph. The set of event sequences that characterizes a phase temporally insulates it from the others if there is a parameterization such that any event that can complete a phase is required before any event that can trigger the next phase. This problem has been specified using a Prolog predicate. The irreversibility property of phase transitions is formalized using a temporal logic and model-checking identifies the parameterizations of a cell cycle model that satisfy this property. We show that this method for formalizing and verifying the fundamental properties of checkpoints allows us to target the inconsistencies of a regulatory graph while getting rid of certain arbitrary choices found in most cell cycle modeling studies.